7+1: provenance

Sources, derivations, corrections and the verification record of the notebook 7+1.nb and of its documentation

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
Generated by tools/make_docs.py on 2026-10-27

Abstract

This record states, for every object of 7+1.nb, where it comes from (the task statement, the notebook 4+4.nb, or the literature), the equation that defines it, how it was derived, and how it was checked: the full list of the 419 checks and their results, the independent Python cross-checks, and the hashes of the files. Sections 1–3 concern the first stage (sections 0–6 of the notebook: octonions and Clifford algebras), section 6 the second stage (sections 7–10: the Dirac equation and the Kohn–Sham theories of the field dirac16complex), section 7 the third stage (sections 11–19: gravity, the self-coupled field in gravitational fields, the Standard Model), section 8 the fourth stage (sections 20–23: the coupled field equations and their solutions in the three gravity regimes), section 9 the analysis of section 24 (dirac16complex as dark matter and dark energy).

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1. Sources

1.1 The task statement

The task asked for a new Mathematica notebook 7+1.nb in `D:\Developer\github\7+1` defining a new realization of the octonions and a new representation of the Clifford algebra of an 8-dimensional vector space that mirror, as far as possible, the split octonions and the pseudo-Clifford algebra of 4+4.nb; real numbers and the real 8×8 and 16×16 matrices of 4+4.nb were to be used if possible, otherwise their complexification; examples were to check all work; notation was to be changed and documented where confusing. It fixed the coordinates

$$\begin{aligned}
\text{4+4.nb:}\quad & X=(x_0,\dots,x_7),\ x_0=\text{hidden space-like},\ x_1,x_2,x_3=\text{3-space},\\
& \qquad x_4=\text{time},\ x_5,x_6,x_7=\text{auxiliary time-like},\\
\text{7+1.nb:}\quad & X=(x_0,\dots,x_7),\ x_0=\text{time (octonion unit)},\ x_1,x_2,x_3=\text{3-space},\\
& \qquad x_4,\dots,x_7=\text{auxiliary space-like},
\end{aligned}$$

and gave two definitions. [1A] the octonions: $e_0=1$, $e_i^2=-1$, Fano plane, Cayley–Dickson doubling $\mathbb{O}=\mathbb{H}\oplus\mathbb{H}$, conjugation $\bar{x}=x_0-\sum_i x_i e_i$ with $\overline{xy}=\bar{y}\bar{x}$, $N(x)=x\bar{x}=\sum_A x_A^2$, $N(xy)=N(x)N(y)$, $x^{-1}=\bar{x}/N(x)$, alternativity, power-associativity. [1B] the Clifford algebra $Cl(V,Q)=\mathcal{T}(V)/\mathcal{I}_Q(V)$ with $v^2=Q(v)1$, $vw+wv=2B(v,w)1$, $B(v,w)=\frac{1}{2}(Q(v+w)-Q(v)-Q(w))$, dimension $2^8=256$, a classification table for $p+q=8$, and $Cl_8(\mathbb{C})\cong\mathbb{C}(16)$. Section 4 lists the statements of [1A] and [1B] that had to be corrected.

1.2 The notebook 4+4.nb

File `4+4.nb`, 9348656 bytes, written by Wolfram 15.0 (header `CreatedBy='Wolfram 15.0'`), SHA-256 `740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5`. It was read with the parser `tools/nbparse.py` (806 cells: 366 input, 366 output, 49 text). Every object used by 7+1.nb was reproduced with the formula of the cell given here, and each reproduction is checked in 7+1.nb section 2:

object	4+4.nb cell	definition in 4+4.nb
$\eta^{(4,4)}$	In[9]	$\mathrm{diag}(1_4,-1_4)$ (`η4488`)
σ	In[18]	$\left(\begin{smallmatrix}0 & 1_4\\ 1_4 & 0\end{smallmatrix}\right)$ (`σ=σ88`)
s_h, t_h	In[29], In[35], In[36]	$(s_h)_{pq}=\epsilon_{hpq4}-(\delta_{p4}\delta_{qh}-\delta_{ph}\delta_{q4})$, $(t_h)_{pq}=\epsilon_{hpq4}+(\dots)$
ζ_A	In[60]–In[63]	$\zeta_0=1_8$, $\zeta_{7-h}=\left(\begin{smallmatrix}0 & t_h\\ -t_h & 0\end{smallmatrix}\right)$, $\zeta_h=\left(\begin{smallmatrix}0 & s_h\\ s_h & 0\end{smallmatrix}\right)$, $\zeta_7=\zeta_1\cdots\zeta_6$
relations of ζ	In[64], In[65], In[67]	$\sigma=\zeta_1\zeta_2\zeta_3$; $\frac{1}{2}\{\zeta_A,\zeta_B\}=-\eta^{(4,4)}_{AB}1_8$; $-\eta^{(4,4)}_{AA}\zeta_A=\zeta_A^{\mathsf{T}}$
τ_A	In[70]	$\tau_0=1_8$, $\tau_A=\zeta_A$
$\bar{\tau}_A$	In[74]	$\bar{\tau}_0=1_8$, $\bar{\tau}_A=\sigma\tau_A^{\mathsf{T}}\sigma$ (symbolized tau-bar)
relations of τ	In[77], In[87], In[89]	$\frac{1}{2}(\tau_A\bar{\tau}_B+\tau_B\bar{\tau}_A)=\eta^{(4,4)}_{AB}1_8$; $\tau_1\cdots\tau_7=1_8$; $\sigma=\tau_4\tau_5\tau_6\tau_7$

object	4+4.nb cell	definition in 4+4.nb
T_A	In[98]	$T_A = \left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$ (`T16^A[A1]`)
T_8, σ_{16}	In[99], In[97], In[106]	$T_8 = T_0 \cdots T_7$; $\sigma_{16} = T_0T_1T_2T_3 = \mathrm{diag}(-\sigma, \sigma)$
Clifford relation	In[110]	$\frac{1}{2}\{T_A, T_B\} = \eta^{(4,4)}_{AB} 1_{16}$
P_L, P_R	In[112]–In[114]	$P_L = \frac{1}{2}(1_{16} - T_8)$, $P_R = \frac{1}{2}(1_{16} + T_8)$
U_h	In[257], In[259]	eigenvectors of σ divided by $\sqrt{2}$
u	In[264], In[266]	`hUSE = 8`, $u = U_8 = (1,0,0,0,1,0,0,0)/\sqrt{2}$
g	In[283]	$g_{AB} = \eta^{(4,4)}$
E_{44}, E_{44}^{-1}	In[294]–In[296]	$(E_{44})_{Ba} = (U_h^{\mathsf T}\sigma\bar\tau_B)_a$; $E_{44}^{-1} = (U_h^{\mathsf T}\sigma U_h)(\eta^{(4,4)}_{BB}\tau_B U_h)_B$
ε_B	In[313]	`EA = Array[eA, 8]`
split table	In[325], Out[325]	$\varepsilon_B\varepsilon_C = \sum_D (g_{BB}E_{44}\tau_{B-1}E_{44}^{-1})_{DC}\varepsilon_D$ (`gridx`)

1.3 Literature

- J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002) 145–205, arXiv:math/0105155, section 2.2 (the page https://math.ucr.edu/home/baez/octonions/node5.html, which the task statement itself cites, was read on 2026-10-27): the Cayley–Dickson product $(a,b)(c,d) = (ac - db^*,\ a^*d + cb)$, where Baez writes x^* for the conjugate $\bar x$ and the conjugate of a pair is $(a,b)^* = (a^*, -b)$. 7+1.nb proves that its octonions satisfy this product with $x = a + \ell b$, $\ell = e_4$.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton (1989): classification of real Clifford algebras, $Cl(p+1, q+1) \cong Cl(p,q) \otimes \mathbb{R}(2)$. 7+1.nb does not rely on it: it computes the classification directly (section 2.10 below).
- Standard facts used in proofs: Schur's lemma (the intertwiners of an irreducible module form a division algebra; over $\mathbb{C}$, the scalars); $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ has a unique irreducible module $\mathbb{C}^{16}$; Hurwitz's theorem (a unital composition algebra of dimension 8 is the octonions if its norm is positive definite and the split octonions if the norm is indefinite); Degen's eight-square identity.

1.4 The signature: the author's explanation of 27 October 2026

Source. On 27 October 2026 the author of the project explained that the xAct list $\{7, 1, 0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1, -1, \ldots, -1)$ is simply the squares of the octonion basis, and asked that this be made clear in all documentation, examples and provenance records. **What it corrected.** The version of 7+1.nb pushed in the commits 52d8729 to dc2f178 of github.com/once-ere/7plus1 declared the xAct metric of section 12 with `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of $\eta^{(1,7)}$; since then section 12 declares `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`, exactly as the author's notebook Einstein-Lovelock-7+1-Nash.nb does. **What was added.** Checks in 7+1.nb — section 4.7: $\eta^{(1,7)}$ is the squares of the octonion basis, $e_Ae_A = \eta^{(1,7)}_{AA}e_0$ and $\mathrm{Re}(e_Ae_B) = \eta^{(1,7)}_{AB}$; section 12.2: `SignatureOfMetric[g71]` $= \{7,1,0\}$ with `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_Ae_B) = \eta^{(1,7)}_{AB}$, and $\mathrm{Re}(x\,x) = \eta^{(1,7)}_{AB}x^Ax^B$; section 17: the flat metric $\mathrm{diag}(+1,-1,-1,-1)$ of the world brane is the squares of the basis e_0, e_1, e_2, e_3 of the quaternion subalgebra — and the same statements in the Python programs `tools/ref71.py`, `tools/ref_gravity.py` and `tools/ref_sm.py`; the explanation itself in the text of 7+1.nb (sections 1.2, 1.4, 4.2, 4.7, 7, 11, 12.2, 13, 15.1, 17) and in every document, symbol file and provenance file of the repository. **What did not change.** No result. Compared output by output with the backup below, every output of 7+1.nb is the same, except the declared list itself, the new checks and the check counts (and, as in every run, the name of the private context and the measured times); in particular every component of every Lovelock tensor is unchanged. The components follow from the component matrix of the metric, and the sign of the determinant, `SignDetOfMetric`, is -1 for both lists. **Backup.** The evaluated notebook as it was before is kept in the workspace as `backups/7+1_2026-10-27_before_signature.nb` (the folder backups/ is excluded from the repository by .gitignore), 1034401 bytes, SHA-256 `060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`; `tools/verify_7p1_nb.py` compares every input, text and section cell of 7+1.nb with it and finds every cell present, each one unchanged or changed by this explanation, and it compares every output in the same way.

The explanation, in full. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta^{(7,1)} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta^{(1,7)} = -\eta^{(7,1)} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta^{(1,7)}_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta^{(1,7)}_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

$\eta^{(1,7)}$ gives the signs of the Clifford relation $\Gamma^{(1,7)}_A \Gamma^{(1,7)}_B + \Gamma^{(1,7)}_B \Gamma^{(1,7)}_A = 2\,\eta^{(1,7)}_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta^{(1,7)} = \det \eta^{(7,1)} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta^{(1,7)}_{AB}$ and $\mathrm{Re}(x\,x) = \eta^{(1,7)}_{AB} x^A x^B$.

2. Derivations: how each new object was obtained

2.1 Octonion matrices τ^O

Origin: the split matrices of 4+4.nb and the observation (checked) that $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ commutes with τ_1, τ_2, τ_3, anticommutes with $\tau_4, \ldots, \tau_7$ and squares to 1_8. Hence

$$\tau^O_0 = 1_8, \quad \tau^O_A = \tau_A \ (A \le 3), \quad \tau^O_k = \tau_k\sigma \ (k \ge 4), \qquad (\tau_k\sigma)^2 = \tau_k\sigma\tau_k\sigma = -\tau_k^2\sigma^2 = -1_8,$$
$$\tau^O_h\tau^O_k = \tau_h\tau_k\sigma = -\tau_k\tau_h\sigma = -\tau_k\sigma\tau_h = -\tau^O_k\tau^O_h, \qquad \tau^O_k\tau^O_l = \tau_k\sigma\tau_l\sigma = -\tau_k\tau_l = \tau_l\tau_k = -\tau^O_l\tau^O_k \quad (k \ne l \ge 4).$$

This is the real counterpart of the complex substitution $\tau_k \to i\tau_k$ that turns split octonions into octonions over $\mathbb{C}$.

2.2 Octonion vielbein and multiplication constants

Origin: In[294] and In[325] of 4+4.nb with the split spinor metric σ replaced by the Euclidean $g^O = 1_8$ and the same unit spinor u:

$$(E^O)_{Ba} = \left(u^{\mathsf{T}} g^O \bar\tau^O_B\right)_a, \qquad L^O_A = g^O_{AA} E^O \tau^O_A (E^O)^{-1}, \qquad e_A e_K = \sum_L (L^O_A)_{LK} e_L.$$

Why the result is the octonion algebra: $(E^O)^{-1} e_0 = u$ makes e_0 a two-sided unit, and $\tau^O(x)\bar\tau^O(x) = N(x) 1_8$ together with the orthogonality of E^O gives $N(xy) = N(x)N(y)$. A unital composition algebra of dimension 8 with a positive definite norm is the octonions (Hurwitz). The notebook checks all the laws directly.

2.3 Cayley–Dickson form

Origin: Baez 2002, section 2.2. With $\ell = e_4$, $x = a + \ell b$, $a = (x_0, \ldots, x_3)$, $b = (x_4, \ldots, x_7)$:

$$(a, b)(c, d) = \left(ac + \gamma\, d\bar b,\ \bar a d + cb\right), \qquad \gamma = -1 \text{ (7+1.nb)}, \qquad \gamma = +1 \text{ (4+4.nb)}.$$

2.4 Euclidean mirror T^O

Origin: In[98] of 4+4.nb with $\tau \to \tau^O$: $T^O_A = \left(\begin{smallmatrix} 0 & \bar\tau^O_A \\ \tau^O_A & 0 \end{smallmatrix}\right)$, so that $\{T^O_A, T^O_B\} = \left(\begin{smallmatrix} \bar\tau^O_A\tau^O_B + \bar\tau^O_B\tau^O_A & 0 \\ 0 & \tau^O_A\bar\tau^O_B + \tau^O_B\bar\tau^O_A \end{smallmatrix}\right) = 2\delta_{AB}1_{16}$. In terms of 4+4.nb: $T^O_A = T_A$ $(A \le 3)$, $T^O_k = T_k S_{44}$ $(k \ge 4)$, $S_{44} = T_4T_5T_6T_7 = \mathrm{diag}(\sigma,\sigma)$.

2.5 Real 7+1 generators $\Gamma^{(1,7)}$

Origin: the standard construction of a Lorentzian Clifford algebra from a Euclidean one, applied to T^O:

$$\Gamma^{(1,7)}_0 = T^O_0, \qquad \Gamma^{(1,7)}_i = T^O_0 T^O_i, \qquad (T^O_0T^O_i)^2 = -(T^O_0)^2(T^O_i)^2 = -1_{16},$$
$$\{T^O_0, T^O_0T^O_i\} = T^O_i + T^O_0T^O_iT^O_0 = T^O_i - T^O_i = 0, \qquad \{T^O_0T^O_i, T^O_0T^O_j\} = -\{T^O_i, T^O_j\} = -2\delta_{ij}1_{16}.$$

Hence $\{\Gamma^{(1,7)}_A, \Gamma^{(1,7)}_B\} = 2\eta^{(1,7)}_{AB}1_{16}$ — the generators square like the octonion basis, since $\eta^{(1,7)}_{AB} = \mathrm{Re}(e_Ae_B)$ ($e_0^2 = +1$, $e_i^2 = -1$) — and $\Gamma^{(1,7)}(x)^2 = \mathrm{Re}(xx)1_{16}$.

2.6 Complexified generators $\Gamma^{(7,1)}$

Origin: the task's instruction to use the complexification of the 4+4 matrices if real numbers fail. Only the time block is multiplied by i: $\tau^C_0 = \bar\tau^C_0 = i1_8$, $\tau^C_k = \tau^O_k$; then $\Gamma^{(7,1)}_A = \left(\begin{smallmatrix} 0 & \bar\tau^C_A \\ \tau^C_A & 0 \end{smallmatrix}\right)$, $(\Gamma^{(7,1)}_0)^2 = i^2\,1_{16} = -1_{16}$, and $\Gamma^{(7,1)}_A = S(i\Gamma^{(1,7)}_A)S^{-1}$ with $S = (1_{16} - i\Gamma^{(1,7)}_0)/\sqrt{2}$.

2.7 The impossibility theorem

Origin: Schur's lemma and the uniqueness of the irreducible module of $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$. The intertwiners M with $M\overline{\Gamma^{(7,1)}_A} = \Gamma^{(7,1)}_A M$ are computed in 7+1.nb with `NullSpace`; they form the line through $M_0 = \Gamma^{(1,7)}_8 = \left(\begin{smallmatrix} 0 & -1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, and $M_0\overline{M_0} = -1_{16}$. A real realization $R_A = T\Gamma^{(7,1)}_A T^{-1}$ would give an intertwiner $M = T^{-1}\overline{T}$ with $M\overline{M} = 1_{16}$, impossible because $M = cM_0$ gives $M\overline{M} = -|c|^2 1_{16}$.

2.8 Real 32×32 generators

Origin: realification. $\widehat\Gamma^{(7,1)}_A = \varepsilon \otimes \Gamma^{(1,7)}_A$ with $\varepsilon^2 = -1_2$ gives $\{\widehat\Gamma^{(7,1)}_A, \widehat\Gamma^{(7,1)}_B\} = \varepsilon^2 \otimes 2\eta^{(1,7)}_{AB}1_{16} = 2\eta^{(7,1)}_{AB}1_{32}$. Commutant: $Q_i = \varepsilon \otimes 1_{16}$, $Q_j = \sigma_1 \otimes \Gamma^{(1,7)}_8$, $Q_k = \sigma_3 \otimes \Gamma^{(1,7)}_8$, found from $\varepsilon\sigma_1 = -\sigma_1\varepsilon$, $\varepsilon\sigma_3 = -\sigma_3\varepsilon$ and $\Gamma^{(1,7)}_8\Gamma^{(1,7)}_A = -\Gamma^{(1,7)}_A\Gamma^{(1,7)}_8$.

2.9 Examples

so(1,7): $\Sigma_{AB} = \frac14[\Gamma^{(1,7)}_A, \Gamma^{(1,7)}_B]$; the commutation relations follow from the Clifford relation. Boost and rotation: $(\Gamma^{(1,7)}_0\Gamma^{(1,7)}_1)^2 = 1_{16}$, $(\Gamma^{(1,7)}_4\Gamma^{(1,7)}_5)^2 = -1_{16}$, so $e^{\beta\Sigma_{01}} = \cosh\frac{\beta}{2} + \sinh\frac{\beta}{2}\Gamma^{(1,7)}_0\Gamma^{(1,7)}_1$ and $e^{\theta\Sigma_{45}} = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}\Gamma^{(1,7)}_4\Gamma^{(1,7)}_5$. Dirac operator: $(\Gamma^A\partial_A)^2 = \frac12\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \eta^{AB}_{(1,7)}\partial_A\partial_B$. Real amplitudes: $p = (3,2,2,0,0,0,0,0)$, $p^{\mathsf T}\eta^{(1,7)}p = 1$, $\not p^{\,2} = 1_{16}$; the real matrix $\not p$ has a real 8-dimensional eigenspace for the eigenvalue $+1$.

Real spinor fields. If J anticommutes with every $\Gamma^{(1,7)}_A$, then $J\Gamma^{(1,7)}_8$ commutes with every $\Gamma^{(1,7)}_A$ (because $\Gamma^{(1,7)}_8$ anticommutes with each of them), so by irreducibility $J\Gamma^{(1,7)}_8 = c\,1_{16}$ and $J = -c\,\Gamma^{(1,7)}_8$ (as $(\Gamma^{(1,7)}_8)^{-1} = -\Gamma^{(1,7)}_8$), whose square is $-c^2\,1_{16}$. Hence the real operator $\Gamma^A\partial_A - mJ$ has the square $\partial_0^2 - \cdots - \partial_7^2 - c^2m^2$ (a tachyon), and a real equation $(G^A\partial_A - m)\Psi = 0$ implies the Klein–Gordon equation only if the real G_A obey the $\eta^{(7,1)}$ relation, impossible for 16×16 matrices (2.7). Massless: for $k = (1,1,0,\ldots,0)$, $\not k^{\,2} = k^{\mathsf T}\eta^{(1,7)}k\,1_{16} = 0$ and $\Gamma^A\partial_A[w\cos(k\cdot x)] = -\sin(k\cdot x)\ \not k\, w$, which vanishes for the 8 real solutions of $\not k\, w = 0$. Massive, with the real 32×32 matrices: $P = \sum_A p^A\widehat\Gamma^{(7,1)}_A$, $P^2 = -m^2\,1_{32}$, and with $\phi = p\cdot x$, $\widehat\Gamma^A = \eta^{AA}_{(7,1)}\widehat\Gamma^{(7,1)}_A$:

$$\widehat\Gamma^A\partial_A\Big(w\cos\phi + \tfrac1m Pw\sin\phi\Big) = -P\Big(-w\sin\phi + \tfrac1m Pw\cos\phi\Big)$$

$$= Pw\sin\phi + m\,w\cos\phi = m\Big(w\cos\phi + \tfrac{1}{m}Pw\sin\phi\Big),$$

using $\partial_A\phi = (\eta^{(1,7)}p)_A$ and $\eta^{AA}_{(7,1)}\eta^{(1,7)}_{AA} = -1$.

2.10 Classification of the nine signatures

For each p, complex generators $G_A = T^O_A$ $(A < p)$, $G_A = iT^O_A$ $(A \geq p)$ satisfy $\{G_A, G_B\} = \pm 2\delta_{AB}1_{16}$ with p plus signs. The indicator sign of $M\overline{M}$ ($M\overline{G_A} = G_A M$) is $+1$ for $p \in \{8,5,4,1,0\}$ ($\mathbb{R}(16)$) and -1 for $p \in \{7,6,3,2\}$ ($\mathbb{H}(8)$). An independent numerical method (`tools/classify_clifford.py`: commutant of the realified 32-dimensional representation, whose traceless part has a negative definite trace form exactly in the quaternionic case) gives the same table. Explicit real generators: T^O (8,0), $\Gamma^{(5,3)}_0 = T_0$, $\Gamma^{(5,3)}_i = T_0T_i$ (5,3) with $\eta^{(5,3)} = \mathrm{diag}(1,-1,-1,-1,1,1,1,1)$, T (4,4), $\Gamma^{(1,7)}$ (1,7), $\Gamma^{(0,8)}_A = T^O_8 T^O_A$ (0,8).

3. Corrections to the task statement (with the evidence in 7+1.nb)

statement in the task	**correction**	**evidence (check in 7+1.nb)**
[1A]: $(a,b)(c,d) = (ac - \bar{d}b,\ \bar{a}d + cb)$ defines the octonions	not alternative, norm not multiplicative; correct form $(ac - d\bar{b},\ \bar{a}d + cb)$ (Baez 2002, 2.2)	"the formula ... of [1A] is NOT alternative", "... its norm is not multiplicative", "Cayley-Dickson, γ = -1 ..."
[1A]: first formula $(ac - \bar{d}b,\ da + cb\bar{x})$	contains a stray "x"; not interpretable	—
[1B]: 1+8+28+56+72+56+28+8+1 = 256	$\binom{8}{4} = 70$	"grades: binomial(8,k) == ..."
[1B]: $Cl(7,1) \cong \mathbb{R}(16)$	$Cl(7,1) \cong \mathbb{H}(8)$ for $v^2 = Q(v)$	"indicator(Γ71) == -1 ...", "classification: ..."
[1B]: $Cl(5,3) \cong \mathbb{C}(16)$, $Cl(3,5) \cong \mathbb{C}(16)$	$Cl(5,3) \cong \mathbb{R}(16)$, $Cl(3,5) \cong \mathbb{H}(8)$; $\mathbb{C}(16)$ has real dimension 512	"classification: ...", "$\mathbb{C}(16)$ cannot occur ..."
[1B]: metric-sign strings for (6,2), (5,3), (4,4), (2,6)	mistyped (wrong number of signs)	table of section 5.6 of 7+1.nb
split octonions are a "pseudo-normed division algebra"	they have zero divisors: $(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = 0$	"split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 ..."

4. Verification record

4.1 The Mathematica run

item	**value**
program	`C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`
$Version	15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)
method	`Get["D:/Developer/github/7+1/tools/run_eval.wl"]`: all cells evaluated (`SelectionEvaluate`) in a fresh kernel (front-end kernel configuration "Eval71"), then `NotebookSave`; earlier stages: `NotebookEvaluate[nb, InsertResults -> True]` in the front end's kernel
finished	Tue 27 Oct 2026 20:36:01
evaluation time	263.848 s
private context of the notebook	`Notebook$$67$147097`
input cells / output cells / message cells	174 / 481 / 0
checks / True	419 / 419 (383 printed where they stand, 36 by `frameChecks` in the summary table)
commutant computation (8192×1024 sparse `NullSpace`)	1.0357279 s
SHA-256 of the evaluated 7+1.nb (1940664 bytes)	`39f64ee81d35c0c9fb94188fd7f8c8e75c96ea198fbf72a5ca6b3c0027426018`

4.2 All checks of 7+1.nb and their results

Each row is one check `check["name", test]` of the notebook, in the order in which the notebook makes them (the order of the summary table of its last section), grouped by the section of the notebook in which it stands; the test is the Wolfram Language expression of that check, and the result is what Wolfram 15.0.1 returned. The 36 checks of the generalized frame matrices are made by the function `frameChecks` of section 13.1, nine for each of four metrics, in sections 13.2 (cosmological metric), 13.3 (pp-wave, inhomogeneous metric) and 13.5 (static radial metric), where they are listed; only the summary table of section 25 prints them.

2.1 σ, the self-dual and anti-self-dual blocks, ζ, τ and τbar

#	check	result
1	4+4: ζ[7] == diag(-1,-1,-1,-1,1,1,1,1)	True
2	4+4: σ == ζ[1].ζ[2].ζ[3]	True
3	4+4: (ζ[A].ζ[B] + ζ[B].ζ[A])/2 == -η4488[[A+1,B+1]] ID8 (A,B = 1..7)	True
4	4+4: -η4488[[A+1,A+1]] ζ[A] == Transpose[ζ[A]] (A = 1..7)	True
5	4+4: (τ[A].τbar[B] + τ[B].τbar[A])/2 == η4488[[A+1,B+1]] ID8	True
6	4+4: (τbar[A].τ[B] + τbar[B].τ[A])/2 == η4488[[A+1,B+1]] ID8	True
7	4+4: τ[1].τ[2]...τ[7] == ID8	True
8	4+4: σ == τ[1].τ[2].τ[3] == τ[4].τ[5].τ[6].τ[7]	True
9	4+4: σ commutes with τ[1..3] and anticommutes with τ[4..7]; σ.σ == ID8	True

2.2 The 4+4 Dirac matrices T16 and the chiral projectors

#	check	result
10	4+4: (T16[A].T16[B] + T16[B].T16[A])/2 == η4488[[A+1,B+1]] ID16	True
11	4+4: T16[A] real (entries 0, ±1)	True
12	4+4: T16[8] == diag(-ID8, ID8)	True
13	4+4: σ16 == diag(-σ, σ)	True
14	4+4: PL44 + PR44 == ID16, PL44.PL44 == PL44, PR44.PR44 == PR44, PL44.PR44 == 0	True

2.3 The split-octonion multiplication constants (gridx of 4+4.nb) and how they come from τ

#	check	result
15	4+4: uBasis == evecs/Sqrt[2] from Eigensystem[σ] (the basis of 4+4.nb)	True
16	4+4: unit == (1,0,0,0,1,0,0,0)/Sqrt[2], σ.unit == unit, unit.unit == 1, unit.σ.unit == 1	True
17	4+4 (In[294]): E44inv[h] == (u.σ.u) Transpose[Table[η4488[[B,B]] τ[B-1].u, {B,1,8}]] for h = 1..8	True
18	4+4: gridx44 is identical to the split-octonion table printed by 4+4.nb	True
19	4+4: LS[A] are signed permutation matrices (constants 0, ±1)	True

2.4 Split octonions: composition algebra, but not a division algebra

#	check	result
20	split: eA[1] is the identity; eA[2..4]^2 == -eA[1]; eA[5..8]^2 == +eA[1]	True
21	split: x x̄ == N(x) eA[1] with N = η4488 (symbolic)	True
22	split: N(xy) == N(x) N(y) (symbolic)	True
23	split: alternative x(xy) == (xx)y and (yx)x == y(xx) (symbolic)	True
24	split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 with both factors nonzero: NOT a division algebra	True

3.1 From the split matrices τ to the octonionic matrices τO

#	check	result
25	τO[A] (A = 1..7): real, antisymmetric, τO[A].τO[A] == -ID8	True
26	τO[A] anticommute: τO[A].τO[B] == -τO[B].τO[A] (A != B, A,B = 1..7)	True
27	mirror of 4+4: (τO[A].τObar[B] + τO[B].τObar[A])/2 == gO[[A+1,B+1]] ID8	True
28	mirror of 4+4: (τObar[A].τO[B] + τObar[B].τO[A])/2 == gO[[A+1,B+1]] ID8	True
29	mirror of 4+4: τO[1].τO[2]...τO[7] == ID8	True
30	mirror of 4+4: τObar[A] == gO.Transpose[τO[A]].gO, as τbar[A] == σ.Transpose[τ[A]].σ	True
31	τO[4..7] are products of three split τ: τO[4] == τ[5].τ[6].τ[7], τO[5] == -τ[4].τ[6].τ[7], τO[6] == τ[4].τ[5].τ[7], τO[7] == -τ[4].τ[5].τ[6]	True

3.2 The octonion vielbein (the same unit spinor u as in 4+4.nb)

#	check	result
32	σ.unit == unit, hence τO[A].unit == τ[A].unit (A = 0..7)	True
33	EO is orthogonal: EOinv == Transpose[EO]	True
34	mirror of 4+4.nb In[294]: EOinv == (u.gO.u) Transpose[Table[gO[[B,B]] τO[B-1].u, {B,1,8}]]	True
35	EO == η4488.E44[hUSE]: the two vielbeins differ only by the signs of rows 5..8	True

3.3 The octonion multiplication constants: gridO, the mirror of gridx

#	check	result
36	gridO equals the octonion table computed independently in Python	True
37	LO[A] are signed permutation matrices (all multiplication constants are 0, ±1)	True
38	oMul reproduces gridO on all 64 pairs of basis units	True

3.4 The properties listed in [1A], verified

#	check	result
39	e[0] is a two-sided identity	True
40	e[i] e[i] == -e[0] (i = 1..7)	True
41	e[i] e[j] == -e[j] e[i] (i != j, i,j = 1..7): non-commutative	True
42	7 Fano lines; every pair of distinct imaginary units lies on exactly one line	True
43	each oriented line (a,b,c) is cyclic: e[a]e[b] == e[c], e[b]e[c] == e[a], e[c]e[a] == e[b]	True
44	each line spans a quaternion subalgebra: {e[0], e[a], e[b], e[c]} closed and associative	True
45	the picture's 7 lines are exactly fanoLines	True
46	left alternative: x(xy) == (xx)y (symbolic)	True
47	right alternative: (yx)x == y(xx) (symbolic)	True
48	flexible: (xy)x == x(yx) (symbolic)	True
49	associator is alternating: [x,y,z] == -[y,x,z] == -[x,z,y] (symbolic)	True
50	power associativity: x(xx) == (xx)x (symbolic)	True
51	Moufang: z(x(zy)) == ((zx)z)y, x(z(yz)) == ((xz)y)z, (zx)(yz) == (z(xy))z (x, y basis units, z symbolic)	True
52	x x̄ == x̄ x == N(x) e[0] (symbolic)	True
53	conjugation is an anti-automorphism: conj(xy) == conj(y) conj(x) (symbolic)	True
54	Re(x) == (x + x̄)/2 == x0	True
55	norm is multiplicative: N(xy) == N(x) N(y) (symbolic; Degen's eight-square identity)	True
56	inverse x^-1 = x̄/N(x): x x^-1 == x^-1 x == e[0] (symbolic)	True
57	every x satisfies x x - 2 Re(x) x + N(x) e[0] == 0, so the subalgebra generated by x is $\mathbb{R}$ or $\mathbb{C}$	True
58	exactly 168 of the 512 ordered basis triples are non-associative (= 7*6*4: distinct imaginary units not on one Fano line)	True

#	check	result
59	example: (e[1]e[2])e[4] == e[7] but e[1](e[2]e[4]) == -e[7]	True
60	numerical example: N(x y) == N(x) N(y) == 15*6 == 90 and x x^-1 == e[0]	True
61	numerical example: x y != y x and (x y) z != x (y z)	True

3.5 Cayley–Dickson doubling: $\mathbb{O} = \mathbb{H} \oplus \ell\mathbb{H}$

#	check	result
62	$\mathbb{H}$ = span{e[0],e[1],e[2],e[3]} is a subalgebra	True
63	ℓ = e[4] doubles $\mathbb{H}$: e[4] e[k] == e[4+k] (k = 1,2,3) and e[4] e[4] == -e[0]	True
64	orientation inherited from 4+4.nb: e[1] e[2] == -e[3]	True
65	Cayley-Dickson, γ = -1: (ac - d $\bar{b}$, $\bar{a}$ d + c b) reproduces the octonion product (symbolic)	True
66	Cayley-Dickson, γ = +1: (ac + d $\bar{b}$, $\bar{a}$ d + c b) reproduces the SPLIT product of 4+4.nb (symbolic)	True
67	the formula (ac - $\bar{d}$ b, $\bar{a}$ d + c b) of [1A] is NOT alternative: x·x·y != x(xy) for x = e[0]+e[4], y = e[1]	True
68	... and its norm is not multiplicative (symbolic)	True

3.6 Octonions versus split octonions

#	check	result
69	e[A]e[B] (octonion) == s_AB e[A]e[B] (split) for all A, B = 0..7	True
70	over $\mathbb{C}$: ϕ(x *split y) == ϕ(x) *oct ϕ(y) (symbolic): split $\otimes$ $\mathbb{C}$ $\cong$ $\mathbb{O}$ $\otimes$ $\mathbb{C}$	True
71	the norms correspond: N_oct(ϕ(x)) == N_split(x) (symbolic)	True

4.1 The exact mirror of T16: the Euclidean Clifford algebra Cl(8,0)

#	check	result
72	Cl(8,0): (T16O[A].T16O[B] + T16O[B].T16O[A])/2 == gO[[A+1,B+1]] ID16	True
73	S44 == diag(σ, σ)	True
74	T16O from the 4+4 matrices: T16O[A] == T16[A] (A = 0..3), T16O[k] == T16[k].S44 (k = 4..7)	True
75	T16O[8] == T16[8] == diag(-ID8, ID8): same chirality matrix and same PL44, PR44 as 4+4.nb	True

4.2 The REAL Clifford algebra of 7+1 spacetime: Γ17, Cl(1,7) $\cong$ $\mathbb{R}$(16)

#	check	result
76	7+1 REAL Clifford relation: (Γ17[A].Γ17[B] + Γ17[B].Γ17[A])/2 == η17[[A+1,B+1]] ID16 == -η71[[A+1,B+1]] ID16	True
77	Γ17[A] are REAL: every entry is 0 or ±1	True
78	block form: Γ17[0] == {{0,ID8},{ID8,0}}, Γ17[i] == {{τO[i],0},{0,τObar[i]}}	True
79	built from the real 4+4 matrices: Γ17[0] == T16[0], Γ17[i] == T16[0].T16[i] (i = 1,2,3), Γ17[k] == T16[0].T16[k].S44 (k = 4..7)	True
80	Γ17[0] symmetric, Γ17[i] antisymmetric, so Transpose[Γ17[A]] == Γ17[0].Γ17[A].Γ17[0] (real Dirac adjoint)	True
81	the 256 products Γ17[A1]...Γ17[Ak] (A1 < ... < Ak) are linearly independent: Cl(1,7) $\cong$ $\mathbb{R}$(16)	True
82	grades: binomial(8,k) == {1,8,28,56,70,56,28,8,1}, total 256 (binomial(8,4) == 70, not 72)	True
83	Tr[product] == 0 for every non-empty product, so Tr[M]/16 is the scalar part of M	True
84	irreducible over $\mathbb{R}$: only multiples of ID16 commute with all Γ17[A] (Schur; real commutant of dimension 1)	True

#	check	result
85	universal property: Γ17(v).Γ17(v) == (v.η17.v) ID16 for a symbolic vector v = X	True

4.3 The complexified 7+1 realization in the exact block form of 4+4.nb: Γ71

#	check	result
86	7+1 COMPLEX Clifford relation, 4+4.nb sign: (Γ71[A].Γ71[B] + Γ71[B].Γ71[A])/2 == η71[[A+1,B+1]] ID16	True
87	Γ71[0] == I T16[0] and Γ71[k] == T16O[k] (k = 1..7): only the time component is complex	True
88	S = (ID16 - I Γ17[0])/Sqrt[2] is unitary and Γ71[A] == S.(I Γ17[A]).Inverse[S]	True

4.4 Theorem: with η71 there is no real 16×16 realization, because Cl(7,1) ≅ ℍ(8)

#	check	result
89	Schur: the intertwiners M with M.Conjugate[Γ71[A]] == Γ71[A].M form a 1-dimensional space	True
90	... spanned by the volume element Γ17[8] == {{0,-ID8},{ID8,0}}, and Γ17[8].Γ17[8] == -ID16	True
91	indicator(Γ71) == -1: Cl(7,1) ≅ ℍ(8), NO real 16×16 realization of the η71 relation	True
92	indicator(Γ17) == +1: Cl(1,7) ≅ ℝ(16), real realization exists (Γ17)	True

4.5 A real realization of Cl(7,1) needs 32×32 matrices

#	check	result
93	REAL 32×32 realization of the η71 relation: (Γ71R32[A].Γ71R32[B] + Γ71R32[B].Γ71R32[A])/2 == η71[[A+1,B+1]] ID32	True
94	Qi, Qj, Qk commute with every Γ71R32[A]; Qi.Qi == Qj.Qj == Qk.Qk == -ID32; Qi.Qj == Qk == -Qj.Qi	True
95	the commutant of {Γ71R32[A]} has dimension exactly 4 and is spanned by ID32, Qi, Qj, Qk: it is ℍ	True

4.6 Volume elements and chirality

#	check	result
96	Γ17[8].Γ17[8] == -ID16 and Γ17[8] anticommutes with every Γ17[A]: no real chirality in 7+1	True
97	Γ71[8].Γ71[8] == -ID16 and χ71 = -I Γ71[8] == T16O[8] == T16[8] (real, diagonal)	True
98	7+1 chiral projectors in the Γ71 basis are exactly PL44, PR44 of 4+4.nb; χ71 anticommutes with every Γ71[A]	True
99	complex chiral projectors for Γ17: PL17 + PR17 == ID16, PL17.PL17 == PL17, PL17.PR17 == 0, rank 8 each	True

4.7 What the octonions do: Γ(x)² = Re(x x), and alternativity

#	check	result
100	η17 is simply the squares of the octonion basis: e[A] e[A] == η17[[A+1, A+1]] e[0] (e[0]^2 = +1, e[i]^2 = -1) and Re(e[A] e[B]) == η17[[A+1, B+1]] (A, B = 0..7)	True
101	Re(x x) == X.η17.X: the octonion square produces the 7+1 interval (symbolic)	True
102	octonion basis: Γ17e[0] == {{0,ID8},{ID8,0}} and Γ17e[i] == {{LO[i],0},{0,-LO[i]}}	True

#	check	result
103	alternativity in action: xImag(xImag ψ) == -|xImag|^2 ψ for the imaginary part xImag of x (symbolic)	True
104	T16O(x)^2 == N(x) ID16 (octonion norm) and T16(x)^2 == (X.η4488.X) ID16 (4+4.nb)	True

5.1 The Lorentz algebra so(1,7)

#	check	result
105	28 real generators Σ17[A,B] (A < B), linearly independent	True
106	[Σ17[A,B], Γ17[K]] == η17[[B+1,K+1]] Γ17[A] - η17[[A+1,K+1]] Γ17[B]	True
107	so(1,7): [Σ_AB, Σ_KL] == η_BK Σ_AL - η_AK Σ_BL - η_BL Σ_AK + η_AL Σ_BK	True
108	the complex structure J = Γ17[8] commutes with all Σ17[A,B] (Spin(1,7) acts complex-linearly on ℝ^16 = ℂ^8)	True

5.2 A boost along x1 and a rotation in the x4–x5 plane

#	check	result
109	S(β) and R(θ) are the exponentials: dS/dβ == Σ17[0,1].S, S(0) == ID16, dR/dθ == Σ17[4,5].R, R(0) == ID16, and S.Sinv == R.Rinv == ID16	True
110	S(β).Γ17[K].S(β)^-1 == Σ_B Λboost[[B+1,K+1]] Γ17[B]; Λboost is the boost along x1	True
111	R(θ).Γ17[K].R(θ)^-1 == Σ_B Λrot[[B+1,K+1]] Γ17[B]; Λrot is the rotation in the x4-x5 plane	True
112	both Λ preserve the 7+1 metric: Transpose[Λ].η17.Λ == η17	True

5.3 The 7+1 Dirac operator squares to the 7+1 wave operator

#	check	result
113	(Γ17^A ∂_A)^2 Ψ == (∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 16 arbitrary functions of x0..x7	True

5.4 Spinors on the mass shell: real amplitudes

#	check	result
114	p is on the mass shell (p.η17.p == m^2) and pslash.pslash == m^2 ID16	True
115	(pslash - m) w == 0 has 8 linearly independent REAL solutions w	True
116	Ψ = w Exp[-I p.x] solves (I Γ17^A ∂_A - m) Ψ == 0	True

5.5 Real spinor fields: massless with Γ17, massive with the real 32×32 matrices

#	check	result
117	the matrices anticommuting with every Γ17[A] are exactly the multiples of Γ17[8], and (c Γ17[8])^2 == -c^2 ID16: no real J with J.J == +ID16	True
118	null momentum k = (1,1,0,0,0,0,0,0): k.η17.k == 0, kslash.kslash == 0, and kslash.w == 0 has 8 REAL solutions	True
119	Ψ = w Cos[k.x] is a REAL solution of the real massless Dirac equation Γ17^A ∂_A Ψ == 0	True
120	(Γ71R32^A ∂_A)^2 Ψ == -(∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 32 arbitrary functions of x0..x7	True
121	p.η71.p == -m^2 and pslash32.pslash32 == -m^2 ID32	True

#	check	result
122	Ψ = w Cos[p.x] + (pslash32.w/m) Sin[p.x], with 32 arbitrary REAL constants w, solves the REAL massive equation (Γ71R32^A ∂_A - m) Ψ == 0	True
123	... and Ψ obeys the Klein-Gordon equation (∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) Ψ == 0	True

5.6 All nine signatures p + q = 8, and the corrected table of [1B]

#	check	result
124	split analogue: Re(x x) of split octonions == X.η53.X, and Γ53 realizes Cl(5,3) with REAL 16×16 matrices	True
125	Γ08[A] = T16O[8].T16O[A] realizes Cl(0,8) with REAL 16×16 matrices	True
126	gens[p] satisfy the Clifford relation of signature (p, 8-p) for p = 0..8	True
127	classification: ℝ(16) for (8,0),(5,3),(4,4),(1,7),(0,8); ℍ(8) for (7,1),(6,2),(3,5),(2,6)	True
128	every ℝ(16) case has explicit REAL 16×16 generators in this notebook (T16O, Γ53, T16, Γ17, Γ08)	True
129	ℂ(16) cannot occur: its real dimension 2*16^2 = 512 differs from 2^8 = 256	True

7.1 The eight real Dirac matrices and the Dirac α, β

#	check	result
130	Dirac matrices: Γ17up[0] == Γ17[0], Γ17up[i] == -Γ17[i]; all REAL, every entry 0 or ±1	True
131	Dirac adjoint: ConjugateTranspose[Γ17up[A]] == Γ17[0].Γ17up[A].Γ17[0] for A = 0..7	True
132	βD == {{0,ID8},{ID8,0}} and αD[i] == {{0,τO[i]},{-τO[i],0}}: real and symmetric	True
133	αD[i].αD[j] + αD[j].αD[i] == 2 δij ID16, αD[i].βD + βD.αD[i] == 0, βD.βD == ID16	True
134	ω17 = Γ17[8] == {{0,-ID8},{ID8,0}} is real, ω17.ω17 == -ID16, and ω17 anticommutes with every Γ17[A]	True

7.2 The field dirac16complex and its Dirac equation

#	check	result
135	(I Γ17up[A] ∂_A + m)(I Γ17up[A] ∂_A - m) dirac16complex == -(∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) dirac16complex (16 arbitrary functions)	True

7.3 Hamiltonian form

#	check	result
136	Γ17[0].(I Γ17up[A] ∂_A - m) ψ == I ∂0 ψ - h ψ with h = -I αD[i] ∂_i + βD m (16 arbitrary functions)	True

7.4 Plane waves and the real spinors u_s, v_s

#	check	result
137	h(p) == {{0, M}, {Transpose[M], 0}} with the REAL 8×8 matrix M = m ID8 + p_i τO[i]	True
138	M.Transpose[M] == Transpose[M].M == E^2 ID8 and h(p).h(p) == E^2 ID16, E^2 = m^2 + p^2	True
139	h(p).u_s == E u_s and h(-p).v_s == -E v_s for s = 1..8	True
140	(pslash - m).u_s == 0 and (pslash + m).v_s == 0, pslash = E Γ17[0] + p_i Γ17[i]	True

#	check	result
141	u_s.u_r == v_s.v_r == 2E δsr, and u_s(p).v_r(-p) == 0	True
142	plane waves: u_s Exp[-I(E x0 - p.x)] and v_s Exp[+I(E x0 - p.x)] solve (I Γ17up[A] ∂_A - m) ψ == 0	True
143	example p = (2,2,4,0,0,0,0), m = 1: E = 5 and the eight u_s are real (entries in Sqrt[5] ℚ) and solve h(p).u_s == 5 u_s	True

7.5 Spin sums, completeness, and the diagonalization of h

#	check	result
144	spin sums: Σ_s u_s ubar_s == pslash + m and Σ_s v_s vbar_s == pslash - m	True
145	completeness: Σ_s u_s(p) u_s(p)^T + Σ_s v_s(-p) v_s(-p)^T == 2E ID16	True
146	U = (u_1..u_8, v_1(-p)..v_8(-p))/Sqrt[2E] is orthogonal and Transpose[U].h(p).U == diag(E ID8, -E ID8)	True

7.6 Grassmann algebra: anticommuting components, Lagrangian, conserved current

#	check	result
147	Grassmann: θ1 θ2 == -θ2 θ1, θ1 θ1 == 0, associativity, and ∂/∂θ2 (θ1 θ2 θ3) == -θ1 θ3	True
148	the three plane waves U1, U2, U3 solve the Dirac equation with m = 1	True
149	Grassmann field: ψ_a ψ_b + ψ_b ψ_a == 0 for all a, b (in particular ψ_a^2 == 0)	True
150	current conservation ∂_A (ψbar Γ17up[A] ψ) == 0 for the Grassmann-valued solution ψ = θ1 U1 + θ2 U2 + θ3 U3	True

7.7 Quantization: mode expansion and canonical anticommutators

#	check	result
151	canonical anticommutator from the mode algebra: {ψ_a(x0,x), ψ_b(x0,y)^†} == δab (1/V) Σ_p Exp[I p.(x - y)] (five momenta, symbolic x, y)	True

7.8 Charge conjugation

#	check	result
152	ψ -> ψc = ω17.ψ^*: (I Γ17up[A] ∂_A - m)(ω17.ψ^*) == ω17.((I Γ17up[A] ∂_A - m) ψ)^* for arbitrary ψ = ψre + I ψim	True
153	(pslash + m).ω17.u_s == 0 and ω17.u_s == Σ_r v_r (-Transpose[M]/E)_rs; ω17.ω17 == -ID16 (no Majorana spinors)	True

7.9 Lorentz covariance

#	check	result
154	boost at β = 2 log 2: S == 5/4 ID16 + 3/4 Γ17[0].Γ17[1], p' = Λ.p == (55/8, -41/8, 2, 4, 0, 0, 0, 0) is on shell, (pslash' - m).S.u_s(p) == 0	True

7.10 Chirality: no real chirality matrix, complex chiral projections

#	check	result
155	χ17 = I ω17: Hermitian, χ17.χ17 == ID16, anticommutes with every Γ17[A], commutes with every Σ17[A,B]	True
156	PL17, PR17 are complex (not real); Γ17[A].PL17 == PR17.Γ17[A]; ConjugateTranspose[PL17].Γ17[0] == Γ17[0].PR17	True
157	EL, ER: χ17.EL == -EL, χ17.ER == ER, EL^†.EL == ER^†.ER == ID8, EL^†.ER == 0, EL.EL^† == PL17, ER.ER^† == PR17, W unitary	True
158	φL == (ψup + I ψdown)/Sqrt[2], φR == (I ψup + ψdown)/Sqrt[2], PL17.ψ == EL.φL, PR17.ψ == ER.φR, ψ == EL.φL + ER.φR	True

7.11 The two coupled 8-component equations

#	check	result
159	σW[i] = I τO[i]: Hermitian, σW[i].σW[j] + σW[j].σW[i] == 2 δij ID8	True
160	chiral (Weyl) form: W^†.h(p).W == {{-σW.p, m ID8}, {m ID8, σW.p}} (symbolic p)	True
161	EL^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 + τO[i] ∂_i) φL - m φR (16 arbitrary functions)	True
162	ER^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 - τO[i] ∂_i) φR - m φL (16 arbitrary functions)	True
163	Klein-Gordon for each 8-component field: (I ∂0 - τO[i] ∂_i)(I ∂0 + τO[j] ∂_j) f == -(∂0^2 - ∂1^2 - ... - ∂7^2) f (8 arbitrary functions)	True
164	plane-wave example p = (2,2,4,0,0,0,0), m = 1, E = 5: (E + σW.p) φL == m φR and (E - σW.p) φR == m φL for all eight u_s	True

8.1 The self-coupled field: Lagrangian, densities, Hamiltonian

#	check	result
165	the densities ρS, ρP, j^A, j5^A are real: Γ17[0].op is Hermitian for op = ID16, I χ17, Γ17up[A], Γ17up[A].χ17	True
166	18 vertex pairs; lower vertex == η17[[A+1,A+1]] upper vertex in the V and A channels	True

8.2 The nonlinear field equation (Grassmann left derivatives)

#	check	result
167	nonlinear field equation: left derivative ∂L_int/∂ψbar_a == -[gS (ψbar ψ) ψ + gP (ψbar Iχ17 ψ) Iχ17 ψ + gV (ψbar Γ17[A] ψ) Γ17up[A] ψ + gA (ψbar Γ17[A]χ17 ψ) Γ17up[A]χ17 ψ]_a (32 Grassmann generators)	True

8.4 The 16-component Kohn–Sham equations

#	check	result
168	h_KS(p) is Hermitian for real S, P, V_A, A_A (symbolic momentum and potentials)	True
169	plane wave: Γ17[0].(pslash(ε,p) - m - S - I χ17 P - Γ17up[A](V_A + χ17 A_A)) == ε ID16 - h_KS(p)	True
170	Hartree potentials: ∂E_H/∂φ_a^* == (v_H.φ)_a, v_H = h_KS - h with S = gS ρS, P = gP ρP, V_A = gV j_A, A_A = gA j5_A (32 variables)	True

8.5 Exchange: Wick's theorem for contact couplings

#	check	result
171	Wick's theorem in the 560-dimensional Fock space: <Φ| :(ψbar o1 ψ)(ψbar o2 ψ): |Φ> == tr(o1 R) tr(o2 R) - tr(o1 R o2 R) for all 18 vertex pairs	True

8.6 The exchange self-energy and the Fierz identities

#	check	result
172	exchange self-energy: d/dt E_x(R + t dR) at t = 0 == tr(Σ_x.dR) and E_x == tr(Σ_x.R)/2 (random complex R, dR; symbolic couplings)	True
173	the 256 products Γ_K are orthogonal matrices with tr(Inverse[Γ_K].Γ_L) == 16 δKL	True
174	Fierz completeness: Σ_K (Γ_K)_ab (Inverse[Γ_K])_cd == 16 δad δcb (as the 256×256 matrix identity Transpose[F].F == 16 ID)	True
175	Fierz rearrangement of every exchange term: tr(o1 R o2 R) == (1/16) Σ_K tr(Inverse[Γ_K].R) tr(o1.Γ_K.o2.R) for all 18 vertex pairs (random complex R)	True

8.7 Example: uniform matter (no sea), Hartree and Hartree–Fock

#	check	result
176	uniform matter: tr R == ρS, tr(Γ17up[0].R) == ρ, and the densities ρP, j^i, j5^A vanish	True
177	the occupied plane waves give this R: (1/2) Σ_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) == (E Γ17[0] + m ID16)/(2E)	True
178	uniform exchange energy: E_x(R) == -gS/32 (ρS^2 + ρ^2) + gP/32 (ρS^2 - ρ^2) - gV/32 (8ρS^2 - 6ρ^2) + gA/32 (8ρS^2 + 6ρ^2)	True
179	uniform exchange self-energy: Σ_x == S_x ID16 + V_x Γ17[0] with S_x == ∂E_x/∂ρS and V_x == ∂E_x/∂ρ	True
180	uniform Hartree-Fock == Hartree with the Fierz-transformed couplings: gS/2 ρS^2 + gV/2 ρ^2 + E_x == G_S/2 ρS^2 + G_V/2 ρ^2 (gP, gA arbitrary)	True
181	c7 == 8 |S^6|/(2 Pi)^7 == 1/(15 Pi^4), |S^6| == 2 Pi^(7/2)/Gamma[7/2] == 16 Pi^3/15, and ρ == c7 kF^7/7 == kF^7/(105 Pi^4)	True
182	closed forms: d/dkF of the integrals gives the integrands, both vanish at kF = 0, and dε_kin/dm* == ρS	True
183	Hartree gap equation m* == m + gS ρS(m*) solved to 45 digits; m* is a stationary point of ε(m*) at fixed kF	True
184	total energy == sum of KS eigenvalues - E_H (double counting) == ε_kin - (m* - m) ρS + E_H	True
185	Hugenholtz-Van Hove at saturation: E/A == ε_F == dε/dρ; E/A - m < 0 and E/A at 0.95 kF0 and 1.05 kF0 is higher	True
186	Hugenholtz-Van Hove away from saturation: μ == dε/dρ == ε_F at kF = 1	True
187	Hartree-Fock at kF = 3/4 (gP = gA = 0): G_S == -98750, G_V == 20000; m* - m == gS ρS + S_x, V_0 == gV ρ + V_x; μ == ε_F	True

9.1 Chiral densities

#	check	result
188	chiral densities: ρS == κ + κ^*, ρP == I(κ - κ^*), j^A == jL^A + jR^A, j5^A == jR^A - jL^A (symbolic chiral orbitals)	True
189	the chiral blocks behind these densities: EL^†.βD.ER == ID8, EL^†.βD.EL == 0, EL^†.αD[i].EL == -σW[i], ER^†.αD[i].ER == σW[i], EL^†.αD[i].ER == 0	True

9.2 The Hartree energy in chiral variables

#	check	result
190	E_H (16-component form, chiral orbitals) == gS/2 (κ + κ^*)^2 - gP/2 (κ - κ^*)^2 + (gV + gA)/2 (jL.jL + jR.jR) + (gV - gA) jL.jR	True

9.3 The chiral Kohn–Sham equations

#	check	result
191	chiral KS Hamiltonian: W^†.h_KS(p).W == {{V^L_0 - σW.(p + V^L), M}, {M^*, V^R_0 + σW.(p + V^R)}}, M = m + S + I P, V^L = V - A, V^R = V + A	True
192	Hartree (Wirtinger): M - m == ∂e_H/∂κ == gS ρS + I gP ρP, V^L_A == ∂e_H/∂jL^A; V^L_A == V_A - A_A and V^R_A == V_A + A_A with V_A = gV j_A, A_A = gA j5_A	True
193	chiral KS potentials are derivatives: ∂E_H/∂φL^* == (V^L_0 - σW[i] V^L_i) φL + (M - m) φR and ∂E_H/∂φR^* == (V^R_0 + σW[i] V^R_i) φR + (M - m)^* φL	True

9.4 Chiral symmetry

#	check	result
194	e^(Iθχ17) == cosθ ID16 - sinθ ω17 is REAL and orthogonal; U.EL == e^(-Iθ) EL, U.ER == e^(Iθ) ER	True
195	U^†.βD.U == βD.(cos 2θ ID16 + I sin 2θ χ17); vector and axial densities invariant	True
196	E_H(e^(Iθχ17) ψ) - E_H(ψ) == (gS - gP)/2 [(e^(4Iθ) - 1) κ^2 + (e^(-4Iθ) - 1) κ^*2]: U(1)_A symmetric iff gP == gS	True

9.5 Uniform matter in chiral form

#	check	result
197	uniform matter in chiral form: κ == ρS/2, jL^0 == jR^0 == ρ/2, jL^i == jR^i == 0	True
198	(h_χ - V_0)^2 == (p^2 + m*^2) ID16: eigenvalues V_0 ± Sqrt[p^2 + m*^2], eight times each	True

9.6 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking

#	check	result
199	|S^6|/(2 Pi)^7 == 1/(120 Pi^4); J(0) == Λ^6/(720 Pi^4); g_c == -1/(8 J(0)) == -90 Pi^4/Λ^6; J(M) -> J(0) and K(M) -> K(0) as M -> 0	True
200	m = 0, g = 6/5 g_c: the gap equation 1 + 8 g J(|M|) == 0 has the root |M| (32 digits); M == 2 g κ^* with κ == -4 M J; E is stationary there and lower than E(0)	True
201	degenerate chiral vacua for m = 0: E(|M| e^(I ϕ)) == E(|M|) for ϕ = 1, 2, 4 (the phase is the Goldstone direction)	True
202	explicit breaking m = 1/100: the real M == m - 8 g J(M) M solves the gap equation, E is stationary there, and every complex phase raises E	True
203	the sea at one momentum: h_χ^2 == E^2 ID16; Π_- is a projector of rank 8; κ_p == tr(Π_- RL block) == -4 M^*/E; tr(Π_-.h0) == -8E + 8 Re[(M - m) M^*]/E	True

11.2 The cosmological metric of 7+1 spacetime

#	check	result
204	cosmological metric: sectional curvatures K_0i = -(Ha'+Ha^2), K_0k = -(Hb'+Hb^2), K_ij = -Ha^2, K_kl = -Hb^2, K_ik = -Ha Hb	True
205	cosmological metric: R^cd_ab is non-zero only for {c,d} = {a,b}	True
206	Einstein tensor: G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2, G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb, G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True
207	scalar curvature R = -2 (3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 Ha Hb)	True

11.3 Symmetries of the Riemann tensor and the contracted Bianchi identity

#	check	result
208	inhomogeneous metric, exactly at (x0, x4) = (1/2, 1/3): R_rsmn = -R_srmn = -R_rsnm = R_mnrs and R^r_[smn] = 0 (first Bianchi identity)	True
209	inhomogeneous metric: contracted Bianchi identity nabla_mu G^mu_nu == 0, exactly at (x0, x4) = (1/2, 1/3)	True
210	cosmological metric: contracted Bianchi identity nabla_mu G^mu_nu = 0 (arbitrary a, b)	True

12.1 The generalized Kronecker delta

#	check	result
211	generalized Kronecker delta: contraction identity delta^{m1..mp r}_{n1..np r} = (8-p) delta^{m1..mp}_{n1..np} (p = 1..4, 60 random index sets each)	True
212	generalized Kronecker delta with 9 upper indices vanishes in 8 dimensions (100 random index sets)	True

12.2 The tensor packages xAct: manifold, metric and chart (as in Einstein-Lovelock-7+1-Nash.nb)

#	check	result
213	xAct declaration as in the reference notebook: SignatureOfMetric[g71] == {7, 1, 0} (seven space-like dimensions, one time-like dimension, none degenerate) and SignDetOfMetric[g71] == -1	True
214	eta17 is simply the squares of the octonion basis: eta17[[A+1, B+1]] == Re(e_A e_B), e_0^2 == +1 (time x0), e_i^2 == -1 (space x_i, i = 1..7)	True
215	the quadratic form of the octonions has the signs of the squares of the octonion basis: Re(x x) == x0^2 - x1^2 - ... - x7^2 == x.eta17.x for x = x^A e_A (symbolic)	True

12.3 The Lovelock tensors with abstract indices: the generalized Kronecker delta kδ

#	check	result
216	xAct: Lovelock1 == R_a^b - (1/2) delta_a^b R, the Einstein tensor (abstract indices)	True
217	xAct: trace of Lovelock2 == -2 L_2 with L_2 = R^2 - 4 R_cd R^cd + R_cdfi R^cdfi (abstract indices)	True

12.4 Components with xAct: CD = LC[g]

#	check	result
218	xAct, cosmological metric: Lovelock1 == EinsteinCD == the Einstein tensor of the curvature engine of section 11.1	True

12.5 An independent computation: the pair-set algorithm

#	check	result
219	pair-set algorithm == brute-force generalized-delta contraction for E^(1) (cosmological metric)	True
220	pair-set algorithm == xAct (section 12.4) for E^(1), E^(2), E^(3) of the cosmological metric (two independent computations)	True

12.6 The Lovelock tensors of the cosmological metric

#	check	result
221	E^(1) == Einstein tensor G and L_1 == R (cosmological metric)	True
222	trace identity E^(k)mu_mu == -((8-2k)/2) L_k for k = 0..4	True
223	E^(4) == 0 identically in 8 dimensions, while the Euler density L_4 != 0	True
224	every E^(k) is diagonal, E^(k)1_1 = E^(k)2_2 = E^(k)3_3 and E^(k)4_4 = ... = E^(k)7_7	True
225	E^(2) == Lanczos tensor and L_2 == R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True
226	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2 Hb + 12 Ha Hb^2 + Hb^3) and E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True
227	divergence-free: d0 E^0_0 + 3Ha (E^0_0 - E^1_1) + 4Hb (E^0_0 - E^4_4) == 0 for k = 1, 2, 3 (arbitrary a, b)	True
228	variation of nL a^3 b^4 L_k: EL_nL = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4 (k = 1..4)	True

12.7 An inhomogeneous metric: the Lovelock tensors are divergence-free (exact check at a point)

#	check	result
229	inhomogeneous metric: nabla_mu E^(k)mu_nu == 0 EXACTLY at (x0, x4) = (1/2, 1/3) for k = 1, 2, 3	True
230	inhomogeneous metric: E^(1) == G at the point, and E^(k)_mu nu symmetric (k = 1, 2, 3)	True
231	inhomogeneous metric: E^(2) == Lanczos tensor at the point (exact)	True

12.8 A plane gravitational wave: only the Einstein tensor survives

#	check	result
232	pp-wave: E^(1)_mu nu == +(1/2)(d2^2 + d3^2) fPP l_mu l_nu	True
233	pp-wave: E^(2) == E^(3) == 0 for every profile fPP	True
234	pp-wave: xAct (section 12.4) == pair-set algorithm for E^(1), E^(2), E^(3) (a metric that is not diagonal; arbitrary profile fPP)	True

12.9 Exponential vacuum solutions: eternal inflation of ordinary space and deflation of the auxiliary space

#	check	result
235	exponential expansion a = e^{H x0}, b = e^{qH H x0}: xAct Lovelock_k == the closed forms of section 12.6 with Ha = H, Hb = qH H, Ha' = Hb' = 0 (k = 1, 2, 3)	True
236	exponential expansion: three distinct vacuum equations, related by 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) == 0	True
237	exponential vacuum solutions: (w2, w3) solve all 64 components for every qH (generic qH) and every Lambda	True
238	eternal inflation (a = e^{H x0}) with deflation (b = e^{-H x0}): w2 == Lambda/4 - w1/2, w3 == w1/24 - Lambda/36; for Lambda = 0: w2/w1 == -1/2, w3/w1 == 1/24	True

13.2 The cosmological metric: frame, spin connection, Dirac equation

#	check	result
239	cosmological metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
240	cosmological metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
241	cosmological metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
242	cosmological metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
243	cosmological metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True
244	cosmological metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
245	cosmological metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
246	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
247	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True
248	cosmological metric: Omega_0 == 0, Omega_i == (a'/2) Gamma17[i].Gamma17[0], Omega_k == (b'/2) Gamma17[k].Gamma17[0]	True
249	cosmological metric: Gamma^mu Omega_mu == (3a'/(2a) + 2b'/b) Gamma17[0]	True
250	current identity: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi] (psi, psibar arbitrary)	True
251	Lichnerowicz formula: (Gamma^mu D_mu)^2 psi == (g^mu nu nabla_mu D_nu - R/4) psi	True
252	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi == (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True
253	mode equation: Gamma17[0].(i Gamma^mu D_mu - m) psi == (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} (i dc/dx0 - h(x0) c) for psi = (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} c(x0)	True

13.3 The pp-wave and the inhomogeneous metric: frames that are not diagonal and not homogeneous

#	check	result
254	pp-wave: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
255	pp-wave: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
256	pp-wave: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
257	pp-wave: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
258	pp-wave: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True
259	pp-wave: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
260	pp-wave: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
261	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
262	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True

#	check	result
263	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
264	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
265	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
266	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
267	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True
268	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
269	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
270	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
271	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True

13.4 Conformal weight 7/2 of the massless field

#	check	result
272	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) == C^(-9/2) Gamma17up[A] d_A f for g = C(x0)^2 eta17	True

13.5 Cartan's first structure equation, a static metric with a radial coordinate, and the coupling to a gauge field

#	check	result
273	Cartan's first structure equation de^A + omega^A_B ^ e^B == 0 (no torsion): cosmological, pp-wave and inhomogeneous frames	True
274	structure coefficients: omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA) with C^A_BC = e_B^mu e_C^nu (d_mu e^A_nu - d_nu e^A_mu): cosmological and inhomogeneous frames	True
275	static radial metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
276	static radial metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
277	static radial metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
278	static radial metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
279	static radial metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True
280	static radial metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
281	static radial metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
282	static radial metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
283	static radial metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True
284	static radial metric: Cartan's first structure equation and omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA)	True
285	static radial metric: Omega_0 == (f'/(4 Sqrt[f] Sqrt[h])) Gamma17[1].Gamma17[0], Omega_1 == 0, Omega_k == (1/(2 Sqrt[h])) Gamma17[1].Gamma17[k] (k = 2..7)	True
286	static radial metric: Gamma^mu Omega_mu == (1/Sqrt[h]) (f'/(4 f) + 3/r) Gamma17up[1]	True
287	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection, Gamma^mu D_mu psi == f^(-1/4) r^(-3) Gamma^mu d_mu chi	True
288	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) == e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A'_mu = A_mu + d_mu alpha	True
289	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True

14.1 Primordial epoch near x0 = 0: Kasner solutions

#	check	result
290	Kasner metric: the Einstein tensor computed directly from the metric equals the closed form (x0^2 G^mu_mu, mu = 0, 1, 4)	True
291	Einstein Kasner: (p, q) = ((1+2Sqrt[2])/7, (2-3Sqrt[2])/14) and ((1-2Sqrt[2])/7, (2+3Sqrt[2])/14) solve E^(1) == 0	True
292	Gauss-Bonnet Kasner: 3 real roots of the cubic; (p, (3-3p)/4) solve E^(2) == 0	True
293	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) == 0; E^(3) == 0 identically for q = 0 and for p = 0	True

14.2 The mode equation in two dimensions

#	check	result
294	2x2 reduction: A, B, betaD square to ID16 and anticommute; Zc = A.B.betaD: Zc.Zc == -ID16, Transpose[Zc] == -Zc, Zc commutes with A, B, betaD	True

14.3 Pair creation in the primordial epoch (Einstein–Kasner branch p = 0.547, q = −0.160)

#	check	result
295	Kasner pair creation: |alpha|^2 + |beta|^2 == 1 (to 10^-8) for (|p|, |q|) = (1,1), (2,1), (1,3)	True
296	Kasner scaling: (|p|, |q|) = (1.4302.., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 10^-6)	True
297	16x16 evolution == 4 copies of the 2x2 evolution with +m sigma3 and 4 with -m sigma3 (Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0 from 0.05 to 40): average |beta|^2 agrees to 10^-8; unitarity; the two classes differ	True
298	isotropic expansion, massless: no pair creation (|beta|^2 < 10^-20)	True

14.4 Modern epoch: a static auxiliary space

#	check	result
299	static auxiliary space (Hb = 0): E^(1) = (3Ha^2, 2Ha' + 3Ha^2, 3Ha' + 6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha' + Ha^2)), E^(3) = 0	True

14.5 Far future: de Sitter expansion and thermal pair creation

#	check	result
300	8D de Sitter: E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (all diagonal entries)	True
301	de Sitter over a static auxiliary space: E^(1) = (3, 3, 6) H^2, E^(2) = (0, 0, -12) H^4, E^(3) = 0	True
302	de Sitter pair creation: |beta|^2 == 1/(Exp[2 Pi M/H] + 1) to 10^-7, M/H = 0.25, 0.5, 1	True
303	integrator benchmark (Landau-Zener): |beta|^2 == Exp[-Pi kappa^2] to 10^-6	True

15.1 The nine channels

#	check	result
304	256 multi-indices in 9 ranks with 1, 8, 28, 56, 70, 56, 28, 8, 1 members	True
305	every bilinear psibar Theta^K psi is real: Gamma17[0].Theta^K Hermitian for all 256 K	True
306	Theta_K.Theta^K == ID16 and Tr[Theta_K.Theta^J] == 16 delta_KJ	True
307	Hodge duality: Theta^K.omega17 == (phase) Theta^(complement of K), |phase| = 1	True
308	the couplings of section 8 are four of the nine: I_0 == (psibar psi)^2, I_1 == (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 == -(psibar i chi17 psi)^2, I_7 == -(axial invariant)	True
309	local Lorentz invariance: I_r(S psi, psibar S^-1) == I_r(psi, psibar) for r = 0..8 (random finite Lorentz transformation), and S^-1 == Gamma17[0].S^dagger.Gamma17[0]	True

15.2 The 9×9 Fierz matrix and Hartree–Fock with nine channels

#	check	result
310	Fierz matrix: closed form == (1/256) Sum_{K in r} Tr[Theta_K.Theta_J.Theta^K.Theta^J] for the first and last J of every rank s	True
311	F.F == IdentityMatrix[9]; eigenvalues: five +1 and four -1	True
312	Fierz identity for commuting spinors (random): Sum_{K in r} (a Theta_K b)(c Theta^K d) == Sum_s F_rs Sum_{J in s} (a Theta_J d)(c Theta^J b), r = 0..8	True
313	Hartree-Fock energy E_H + E_x == Sum_s (G_s/2) D_s with G = (1 - Transpose[F]) g (random density matrix and couplings)	True
314	uniform density matrix (rhoS ID16 + rho Gamma17[0])/16: D_0 == rhoS^2, D_1 == rho^2, D_2 .. D_8 == 0	True
315	section 8 recovered: with g = (gS, gV, 0, 0, 0, 0, 0, -gA, -gP) the exchange energy of uniform matter is that of section 8.7	True
316	the Hartree-Fock couplings of section 8.7: gS = -100000, gV = 10000 give G_0 = -98750, G_1 = 20000	True

15.3 The covariant nonlinear field equation

#	check	result
317	nonlinear field equation: dL_self/d psibar_a == -Sum_r g_r Sum_K (psibar Theta_K psi) (Theta^K psi)_a	True

15.4 Chirality in a gravitational field

#	check	result
318	chi17 commutes with every generator (Gamma17up[A].Gamma17up[B] - Gamma17up[B].Gamma17up[A])/4, hence with every Omega_mu	True
319	curved chiral equations: E_L^dag.Gamma17[0].(i Gamma^mu D_mu - m) psi == (i d0 + i(3Ha/2 + 2Hb) + tauO.d) phi_L - m phi_R, and the same for E_R with -tauO.d	True

16.1 Homogeneous Kohn–Sham matter in the cosmological spacetime

#	check	result
320	flat limit a = b = 1, kF = kFsat: m* == 0.3480973401... and E/A - m == -0.0283214279... (section 8.7), to 10^-11	True
321	flat limit: pa == pb == rho eps_F - eps (Gibbs-Duhem), which is 0 at the saturation density	True

#	check	result
322	energy-momentum conservation of the Kohn-Sham matter: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) == 0 (a = 1 + 0.3x + 0.05x^2, b = 1/(1 + 0.2x), x0 = 0.7; relative 10^-8)	True

16.2 The three epochs

#	check	result
323	primordial: rho x0 == kFsat^7/(105 Pi^4) at every x0, and m* decreases monotonically to below 0.05 at x0 = 10^-4	True
324	modern (b = 1): eps a^3 tends to a constant and pa/eps -> 0 (pressureless in ordinary space)	True
325	far future (de Sitter, H = 0.1): rho == rho0 e^{-3 H x0} and 1 - m* decays like rho	True

16.3 Self-consistent evolution: Einstein–Lovelock gravity driven by the Kohn–Sham matter

#	check	result
326	the 11 and 44 Einstein-Lovelock equations are linear in (Ha', Hb'); the 00 equation contains neither (constraint)	True
327	self-consistent evolution: the constraint Sum alpha_k E^(k)0_0 - Lambda == kappa eps is preserved to 10^-7 (relative) for Einstein and for Einstein-Lovelock (alpha2 = 20, alpha3 = 500)	True

17. The Standard Model on the world brane of 7+1 spacetime

#	check	result
328	brane frame metric diag(+1, -1, -1, -1) == eta17 restricted to x0..x3: the squares of the octonion basis e[0], e[1], e[2], e[3] of the quaternion subalgebra (Re(e[A] e[B]), A, B = 0..3), which is closed under the octonion product	True

17.1 The fields, their quantum numbers, and anomaly cancellation

#	check	result
329	anomaly cancellation per generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3, grav^2U(1), SU(3)^3 all vanish; doublets per generation even	True
330	electric charges Q = T3 + Y: (u, d) = (2/3, -1/3), (nu, e) = (0, -1), u_R 2/3, d_R -1/3, e_R -1, nu_R 0	True
331	g* = 28 + (7/8) 90 = 106.75 for the Standard Model fields (without nu_R)	True

17.3 Electroweak symmetry breaking

#	check	result
332	gauge-boson mass matrix: eigenvalues g^2 v^2/4 (twice), (g^2 + g'^2) v^2/4 and 0 (photon)	True
333	tree level: m_W == g v/2, m_Z == Sqrt[g^2 + g'^2] v/2, rho == m_W^2/(m_Z^2 cos^2 theta_W) == 1, m_H^2 == 2 lambda v^2	True

17.4 Quark and lepton masses; CKM and PMNS mixing

#	check	result
334	CKM: unitary; Jarlskog invariant J = 3.12 x 10^-5 (PDG 2024: 3.12 +0.13 -0.12 x 10^-5); A, rhobar, etabar within the PDG uncertainties	True
335	Yukawa matrices: singular values x v/Sqrt[2] are the quark masses and |U_uL^dag.U_dL| == |V_CKM|	True
336	PMNS unitary; Dirac neutrino Yukawa Sqrt[2] U.diag(m)/v reproduces the masses (singular values)	True

17.5 The modern epoch on the brane

#	check	result
337	age of the universe from the brane Friedmann equation: 13.79 Gyr (Planck 2018: 13.787 +- 0.020 Gyr)	True
338	ADD relation: kappa8 rho4/V_perp == 8 Pi G4 rho4 with G4 = G8/V_perp (kappa8 = 8 Pi G8)	True

18. dirac16complex at the world brane: the 16 components as four brane Dirac spinors, and the portals

#	check	result
339	brane representation: R_A R_B + R_B R_A == 2 eta17_AB ID16 for R = (gamma_mu (x) 1_4, gamma5 (x) rho_k)	True
340	intertwiner: Gamma17[A].U == U.R_A for A = 0..7, and U is unitary	True
341	chi17 == U.(gamma5 (x) rho4567).U^dag; rho4567 Hermitian with eigenvalues {-1, -1, 1, 1}	True
342	bilinears at the brane: psibar psi' == Sum_a psibar_a psi'_a and psibar Gamma^mu psi' == Sum_a psibar_a gamma^mu psi'_a (mu = 0..3)	True

18.1 Kaluza–Klein masses on the auxiliary torus

#	check	result
343	KK operator: X.X == -|q|^2/b0^2 ID16 and X anticommutes with gamma_mu (x) 1_4	True
344	KK masses: Det[gamma.p (x) 1 + X - m] vanishes on p0^2 - |p|^2 == m^2 + |q|^2/b0^2 (40-digit arithmetic) and not off shell	True

18.2 The portals

#	check	result
345	portal mixing: 1 - N.N^dag == mu.mu^dag/m^2 to second order (non-unitary light mixing)	True

19. Neutrino flavour and flavour conversion

#	check	result
346	vacuum: P(a->b) doubly stochastic; equals delta - 4 Sum Re X sin^2 Delta + 2 Sum Im X sin 2Delta; CPT P(a->b) == P(bbar->abar)	True

#	check	result
347	V_CC(rho Y_e = 1 g/cm^3) == 7.63 x 10^-14 eV; matter evolution doubly stochastic; a flavour-universal potential has no effect	True
348	two-flavour MSW: sin^2 theta_m from diagonalization == formula, and sin^2 2theta_m == 1 at 2 E V == dm^2 cos 2theta	True
349	Sun: numerical P_ee == adiabatic MSW formula to 2 x 10^-3 at 0.3, 1, 5 and 10 MeV; P_ee(0.3 MeV) within 0.03 of the vacuum average; P_ee falls with energy	True
350	cosmic neutrino background: T_nu == 1.945 K and n == 112 cm^-3 per flavour	True
351	dirac16complex portal: non-unitary oscillations, Sum_b P(a->b) < 1 (zero-distance effect); decoupled limit == three-flavour vacuum result	True

20.1 The action, term by term

#	check	result
352	cubic Lovelock density in invariant form: L_3 == R^3 - 12 R Ric.Ric + 16 Ric^3 + 24 R^ac R^bd R_abcd + 3 R Riem.Riem - 24 R^ab R_a^cde R_bcde + 4 R^ab_cd R^cd_ef R^ef_ab - 8 R_a^c_b^d R_c^e_d^f R_e^a_f^b (exact, four random integer algebraic curvature tensors)	True
353	Gauss-Bonnet density in invariant form: L_2 == R^2 - 4 Ric.Ric + Riem.Riem (the same tensors, exact)	True

20.2 The coupled field equations and the energy–momentum tensors

#	check	result
354	stabilizing potential: W = a^3 b^4 U(b) gives eps = U, pa = -U, pb = -U - (b/4) U'(b)	True
355	brane tension smeared over the torus (V_perp = b^4 Lc^4): eps = f_W/V_perp, pa = -f_W/V_perp, pb = 0	True
356	brane fluid rho_4 = N a^(-3(1+w)), smeared: eps = rho_4/V_perp, pa = w eps, pb = 0	True
357	massless dirac16complex particle (comoving momentum (k_a, k_b)): eps V = omega, pa V = k_a^2/(3 a^2 omega), pb V = k_b^2/(4 b^2 omega)	True
358	every such source is conserved: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) == 0 for arbitrary W(a, b), a(x0), b(x0)	True
359	minisuperspace: EL_n == (a^3 b^4/kappa)(S00 - kappa eps), EL_a == (3 a^2 b^4/kappa)(S11 + kappa pa), EL_b == (4 a^3 b^3/kappa)(S44 + kappa pb) with S = Sum alpha_k E^(k) (identities of section 12.6)	True

20.3 The homogeneous equations, the radion and the four-dimensional constants

#	check	result
360	radion equation of Einstein gravity: Hb' + 3 Ha Hb + 4 Hb^2 == kappa (eps/6 - pa/2 + pb/3) + Lambda/3 (constraint used)	True
361	radion mass: U = U0 + (K_U/2) beta^2, b U' = K_U beta gives the restoring term -(kappa K_U/12) beta, m_r^2 = kappa K_U/12	True
362	static auxiliary space: (kappa/4) b U' == Lambda + kappa U + 2 kappa f_W/V_perp + (kappa/2)(eps - 3pa) + kappa pb - 12 alpha2 Ha^2 (Ha' + Ha^2)	True
363	static auxiliary space: Ha' == -(kappa/2)(eps + pa) and E^(3) does not contribute (Hb = 0)	True
364	brane Friedmann equation: 3 Ha^2 == (kappa/V_perp) rho_4 + Lambda4 with Lambda4 = Lambda + kappa U0 + kappa f_W/V_perp, i.e. 8 Pi G4 = kappa/V_perp	True
365	Einstein-frame form of the constraint: 3 (Ha + 2 Hb)^2 == 6 Hb^2 + (E^(1)0_0 of section 12.6), i.e. 3 M_Pl^2 H_E^2 == phi'^2/2 + rho with phi = Sqrt[12] M_Pl beta	True

21.1 The exact vacuum solution: an algebraic curve and one quadrature

#	check	result
366	vacuum constraint in u = Hb/Ha, X = Ha^2 (alpha2 = 1, alpha3 = xi): P_E + X P_G + xi X^2 P_C == 0 with P_E = 3(2u^2 + 4u + 1), P_G = -12u(u^3 + 12u^2 + 18u + 4), P_C = 360 u^3 (3u + 4)	True
367	Kasner constants: P_E(u1) == 0, p1 == (1 + 2 Sqrt[2])/7, q1 == (2 - 3 Sqrt[2])/14; u2 in (-1/3, 0) with P_G(u2) == 0, 3 p2 + 4 q2 == 3	True
368	the explicit curve solves the vacuum constraint: residual < 10^-25 (relative) at u = u1 + 10^-6, -0.28, u2, -0.1, -10^-4 (xi = 10^-6, 40 digits)	True
369	the effective exponent along the curve: p_eff -> p1 (u -> u1), about p2 at u = u2 (within 10^-2), -> 1 (u -> 0)	True

21.2 Accurate approximate solutions in the three Lovelock regimes

#	check	result
370	cubic regime: Ha = 1/tau + A0 + A1 tau, Hb = h0 + h1 tau + h2 tau^2 solves the 00, 11, 44 equations through order tau^-1 (11: checked, not used)	True
371	cubic regime coefficients: h1 == alpha2/(16 alpha3) - 1/(32 alpha2), A0 == Sqrt[30](58 alpha2^2 - 45 alpha3)/(1920 alpha2^(3/2) alpha3^(1/2)), A1 and h2 as in the text	True
372	Einstein regime: u = u1 + d1 Ha^2, d1 == alpha2 (16 - 11 Sqrt[2])/4; p_eff = p1 + e1 Ha^2, e1 == alpha2 (1 + 9 Sqrt[2])/7	True
373	Gauss-Bonnet regime: p_eff = p2 + f1/Ha^2 + f2 xi Ha^2 with f1 = -0.69887346..., f2 = -0.7853407... (differences give the same values at Ha = 10^4 and 10^-6 xi Ha^2 to 10^-5)	True
374	cubic-series data at tau = 10^-8 satisfy the vacuum constraint to < 10^-12 (relative to 48 Ha^3 |Hb|)	True
375	the ODE solution agrees with the exact quadrature: tau at Ha(u2) to 10^-10, and tau - p1/Ha + e1 Ha at Ha = 3 10^-3 equals s1 to 10^-6	True
376	regime approximations: cubic error < 10^-5 for Ha >= 10^4, Gauss-Bonnet < 10^-3 for 3 <= Ha <= 300, Einstein < 10^-3 for Ha <= 0.1	True

21.3 Coupled to dirac16complex: the matter-dominated cubic regime and decompactification

#	check	result
377	matter-dominated cubic regime: p_eff -> 3/4, Hb -> -Sqrt[alpha2/(54 alpha3)], kappa eps tau^3 -> 9 alpha2 |h*|, alpha2 E2 : alpha3 E3 : kappa eps -> 9/4 : -5/4 : 1 (Ha = 10^15)	True
378	coupled: where the gas is a small fraction the effective exponents are those of the vacuum (1, p2, p1 within 10^-2)	True
379	without the stabilizing potential the gas-dominated Einstein regime decompactifies: tau Ha -> 0, tau Hb -> 2/5, pb/eps -> 1/4 (Ha = 10^-14)	True
380	the gas fraction grows like tau^(2 - 3 p1 - 5 q1) = tau^1.160 in the Einstein regime (to 2 10^-2)	True
381	Einstein + fluid power laws a ~ tau^P, b ~ tau^Q: (wa, wb) = (0, 1/4) -> (0, 2/5); (1/3, 0) -> (1/2, 0); (1/7, 1/7) -> (1/4, 1/4) (tau^2 E^(1) from EKas of section 14.1)	True
382	matter-dominated cubic regime (any fluid, b finite): leading 44 term p^3 (p - 1)(648 alpha3 h^2 - 12 alpha2); with h^2 = alpha2/(54 alpha3): p = 1/(1 + w_a) and kappa eps tau^3 = (64/3) p^3 alpha2 |h|	True

21.4 Pair creation in the primordial regime and the brane

#	check	result
383	Einstein-Kasner pair creation: F(1) == 0.0107493 (section 14.3) and the tail F ~ K^(-(3 Sqrt[2] + 2)/2) (exponent between K = 8 and 16 to 2 10^-2)	True
384	brane plasma at T = M8, H = M8: fraction (Pi^2 g*/30) M8^4/(3 M_Pl^2 M8^2) < 10^-27	True

22.1 The stabilizing potential catches the auxiliary space: radion oscillations

#	check	result
385	capture: the radion oscillates like a four-dimensional massive field in a matter-dominated universe: with tau_E from <H_E> = 2/(3 (tau - tau_E)) over the last period, beta = A sin(m_r tau + vartheta)/(m_r (tau - tau_E)) to 2 10^-2, <H_E>(tau - tau_E) = 2/3 to 10^-2 at tau = 10^5 and 2 10^5, the gas share constant to 2 10^-2	True

22.2 Radion decay, the moduli problem and the Kaluza–Klein channels

#	check	result
386	through brane fields alone a 10 GeV radion would decay far too late: T_RH < 10^-6 GeV (nucleosynthesis needs 4 10^-3 GeV); 10^13 Kaluza-Klein channels below m_r/2	True
387	the number of lattice vectors with |n|^2 = k is r_4(k) = 8 sigma(k) - 32 sigma(k/4) (direct count for k <= 60)	True
388	the radion decays into the bulk: the lattice sum over the Kaluza-Klein modes of dirac16complex (m << m_r/2) equals its continuum limit to 10^-6, the number of momenta to 10^-5, and Gamma_bulk/Gamma_brane > 10^12	True

22.3 The Standard Model plasma on the brane: g*(T), g*s(T) and the time–temperature relation

#	check	result
389	g* limits: 106.75 above the electroweak scale (to 0.1), 10.75 near 3 MeV (to 0.1), 2 + (21/4)(4/11)^(4/3)(3.044/3) = 3.3855 after e+e- annihilation (to 10^-3)	True
390	the time-temperature quadrature is converged: halving the grid changes t(10 keV) by less than 10^-5	True
391	radiation era: t(1 MeV) = 0.74 s within 5% (1/(2 Ha) with g* = 10.75)	True

22.4 The late universe: an accurate analytic solution, the age, and the radion today

#	check	result
392	age of the universe from the brane Friedmann equation: 13.79 Gyr (Planck 2018: 13.787 +- 0.020)	True
393	composite analytic solution t(a) = t_rm(a) + t_mLambda(a) - t_m(a) accurate to < 3 10^-4 for 10^-8 <= a <= 10^0.5	True

23.1 The de Sitter attractor

#	check	result
394	far future: Ha(t) = H_inf Coth[3 H_inf t/2] agrees with the full brane solution at a = 3 and a = 10 to 10^-4	True

23.2 The radion in de Sitter: Gauss–Bonnet corrections and relaxation

#	check	result
395	radion in de Sitter: gamma/(3H) and m_eff^2/m_r^2 as in the text (beta_eq -> 0); gamma = 3H and m_eff^2 = m_r^2 for alpha2 = alpha3 = 0	True
396	radion equilibrium in de Sitter: 3 m_r^2 beta_eq == 3 H^2 - 12 alpha2 H^4 + O(beta_eq^2)	True
397	the radion relaxes to the equilibrium of the two algebraic equations: beta(3000) to 10^-6, Ha(3000) to 10^-8 (relative)	True
398	eight-dimensional de Sitter (no stabilizer): all three equations read 21 H^2 - 420 alpha2 H^4 + 2520 alpha3 H^6 == Lambda	True

23.3 The size of every interaction in the three regimes

#	check	result
399	the shares of the four terms of the 00 equation: Einstein > 0.99 at Ha = 10^-2, Gauss-Bonnet > 0.9 at Ha = 30, Gauss-Bonnet and cubic about 1/2 each at Ha = 3 10^5, gas > 0.1 at Ha = 10^15	True

24.1 What the observations require

#	check	result
400	required abundance: m Y = Omega_c h^2 (rho_c/h^2)/s0 = 4.37 10^-10 GeV with rho_c/h^2 = 1.053 10^-5 GeV/cm^3 and s0 = 2891 cm^-3 (to 10^-3)	True
401	collider bound: with M_Pl^2 = R^n M_D^(n+2) and V_perp = (2 Pi R)^4 = M_Pl^2/M8^6, M_D = (2 Pi)^(2/3) M8 = 34 TeV, above the strongest bound 11.2 TeV	True

24.2 dirac16complex as cold matter in the solutions of sections 21–23 (DERIVED)

#	check	result
402	a massive dirac16complex particle: eps V == omega, pa V == k_a^2/(3 a^2 omega), pb V == k_b^2/(4 b^2 omega), omega = Sqrt[m^2 + k_a^2/a^2 + k_b^2/b^2]	True
403	the lightest mode (k_b = 0) is cold matter: pa/eps == k_a^2/(3 (m^2 a^2 + k_a^2)) -> 0, pb == 0, and eps ~ a^-3 at fixed b for k_a = 0	True
404	dark matter displaces the radion today by H0^2 Omega_c/(2 m_r^2) < 10^-80 (m_r = 10 GeV): the four-dimensional Newton constant is constant	True
405	regime III: Omega_m a^-3/Omega_Lambda == Csch[3 H t/2]^2 for a = (Omega_m/Omega_Lambda)^(1/3) Sinh[3 H t/2]^(2/3), and -> 4 Exp[-3 H t]	True

24.3 The lightest mode: darkness, collisionlessness and stability (ESTIMATE)

#	check	result
406	couplings of the lightest mode: with V_perp = M_Pl^2/M8^6, lambda_P/V_perp == c_P M8/M_Pl^2, y/Sqrt[V_perp] == c_y M8/M_Pl, G_P/V_perp == c_V/M_Pl^2, g_r/V_perp == c_r/M_Pl^2	True
407	natural couplings make the lightest mode dark and collisionless: sigma_SI(c_P = 1, m = 40 GeV) < 10^-50 x (LZ limit 2.2 10^-48 cm^2), sigma/m(c_r = 1, m = 1 GeV) < 10^-70 cm^2/g (Bullet Cluster: 0.7 cm^2/g)	True
408	stability through the Yukawa portal: for c_y = 1 the lifetime chi -> 3 nu equals the age of the universe at m = 40 MeV; it scales as 1/(c_y^2 m^3), and Gamma(nu gamma)/Gamma(3 nu) == 27 alpha/(8 Pi)	True

24.4 The abundance (ESTIMATE; the outcome is OPEN)

#	check	result
409	primordial gas: overabundance O = (f/(1 - f)) T_RH/((4/3) m Y) = 1.5 10^7 T_RH/GeV (6 10^4 at T_RH = 4 MeV)	True
410	the radion puts more than 1 - 10^-12 of its decays into the bulk (Kaluza-Klein pairs of dirac16complex)	True
411	Boltzmann factor of production below T_RH: Int_0^Infinity t^3 K_1(t) dt == 3 Pi/2 (exact), S(0) = 1, S(m_h/T*) < 10^-2 with T* = (M8^6/M_Pl)^(1/5) = 13.3 GeV	True
412	zero mode through the Yukawa portal: the abundance (~ c_y^2 m) and the stability (lifetime ~ 1/(c_y^2 m^3) > age) are compatible only for m < m*, and m* < 5.7 keV (Lyman-alpha) for N = 1 and N = 12 channels even with S = 1	True
413	Kaluza-Klein tower: Sum_n (1 - mu_n^2/m_h^2)^2 -> V_perp m_h^4/(192 Pi^2) and Sum_n mu_n (1 - mu_n^2/m_h^2)^2 -> V_perp m_h^5/(315 Pi^2) (d^4n = 2 Pi^2 |n|^3 d|n|: exact integrals); the lattice sums with r_4(k) for R = 100 equal Pi^2 R^4/12 and 16 Pi^2 R^4/315 to 10^-8	True
414	Kaluza-Klein tower through the Yukawa portal: 7.7 10^17 states below m_h with mean mass (192/315) m_h; at T_RH = T* it carries more energy than all the dark matter unless c_y < 2 10^-5 (N = 1)	True
415	Higgs portal: the zero mode gets Y < 10^-40 (c_P = 1); the pairs of Kaluza-Klein levels (lambda(1, u1^2, u2^2) == (1 - (u1 + u2)^2)(1 - (u1 - u2)^2)) give the observed abundance only for c_P between 10 and 20 (T_RH >> m_h, S -> 1) or between 100 and 200 (T_RH = T*)	True
416	vector portal: gravitational strength per pair; the production rate of all pairs lighter than T divided by H ~ (T/T_V)^11 with T_V = (M8^12/M_Pl)^(1/11) = 490 GeV is below 10^-17 at T*	True

24.5 dirac16complex and dark energy

#	check	result
417	Kohn-Sham matter of section 8: p > 0 at low density, p < 0 below saturation, p = 0 at kF0, a spinodal region (dp/dkF < 0) in between, and w = p/eps >= -0.06 everywhere	True
418	Epstein sum over Z^4 without 0: Sum |n|^-8 == 8 (1 - 4^-3) zeta(4) zeta(3), compared with the shell sum Sum r_4(k)/k^4 to k = 20000 (tail < 3 10^-8)	True
419	Casimir energy of dirac16complex on the auxiliary torus: massless 16 Gamma(4) Z/(Pi^4 l^4) > 10^34 rho_Lambda, the massive sum -> the massless value for m l -> 0 (to 10^-4), and equals rho_Lambda at m l = 92.2	True

4.3 Independent Python cross-checks

tools/verify_7p1_nb.py (parses the evaluated 7+1.nb; exit code 0)

```
input cells
    ...: 174
output cells
    ...: 481
message cells (must be 0)
    ...: 0
check outputs
    ...: 383
check outputs that are True
    ...: 383
check["..."] calls in the input cells
    ...: 383
$Version output
    ...: ['{"15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)","Notebook$$67$147097`"}']
private notebook context
    ...: True
Length[checks]
    ...: 419
failedChecks == {}
```

```
    ...: True
allChecksPassed output True
    ...: True
tau[A] printed == 4+4 tau[A] (python)
    ...: True
tauO[A] printed == python tauO[A]
    ...: True
Gamma17[A] printed == python G17[A]
    ...: True
gridO printed == python octonion table
    ...: True
gridx44 printed == python split table (4+4.nb)
    ...: True
Lambda boost/rotation printed
    ...: True
8 real on-shell spinors printed solve (pslash - m) w = 0 (python check)
    ...: True
classification table printed
    ...: True
section 7: check outputs (all True)
    ...: 35 (True)
section 8: check outputs (all True)
    ...: 23 (True)
section 9: check outputs (all True)
    ...: 16 (True)
section 7: printed Sqrt[5] u_s(p) == python (5 e_s; M^T e_s) and (pslash - m) u_s == 0
    ...: True
section 8: printed closed forms of int k^6 m*/E and int k^6 E: d/dkF == integrands, zero at kF
    ...= 0 (sympy)
    ...: True
sections 8-9: 23 printed numbers agree with tools/ref_dirac16.py to 17 digits
    ...: True
sections 8-9: G_S == -98750, G_V == 20000 printed
    ...: True
tools/ref_dirac16.py: all its boolean checks True
    ...: True
sections 0-6: the 166 input and text cells of the backup are all present, 159 unchanged, 7
    ...changed and 1 added by the signature explanation of 27 October 2026
    ...: True
section 11: check outputs (all True)
    ...: 7 (True)
section 12: check outputs (all True)
    ...: 28 (True)
section 13: check outputs (all True)
    ...: 15 (True)
section 14: check outputs (all True)
    ...: 14 (True)
section 15: check outputs (all True)
    ...: 16 (True)
section 16: check outputs (all True)
    ...: 8 (True)
section 17: check outputs (all True)
    ...: 11 (True)
section 18: check outputs (all True)
    ...: 7 (True)
section 19: check outputs (all True)
    ...: 6 (True)
summary table: entries == Length[checks], all True
    ...: True
summary table: every printed check is listed
    ...: True
sections 0-10: the 368 input and text cells of the backup taken before sections 11-20 are all
    ...present, 360 unchanged, 8 changed and 1 added by the signature explanation of 27 October
    ...2026
    ...: True
```

```
sections 0-20: the 601 input and text cells of the backup taken before the signature
    ...explanation are all present, 586 unchanged, 15 changed and 4 added by that explanation
    ...: True
sections 0-20, outputs: of the 382 outputs of the backup taken before the signature explanation
    ...373 are identical; the others differ only in the check counts and tables of the summary
    ...sections, the declared list {7,1,0}, the private context name and measured times; 5 new
    ...check outputs, all True : True
sections 14-19: 27 groups of printed numbers agree with the Python programs (relative 1e-6;
    ...rounded prints 1e-4; ODE and ill-conditioned results 1e-5)
    ...: True
section 20: check outputs (all True)
    ...: 14 (True)
section 21: check outputs (all True)
    ...: 19 (True)
section 22: check outputs (all True)
    ...: 9 (True)
section 23: check outputs (all True)
    ...: 6 (True)
stage 4: without sections 20-23 (and with the old title of the summary section) the 605 input
    ...and text cells equal those of the backup taken before sections 20-23 were appended
    ...: True
stage 4: the 387 outputs of sections 0-19 and of the summary equal those of the backup (summary
    ...counts and tables, the private context name and measured times excepted)
    ...: True
sections 20-23: 34 groups of printed numbers agree with tools/ref_coupled.py (tolerances per
    ...group, from 1e-14 for exact constants to 2e-3 for numbers read off numerical solutions)
    ...: True
section 24: check outputs (all True)
    ...: 20 (True)
stage 5: without section 24 (and with the old title of the summary section) the 732 input and
    ...text cells equal those of the backup taken before section 24 was appended, after undoing
    ...the 1 documented correction(s) of section 23.3
    ...: True
stage 5: the 453 outputs of sections 0-23 and of the summary equal those of the backup (summary
    ...counts and tables, the private context name and measured times excepted; 1 output(s)
    ...differ only by the documented correction)
    ...: True
section 24 numbers: 29 groups of printed numbers agree with tools/ref_dark.py (tolerances per
    ...group, from 1e-14 for closed forms to 1e-5 for a number read off a numerical solution)
    ...: True

VERDICT: ALL CROSS-CHECKS PASSED
```

tools/ref44.py (rebuilds 4+4.nb; exit code 0)

```
ζ7 = [-1, -1, -1, -1, 1, 1, 1, 1]
σ == ζ1ζ2ζ3 : True
{ζA,ζB}/2 == -η ID8 (A,B=1..7): True
-η_AA ζA == ζA^T : True
(τA τB+τB τA)/2 == η ID8 : True
(τA τB+τB τA)/2 == η ID8 : True
τ1..τ7 product == ID8: True
σ == τ4τ5τ6τ7: True
{T16A,T16B}/2 == η ID16 : True
T16[8] diag: [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1]
σ16 == diag(-σ,σ): True
unit: [sqrt(2)/2, 0, 0, 0, sqrt(2)/2, 0, 0, 0]  unit.σ.unit = 1
E^a_A[h] == (u σ u) Transpose[Table[η_BB τ[B-1].u]] for all h: True
eA[1] : ['eA[1]', 'eA[2]', 'eA[3]', 'eA[4]', 'eA[5]', 'eA[6]', 'eA[7]', 'eA[8]']
eA[2] : ['eA[2]', '-eA[1]', '-eA[4]', 'eA[3]', '-eA[6]', 'eA[5]', 'eA[8]', '-eA[7]']
eA[3] : ['eA[3]', 'eA[4]', '-eA[1]', '-eA[2]', '-eA[7]', '-eA[8]', 'eA[5]', 'eA[6]']
eA[4] : ['eA[4]', '-eA[3]', 'eA[2]', '-eA[1]', '-eA[8]', 'eA[7]', '-eA[6]', 'eA[5]']
eA[5] : ['eA[5]', 'eA[6]', 'eA[7]', 'eA[8]', 'eA[1]', 'eA[2]', 'eA[3]', 'eA[4]']
```

```
eA[6] : ['eA[6]', '-eA[5]', 'eA[8]', '-eA[7]', '-eA[2]', 'eA[1]', '-eA[4]', 'eA[3]']
eA[7] : ['eA[7]', '-eA[8]', '-eA[5]', 'eA[6]', '-eA[3]', 'eA[4]', 'eA[1]', '-eA[2]']
eA[8] : ['eA[8]', 'eA[7]', '-eA[6]', '-eA[5]', '-eA[4]', '-eA[3]', 'eA[2]', 'eA[1]']
```

tools/ref71.py (rebuilds 7+1.nb objects; exit code 0)

```
tauO antisymmetric (A>=1)                          : True
tauO[A]^2=-1 (A>=1)                                : True
tauO Euclid relation                               : True
tauObar tauO relation                              : True
tauO1..7 product                                   : [1, 1, 1, 1, 1, 1, 1, 1]
sigma commutes tau1..3, anticommutes tau4..7       : True
E orthogonal                                       : True
oct: unit two-sided                                : True
oct: e_i^2                                         : [-1, -1, -1, -1, -1, -1, -1]
oct: x xbar = N(x) 1                               : True
oct: N(xy)=N(x)N(y)                                : True
oct: conj(xy)=conj(y)conj(x)                       : True
oct: alternative (left,right,flexible)             : True
oct: Moufang identities                            : True
oct: #nonassociative basis triples                 : 168
oct: #noncommuting ordered basis pairs             : 42
e_A*e_B (O) = s_AB e_A*e_B (S)                     : True
T16O Euclid Cl(8,0)                                : True
T16O[A]==T16A44[A] (A<=3)                          : True
T16O[k]==T16A44[k].S44 (k>=4)                      : True
S44==diag(sigma,sigma)                             : True
T16O[8] diag                                       : [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1,
    ...1, 1, 1, 1, 1]
eta17 = squares of the octonion basis: Re(e_A e_B) = eta17_AB: True
G17 Cl(1,7)                                        : True
G17 real                                           : True
G17[0] sym, G17[i] antisym                         : True
G17[8]^2                                           : -1
G17 blocks: G17[i]==diag(tauO,tauObar)             : True
G71 Cl(7,1) eta71                                  : True
G71[8]^2                                           : -1
G71 = Ssim (I G17) Ssim^-1                         : True
G71R32 Cl(7,1) real                                : True
```

tools/extra71.py (further identities; exit code 0)

```
Lambda boost (rows B = 0..7):
    ['cosh(beta)', '-sinh(beta)', '0', '0', '0', '0', '0', '0']
    ['-sinh(beta)', 'cosh(beta)', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', '1', '0', '0', '0']
    ['0', '0', '0', '0', '0', '1', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
Lambda rotation (rows B = 0..7):
    ['1', '0', '0', '0', '0', '0', '0', '0']
    ['0', '1', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', 'cos(theta)', '-sin(theta)', '0', '0']
    ['0', '0', '0', '0', 'sin(theta)', 'cos(theta)', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
indicator classification (signature, number of intertwiners, M.conj(M) for the normalized M,
```

```
    ...algebra):
    Cl(8,0): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(7,1): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(6,2): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(5,3): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(4,4): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(3,5): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(2,6): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(1,7): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(0,8): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
EO == eta4488 . E44[8]                                                  : True
G53 Cl(5,3) real                                                        : True
G08 Cl(0,8) real                                                        : True
so(1,7): [S_AB,S_CD]=eta_BC S_AD - eta_AC S_BD - eta_BD S_AC + eta_AD S_BC: True
[S_AB,G_C]=eta_BC G_A - eta_AC G_B                                      : True
S Sinv = 1                                                              : True
S G_C S^-1 == sum_B L[B,C] G_B                                          : True
L^T eta L == eta                                                        : True
p on shell p.eta17.p == 1                                               : True
pslash^2 == 1                                                           : True
8 real solutions                                                        : True
classification                                                          : True
explicit quaternion commutant of G71R32                                 : True
G17e[0]==[[0,1],[1,0]]x1, G17e[i]==diag(LO[i],-LO[i])                   : True
Gamma17(x)^2 == (x.eta17.x) 1                                           : True
anticommutant of G17: dimension 1 (numerical rank 255 of 2048x256 system): True
G17[8] anticommutes with all G17[A], G17[8]^2 == -1                     : True
null k: k.eta17.k == 0, kslash^2 == 0, 8 rational solutions of kslash w = 0: True
Psi = w cos(k.x) is a REAL solution of G17^A d_A Psi = 0                : True
p.eta71.p == -m^2 and P32^2 == -m^2                                     : True
Psi32 (32 arbitrary real w) solves (G71R32^A d_A - m) Psi = 0           : True
Psi32 obeys the Klein-Gordon equation (box17 + m^2) Psi = 0             : True
(G71R32^A d_A)^2 == -box17 on 32 arbitrary functions                    : True
```

tools/ref_dirac16.py (independent checks of sections 7-9; exit code 0)

```
Gamma17 real, entries 0,+-1 : True
{Gamma_A,Gamma_B} = 2 eta17_AB : True
Gamma0 symmetric, Gamma_i antisymmetric : True
Gamma^A dagger = Gamma0 Gamma^A Gamma0 : True
alpha^i, beta real symmetric : True
{alpha_i,alpha_j}=2delta, {alpha_i,beta}=0, beta^2=1 : True
block forms alpha^i = [[0,tauO_i],[-tauO_i,0]], beta = [[0,1],[1,0]] : True
omega = Gamma17[8] = [[0,-1],[1,0]], omega^2 = -1 : True
(i Dslash + m)(i Dslash - m) psi = -(box + m^2) psi : True
Gamma0 (i d0 - h) psi = (i Dslash - m) psi : True
h(p) = [[0,M],[M^T,0]] : True
h(p)^2 = E^2 1 : True
M M^T = M^T M = E^2 1_8 : True
h(p) u_s = E u_s,  h(-p) v_s = -E v_s : True
(pslash - m) u_s = 0, (pslash + m) v_s = 0 : True
u_s^T u_r = 2E delta, v_s^T v_r = 2E delta : True
u_s(p)^T v_r(p~) = 0 : True
sum_s u_s ubar_s = pslash + m, sum_s v_s vbar_s = pslash - m : True
completeness sum u u^T + sum v(p~) v(p~)^T = 2E 1_16 : True
U_p orthogonal and U_p^T h(p) U_p = diag(E 1_8, -E 1_8) : True
charge conjugation: (pslash + m) omega u_s = 0 : True
omega u_s = sum_r v_r (-M^T/E)_rs : True
S Gamma_K S^-1 = sum_B Lambda_BK Gamma_B (boost) : True
Lorentz covariance: (pslash(Lambda p) - m) S u_s(p) = 0 : True
real matrices anticommuting with all Gamma17[A]: dimension 1 (rank 255) : True
omega anticommutes with all Gamma17[A] : True
chi = i omega: Hermitian, chi^2 = 1, {chi,Gamma_A} = 0 : True
```

```
[chi, Sigma_AB] = 0 (chirality is Lorentz invariant) : True
P_L, P_R not real : True
P_L+P_R = 1, P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0, rank 8 each : True
Gamma_A P_L = P_R Gamma_A,  P_L^dagger Gamma0 = Gamma0 P_R : True
E_L, E_R isometries: E^dag E = 1_8, E_L^dag E_R = 0, E_L E_L^dag = P_L, E_R E_R^dag = P_R : True
chi E_L = -E_L, chi E_R = E_R, W unitary : True
W^dag h(p) W = [[-sigma.p, m],[m, sigma.p]] : True
sigma_i Hermitian, {sigma_i,sigma_j} = 2 delta : True
chiral blocks: E_L^dag beta E_L = 0, E_L^dag beta E_R = 1, E_L^dag alpha E_L = -sigma, E_R^dag
    ...alpha E_R = sigma, E_L^dag alpha E_R = 0 : True
E_L^dag Gamma0 (i Dslash - m) psi = (i d0 + tauO.d) phi_L - m phi_R : True
E_R^dag Gamma0 (i Dslash - m) psi = (i d0 - tauO.d) phi_R - m phi_L : True
(i d0 - tauO.d)(i d0 + tauO.d) f = -box f (8 components) : True
psi = E_L phi_L + E_R phi_R : True
chiral form of the bilinears (beta i chi, chi, alpha chi) : True
{psi_a(t,x), psi_b^dag(t,y)} = delta_ab (1/V) sum_q e^{iq(x-y)} (5 momenta, abstract mode
    ...algebra) : True
{psi_a(t,x), psi_b(t,y)} = 0 : True
the three U_k solve the Dirac equation (m = 1) : True
Grassmann components anticommute: psi_a psi_b + psi_b psi_a = 0, psi_a^2 = 0 : True
current conservation d_A (psibar Gamma^A psi) = 0 (Grassmann field) : True
dL_int/dpsibar_a = -[gS(psibar psi) psi + gP (psibar i chi psi) i chi psi + gV (psibar G_A psi)
    ...G^A psi + gA (psibar G_A chi psi) G^A chi psi]_a : True
h_KS Hermitian for real S, P, V_A, A_A : True
Gamma0 [pslash(eps,p) - m - S - i chi P - Gamma^A (V_A + chi A_A)] = eps - h_KS(p)  (plane wave
    ...e^{-i(eps t - p.x)}) : True
W^dag h_KS W = [[V^L_0 - sigma.(p+V^L), M],[M^*, V^R_0 + sigma.(p+V^R)]], V^L = V - A, V^R = V
    ...+ A, M = m + S + iP : True
Hartree potentials: dE_H/dphi*_a = (v_KS phi)_a with S = gS rhoS, P = gP rhoP, V_A = gV j_A,
    ...A_A = gA j5_A : True
densities real: Gamma0 O Hermitian for O = 1, i chi, Gamma^A, Gamma^A chi : True
chiral densities: rhoS = kappa + kappa^*, rhoP = i(kappa - kappa^*), j = jL + jR, j5 = jR - jL
    ...: True
E_H = gS/2 (k+k*)^2 - gP/2 (k-k*)^2 + (gV+gA)/2 (jL.jL + jR.jR) + (gV-gA) jL.jR : True
V^L_A = V_A - A_A, V^R_A = V_A + A_A (Hartree, chiral currents) : True
M - m = dE_H/dkappa = gS(k+k*) - gP(k-k*) = S + iP : True
dE_H/dphi_L* = (V^L_0 - sigma_i V^L_i) phi_L + (M - m) phi_R ; dE_H/dphi_R* = (V^R_0 + sigma_i
    ...V^R_i) phi_R + (M - m)^* phi_L : True
U_theta = cos(theta) - sin(theta) omega is real orthogonal; U E_L = e^{-i theta} E_L, U E_R =
    ...e^{i theta} E_R : True
U^dag beta U = beta (cos 2theta + i sin 2theta chi);  vector and axial currents invariant : True
E_H(e^{i theta chi} psi) - E_H(psi) = (gS-gP)/2 [(e^{4i theta}-1) k^2 + (e^{-4i theta}-1) k*^2]
    ...->  U(1)_A symmetric iff gP = gS : True
256 products Gamma_K are orthogonal: Gamma_K^-1 = Gamma_K^T : True
tr(Gamma_K^-1 Gamma_L) = 16 delta_KL (256 x 256) : True
Fierz completeness: sum_K (Gamma_K)_ab (Gamma_K^-1)_cd = 16 delta_ad delta_cb : True
delta E_x = tr[Sigma_x delta R] (random R, dR), Sigma_x = -gS R - gP (i chi) R (i chi) - gV G_A
    ...R G^A - gA G_A chi R G^A chi : True
E_x = (1/2) tr[Sigma_x R] : True
E_H = (gS/2)(tr R)^2 + ... = sum_c (g_c/2) tr(O_c R) tr(O^c R) and dE_H = tr[Sigma_H dR] : True
Fierz: tr[O1 R O2 R] = (1/16) sum_K tr(Gamma_K^-1 R) tr(O1 Gamma_K O2 R) for all channel
    ...vertices : True
uniform R: tr R = rho_S, tr(Gamma^0 R) = rho : True
(1/2) sum_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) = (E Gamma_0 + m)/(2E) : True
E_x(uniform) = -gS/32 (rS^2+rho^2) + gP/32 (rS^2-rho^2) - gV/32 (8rS^2-6rho^2) + gA/32
    ...(8rS^2+6rho^2) : True
Sigma_x(uniform) = S_x + V_x Gamma_0, S_x = rS/16(-gS+gP-8gV+8gA), V_x = rho/16(-gS-gP+6gV+6gA)
    ...: True
S_x = dE_x/drho_S, V_x = dE_x/drho : True
uniform Hartree-Fock = Hartree with G_S = gS + (-gS+gP-8gV+8gA)/16, G_V = gV +
    ...(-gS-gP+6gV+6gA)/16 : True
uniform matter: Hartree terms of P and A vanish (rho_P = 0, j5 = 0, j^i = 0) : True
Wick (3 particles, 16 modes): <:(psibar O psi)(psibar O' psi):> = tr(O R) tr(O' R) - tr(O R O'
    ...R) for all 18 vertex pairs : True
c7 = 8 |S^6|/(2pi)^7 = 1/(15 pi^4) and rho = c7 kF^7/7 = kF^7/(105 pi^4) : True
```

```
NUM Hartree saturation kF0 : 0.75663615070149289469
NUM Hartree rho0 : 0.00001388100213231687719 7
NUM Hartree m* : 0.34809734014624326514
NUM Hartree rhoS : 6.5190265985375673486e-6
NUM Hartree V0 : 0.13881002132316877197
NUM Hartree E/A - m : -0.028321427964738575854
NUM Hartree eps_F : 0.97167857203526142415
Hartree gap: m* - m - gS rhoS(m*) = 0 : True
m* is a stationary point of eps(m*) = ekin(m*) - (m*-m)^2/(2gS) + gV rho^2/2 : True
total energy = sum eps_k - E_H (double counting) = ekin - (m*-m) rhoS + E_H : True
Hugenholtz-Van Hove at saturation: E/A = eps_F = d eps/d rho : True
d(eps/rho)/dkF = 0 at kF0 (minimum: E/A(0.95 kF0), E/A(1.05 kF0) larger) : True
Hugenholtz-Van Hove mu = d eps/d rho = eps_F at kF = 1 (not saturated) : True
NUM Hartree kF=1: m* : 0.080964430277606126329
NUM Hartree kF=1: E/A - m : 0.41086848863433217115
NUM Hartree kF=1: eps_F : 1.9809848613081542876
NUM HF couplings G_S, G_V : -98750.0, 20000.0
HF self-consistency with the exchange self-energy: m* - m = gS rhoS + S_x, V0 = gV rho + V_x :
    ...True
HF energy: ekin - (m*-m) rhoS + E_H + E_x = eps(G_S, G_V) : True
HF Hugenholtz-Van Hove: d eps/d rho = eps_F at kF = 3/4 : True
NUM HF kF=3/4: m* : 0.36643630198154392124
NUM HF kF=3/4: V0 : 0.2610177547316155257
NUM HF kF=3/4: E/A - m : 0.039146338016119861526
NUM Hartree kF=3/4: m* : 0.36297394764803019822
NUM Hartree kF=3/4: E/A - m : -0.028065375846872960879
J(0) = Lambda^6/(720 pi^4), g_c = -1/(8 J(0)) = -90 pi^4/Lambda^6 : True
NUM NJL g_c : -8766.8181930602193513
NUM NJL g = 6/5 g_c : -10520.181831672263222
NUM NJL |M| (m = 0) : 0.55369385182924169122
NUM NJL kappa = -4 M J(|M|) (M real) : -0.00002631579285836487 42
NUM NJL E(|M|) - E(0) : -1.3102449629010022473e-6
NJL m = 0: gap 1 + 8 g J(|M|) = 0 solved, M = 2 g kappa^*, stationary and lower than M = 0 :
    ...True
NJL m = 0: E(|M| e^{i theta}) independent of theta (degenerate chiral vacua) : True
NUM NJL M (m = 1/100) : 0.58531284674622051123
NJL m = 1/100: M = m - 8 g J(M) M real, stationary; E(M e^{i theta}) > E(M) for theta != 0 :
    ...True
sea at one momentum: eigenvalues -E (x8), +E (x8); kappa_p = tr[(Pi_-)_{RL}] = -4 M^*/E;
    ...tr[Pi_- h_0] = -8E + 8 Re[(M-m) M^*]/E : True
boolean checks: 90   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_gravity.py (independent checks of sections 11-14; exit code 0)

```
eta17 is simply the squares of the octonion basis: eta_AB = Re(e_A e_B), e_0^2 = +1, e_i^2 = -1
    ...(i = 1..7) : True
the quadratic form of the octonions has the signs of the squares of the octonion basis: Re(x x)
    ...= x0^2 - x1^2 - ... - x7^2 = x.eta17.x : True
cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2,
    ...-Ha Hb : True
cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b} : True
G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2 : True
G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb : True
G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb : True
scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb) : True
E^(1) = Einstein tensor G (cosmological metric) : True
L_1 = R : True
E^(0) = -delta/2 : True
trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4 : True
E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0 : True
E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ...
    ...= E^(k)7_7 : True
```

```
E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3) : True
E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb) : True
E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2,
    ...and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric) : True
closed forms : {(1, 0): 3*H_a**2 + 12*H_a*H_b + 6*H_b**2, (1, 1): 3*H_a**2 + 8*H_a*H_b +
    ...10*H_b**2 + 2*dH_a + 4*dH_b, (1, 4): 6*H_a**2 + 9*H_a*H_b + 6*H_b**2 + 3*dH_a + 3*dH_b,
    ...(2, 0): -48*H_a**3*H_b - 216*H_a**2*H_b**2 - 144*H_a*H_b**3 - 12*H_b**4, (2, 1):
    ...-32*H_a**3*H_b - 136*H_a**2*H_b**2 - 16*H_a**2*dH_b - 192*H_a*H_b**3 - 32*H_a*H_b*dH_a -
    ...96*H_a*H_b*dH_b - 60*H_b**4 - 48*H_b**2*dH_a - 48*H_b**2*dH_b, (2, 4): -12*H_a**4 -
    ...108*H_a**3*H_b - 180*H_a**2*H_b**2 - 12*H_a**2*dH_a - 36*H_a**2*dH_b - 108*H_a*H_b**3 -
    ...72*H_a*H_b*dH_a - 72*H_a*H_b*dH_b - 12*H_b**4 - 36*H_b**2*dH_a - 12*H_b**2*dH_b, (3, 0):
    ...1440*H_a**3*H_b**3 + 1080*H_a**2*H_b**4, (3, 1): 576*H_a**3*H_b**3 + 1368*H_a**2*H_b**4
    ...+ 864*H_a**2*H_b**2*dH_b + 576*H_a*H_b**5 + 576*H_a*H_b**3*dH_a + 576*H_a*H_b**3*dH_b +
    ...144*H_b**4*dH_a, (3, 4): 648*H_a**4*H_b**2 + 1224*H_a**3*H_b**3 + 432*H_a**3*H_b*dH_b +
    ...648*H_a**2*H_b**4 + 648*H_a**2*H_b**2*dH_a + 648*H_a**2*H_b**2*dH_b +
    ...432*H_a*H_b**3*dH_a}
divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b) : True
variation of sqrt|g| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4
    ...E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4 : True
inhomogeneous metric: E^(3)0_4 at the point (exact) :
    ...353743050254866364332989413672793316453216419840002457773688891322171130362900905756260763178144678501
inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point : True
inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3 : True
inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3 : True
pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1 : True
pp-wave: E^(2) = E^(3) = 0 for every profile F : True
Kasner E^(k) components x0^(2k) (00, 11, 44) : {1: [3*(p**2 + 4*p*q + 2*q**2), 3*p**2 + 8*p*q -
    ...2*p + 10*q**2 - 4*q, 3*(2*p**2 + 3*p*q - p + 2*q**2 - q)], 2: [-12*q*(4*p**3 + 18*p**2*q
    ...+ 12*p*q**2 + q**3), -4*q*(8*p**3 + 34*p**2*q - 12*p**2 + 48*p*q**2 - 36*p*q + 15*q**3 -
    ...12*q**2), -12*(p**4 + 9*p**3*q - p**3 + 15*p**2*q**2 - 9*p**2*q + 9*p*q**3 - 9*p*q**2 +
    ...q**4 - q**3)], 3: [360*p**2*q**3*(4*p + 3*q), 72*p*q**3*(8*p**2 + 19*p*q - 20*p + 8*q**2
    ...- 10*q), 72*p**2*q**2*(9*p**2 + 17*p*q - 15*p + 9*q**2 - 15*q)]}
Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2
    ...= 1 : True
Gauss-Bonnet Kasner cubic (64 x e4 at q = (3-3p)/4) : -203*p**3 + 81*p**2 + 351*p + 27
Gauss-Bonnet Kasner exponents p (three real roots) : (-1.083501787612105742423976294,
    ...-0.078631058418927766227861052109, 1.561147624356156661361196952 75)
Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3 : True
cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate
    ...families q = 0 and p = 0 : True
Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0) : True
8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times
    ...delta) : True
4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) =
    ...diag(0,0,-12) H^4, E^(3) = 0 : True
static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2
    ...(Ha'+Ha^2)), E^(3) = 0 : True
exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha =
    ...H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3) : True
exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three
    ...vacuum equations have rank 2) : True
exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all
    ...three equations : True
eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 -
    ...Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24) : True
NUM exponential vacuum solution (w2, w3) for general q : ((-2*Lambda*q - 3*Lambda + 12*q**3*w1
    ...+ 27*q**2*w1 + 22*q*w1 + 9*w1)/(4*q*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)),
    ...(-3*Lambda*q**2 - 6*Lambda*q - Lambda + 15*q**4*w1 + 36*q**3*w1 + 33*q**2*w1 + 18*q*w1 +
    ...3*w1)/(72*q**3*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)))
NUM eq00, eq11, eq44 for q = -1 : (-Lambda - 3*w1 - 36*w2 - 360*w3, -Lambda + 5*w1 + 28*w2 +
    ...216*w3, -Lambda + 3*w1 + 12*w2 + 72*w3)
cosmological metric: eta e e = g : True
cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
cosmological metric: Gamma_mu real : True
```

```
cosmological metric: omega_mu AB = -omega_mu BA : True
cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu =
    ...0 : True
cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu
    ...nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 :
    ...True
cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0] : True
cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2)
    ...Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0 : True
pp-wave: eta e e = g : True
pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
pp-wave: Gamma_mu real : True
pp-wave: omega_mu AB = -omega_mu BA : True
pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l
    ...+ [Omega_mu, Gamma_nu] = 0 : True
pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
inhomogeneous metric N, A, B (x0, x4): eta e e = g : True
inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
inhomogeneous metric N, A, B (x0, x4): Gamma_mu real : True
inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA : True
inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l +
    ...omega_mu^A_B e^B_nu = 0 : True
inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] =
    ...-Omega_mu : True
inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu
    ...Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B +
    ...[Omega_mu, Gamma^A] = 0 : True
inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A
    ...Gamma^B : True
inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0
    ...: True
inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the
    ...cosmological frame : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} =
    ...2 g_mu nu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB =
    ...-omega_mu BA : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate
    ...d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real,
    ...Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant
    ...derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu,
    ...Gamma_nu] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame
    ...matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu
    ...= (1/4) R_AB mu nu Gamma^A Gamma^B : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l
    ...F_mn + [Omega_l, F_mn] + cyclic = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu
```

```
    ...Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation for the static radial metric : True
structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) =
    ...-(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric) : True
static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k
    ...= (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7) : True
static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i
    ...Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) +
    ...3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi : True
static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi
    ...= f^(-1/4) r^(-3) Gamma^mu d_mu chi : True
U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu
    ...(D_mu - i q A_mu) psi with A' = A + d alpha : True
current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu
    ...psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu : True
current identity d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi +
    ...psibar Gamma^mu D_mu psi] : True
Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi : True
rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3
    ...b^4)^(-1/2) Gamma^mu d_mu chi : True
mode equation: (i Gamma^mu D_mu - m) psi = 0  <=>  i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) +
    ...beta m : True
conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta :
    ...True
2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B
    ...beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta : True
Kasner (Einstein branch), massless: |beta|^2 for (|p|, |q|) = (1,1), (2,1), (1,3) :
    ...(np.float64(0.01074934616777871), np.float64(0.00013294108220674155),
    ...np.float64(0.06744202228393215))
Kasner scaling variable K = |p| lambda^(1-p), lambda = (|q|/|p|)^(1/(q-p)) : (1.0,
    ...3.11828865465698, 0.4946322370571032)
Kasner: |alpha|^2 + |beta|^2 = 1 (to 1e-8) : True
Kasner scaling: (|p|, |q|) = (1.430..., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to
    ...1e-6) : True
Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0: 0.05 -> 40: |beta|^2 of the 16x16 evolution (average),
    ...of the sector Z = +i (+m sigma3) and of the sector Z = -i (-m sigma3) :
    ...(np.float64(0.027052647156543617), np.float64(0.02528659275098409),
    ...np.float64(0.028818701562187667))
16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3:
    ...average agrees (to 1e-8), unitarity; the two sectors differ : True
isotropic expansion, massless: no pair creation (|beta|^2 < 1e-20) : True
far future (de Sitter, H = 1): |beta|^2 for m/H = 0.25, 0.5, 1 :
    ...(np.float64(0.1721029191835808), np.float64(0.04142388447252214),
    ...np.float64(0.001863977463333981))
de Sitter: |beta|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature
    ...H/2pi), to 1e-7 : True
integrator benchmark: Landau-Zener |beta|^2 = exp(-pi kappa^2) (to 1e-6) : True
boolean checks: 101   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_ks_gravity.py (independent checks of sections 15-16; exit code 0)

```
256 products in 9 ranks: 1, 8, 28, 56, 70, 56, 28, 8, 1 : True
every bilinear psibar Theta^K psi is real: Gamma17[0] Theta^K Hermitian (Theta^K =
    ...i^((r(r-1)/2) mod 2) Gamma^K) : True
Theta_K Theta^K = 1_16 and tr(Theta_K Theta^J) = 16 delta_KJ : True
Fierz completeness: sum_K (Theta_K)_cd (Theta^K)_ab = 16 delta_ad delta_bc : True
Hodge duality: Theta^K omega17 = +- Theta^(complement of K) (up to a phase) : True
stage-2 channels: I_0 = (psibar psi)^2, I_1 = (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 =
    ...-(psibar i chi psi)^2, I_7 = -(axial) : True
local Lorentz invariance: I_r(S psi, psibar S^-1) = I_r(psi, psibar), r = 0..8 (random finite
    ...boost-rotation) : True
Fierz matrix F_rs = (1/256) sum_{K in r} tr(Theta_K Theta_J Theta^K Theta^J), independent of
```

```
    ...the representative J of rank s : True
Fierz matrix closed form F_rs = (1/16) sum_l (-1)^(rs-l) C(s,l) C(8-s,r-l) : True
F^2 = 1 (the Fierz transformation is an involution) : True
eigenvalues of F: number of +1, number of -1 : (5, 4)
Fierz matrix (rows r, columns s) times 16 : [[1, 1, 1, 1, 1, 1, 1, 1, 1], [8, -6, 4, -2, 0, 2,
    ...-4, 6, -8], [28, 14, 4, -2, -4, -2, 4, 14, 28], [56, -14, -4, 6, 0, -6, 4, 14, -56],
    ...[70, 0, -10, 0, 6, 0, -10, 0, 70], [56, 14, -4, -6, 0, 6, 4, -14, -56], [28, -14, 4, 2,
    ...-4, 2, 4, -14, 28], [8, 6, 4, 2, 0, -2, -4, -6, -8], [1, -1, 1, -1, 1, -1, 1, -1, 1]]
Fierz identity for commuting spinors: sum_{K in r} (a Theta_K b)(c Theta^K d) = sum_s F_rs
    ...sum_{J in s} (a Theta_J d)(c Theta^J b) : True
Hartree-Fock energy E_H + E_x = sum_s (G_s/2) D_s with G = (1 - F^T) g (random density matrix,
    ...random couplings) : True
uniform matter R = (rho_S + rho Gamma17[0])/16: only D_0 = rho_S^2, D_1 = rho^2; with g_0 = gS,
    ...g_1 = gV, g_7 = -gA, g_8 = -gP the exchange energy is that of stage 2 (section 8.7) :
    ...True
uniform density matrix: D_0 = rho_S^2, D_1 = rho^2, D_2..D_8 = 0 : True
stage-2 couplings gS = -100000, gV = 10000: G_0 = -98750, G_1 = 20000 (the HF couplings of
    ...section 8.7) : True
effective couplings G_0, G_1 in terms of all nine g_r : (15*g0/16 - g1/2 - 7*g2/4 - 7*g3/2 -
    ...35*g4/8 - 7*g5/2 - 7*g6/4 - g7/2 - g8/16, -g0/16 + 11*g1/8 - 7*g2/8 + 7*g3/8 - 7*g5/8 +
    ...7*g6/8 - 3*g7/8 + g8/16)
example g = (-100000, 10000, 3000, 0, ..., 0): G_0, G_1 : (-104000, 17375)
nonlinear field equation: dL_int/d psibar_a = -sum_r g_r sum_K (psibar Theta_K psi)(Theta^K
    ...psi)_a : True
the chirality chi17 commutes with every generator Sigma^AB, hence with every spin connection
    ...Omega_mu : True
curved chiral equations: E_L^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) +
    ...tauO.d/(a,b)) phi_L - m phi_R : True
curved chiral equations: E_R^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) -
    ...tauO.d/(a,b)) phi_R - m phi_L : True
flat limit a = b = 1, kF = kF0 (stage-2 saturation): m* = 0.3480973401..., E/A - m =
    ...-0.0283214279... : True
flat limit: isotropic pressure pa = pb = rho eps_F - eps (Gibbs-Duhem, eps_F = V0 + sqrt(kF^2 +
    ...m*^2)) : True
NUM flat: m*, rho, eps, p : (0.3480973401462447, np.float64(1.3881002132316779e-05),
    ...np.float64(1.3487872330347983e-05), np.float64(-1.588186776101813e-20))
energy-momentum conservation eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for the KS matter
    ...(arbitrary a, b; relative residual) : True
NUM conservation residual / eps' : -5.166108748755716e-12
primordial (Einstein-Kasner background, comoving kF = kF0 at x0 = 1): m*(x0) for x0 = 1e-4 ...
    ...10 : (0.003824179379430487, 0.010783989711922532, 0.030238519612042748,
    ...0.08844464116180238, 0.3480973401462447, 0.8917637493589158)
primordial: rho(x0) x0 = kF0^7/(105 pi^4) (density grows like 1/x0) : True
primordial: m* decreases toward 0 as x0 -> 0 (blue-shifted ordinary momenta, attractive scalar
    ...coupling) : True
primordial: anisotropic pressure pa/pb at x0 = 1e-3, 1e-1, 10 :
    ...(np.float64(1.0868869169150372), np.float64(1.6461737747098435),
    ...np.float64(-0.4036333858521348))
modern (static auxiliary space b = 1): m*(a) for a = 1, 2, 4, 10, 100 : (0.3480973401462447,
    ...0.853540809048541, 0.9807410717586909, 0.9987576007278518, 0.9999987563024887)
modern: eps a^3 -> constant and pa/eps -> 0 (dirac16complex behaves as pressureless matter in
    ...ordinary space) : True
modern: pb/eps at a = 100 (stress along the auxiliary space, balanced by the stabilizing
    ...stress) : 0.049773512087215956
far future (de Sitter a = e^{0.1 x0}, b = 1): m*(x0) for x0 = 0, 10, 20, 40, 80 -> m = 1 :
    ...(0.3480973401462447, 0.9395796910293632, 0.996923581335845, 0.9999923586141617,
    ...0.9999999999530483)
far future: rho = rho0 e^{-3 H x0} and 1 - m* -> 0 like rho : True
the 00 closed forms of E^(1), E^(2), E^(3) (pair-set algorithm, tools/lovelock71.py) agree with
    ...the closed forms of section 12.6 of 7+1.nb : True
Einstein-Lovelock equations are linear in (Ha', Hb') (quasi-linear second-order system) : True
Einstein + KS matter (kappa = 1000, Lambda = 0, Ha(0) = 0.05): Hb(0) from the constraint, (a,
    ...b, Ha, Hb) at x0 = 20, max relative constraint violation : (0.009143712444558836,
    ...(np.float64(1.5817386798469366), np.float64(1.1975585618423046),
    ...np.float64(0.011877999367582402), np.float64(0.007419107766685833)),
```

```
    ...np.float64(8.415965253866209e-12))
Einstein-Lovelock (alpha2 = 20, alpha3 = 500) + KS matter: Hb(0), (a, b, Ha, Hb) at x0 = 20,
    ...max relative constraint violation : (0.014980703942304356,
    ...(np.float64(1.4773514780943342), np.float64(1.276084053398673),
    ...np.float64(0.010175914492223711), np.float64(0.008488764693542736)),
    ...np.float64(4.387597609521105e-12))
the constraint Sum alpha_k E^(k)0_0 - Lambda = kappa eps is preserved by the evolution
    ...(relative violation < 1e-7) in both cases : True
boolean checks: 29   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_sm.py (independent checks of sections 17-19; exit code 0)

```
anomaly cancellation in each generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3,
    ...grav^2U(1), SU(3)^3 all 0 : True
Witten SU(2) anomaly: number of doublets per generation (3 colours of Q + 1 L) = 4, even : True
electric charges Q = T3 + Y: Q -> (2/3, -1/3), u_R 2/3, d_R -1/3, L -> (0, -1), e_R -1, nu_R 0
    ...: True
effective number of relativistic degrees of freedom g* = 28 + (7/8) 90 = 106.75 (106.75 + 5.25
    ...if nu_R were thermal) : True
gauge-boson mass matrix from |D<Phi>|^2: eigenvalues g^2v^2/4 (twice, W+-), (g^2+g'^2)v^2/4
    ...(Z), 0 (photon) : True
NUM v = (sqrt2 G_F)^(-1/2) [GeV], g, g', sin^2 theta_W (on shell), e, 1/alpha(tree), lambda =
    ...m_H^2/(2v^2) : (246.21964023926205, 0.6528252573344827, 0.34994623401223707,
    ...0.22320949284344327, 0.3084276316025824, 132.10014858375337, 0.1292805756612353)
tree-level relations: m_W = g v/2, m_Z = sqrt(g^2+g'^2) v/2, rho = m_W^2/(m_Z^2 cos^2) = 1,
    ...m_H^2 = 2 lambda v^2 : True
CKM unitarity V V^dag = 1 : True
NUM CKM |V| (rows ud us ub / cd cs cb / td ts tb) : [[0.97435, 0.225008, 0.003732], [0.224876,
    ...0.973489, 0.04183], [0.008583, 0.041109, 0.999118]]
NUM Jarlskog J = Im(V_us V_cb V_ub* V_cs*) : 3.1169935287554706e-05
J = c12 c23 c13^2 s12 s23 s13 sin(delta) = 3.12e-5 (PDG 2024: 3.12 +0.13 -0.12 e-5) : True
NUM Wolfenstein lambda = s12, A = s23/s12^2, rhobar + i etabar = -V_ud V_ub*/(V_cd V_cb*) :
    ...(0.22501, 0.8261981634695225, np.float64(0.15918443594247167),
    ...np.float64(0.35227461890586964))
Wolfenstein parameters agree with the PDG fit (A 0.826, rhobar 0.1591, etabar 0.3523) within
    ...their quoted uncertainties : True
Yukawa matrices Y_u = sqrt2 diag(m_u,m_c,m_t)/v, Y_d = sqrt2 V diag(m_d,m_s,m_b)/v: singular
    ...values x v/sqrt2 = masses, |U_uL^dag U_dL| = |V| : True
NUM top Yukawa y_t = sqrt2 m_t/v : 0.9911341438317451
PMNS unitarity : True
NUM PMNS |U| and J_PMNS : ([[0.82328, 0.54796, 0.14816], [0.27491, 0.61298, 0.74073], [0.49663,
    ...0.56921, 0.65526]], np.float64(0.0017359130944756358))
NUM Dirac neutrino masses (lightest m1 = 0, eV) and their sum : (0.0, 0.008654478609367522,
    ...0.05033885179461287, 0.058993330403980394)
neutrino Yukawa matrix Y_nu = sqrt2 U diag(m1,m2,m3)/v: masses and |U| recovered (singular
    ...value decomposition) : True
NUM largest neutrino Yukawa sqrt2 m3/v : 2.891316340688847e-13
vacuum: probabilities P(a -> b) form a doubly stochastic matrix (rows and columns sum to 1) :
    ...True
vacuum: amplitude result = delta_ab - 4 sum_{i>j} Re(X_ij) sin^2 Delta_ij + 2 sum_{i>j}
    ...Im(X_ij) sin 2Delta_ij, X_ij = U_ai* U_bi U_aj U_bj* : True
NUM DUNE vacuum (L = 1300 km, E = 2.5 GeV): P(mu->e), P(mubar->ebar), P(mu->mu) :
    ...(np.float64(0.04883685296656589), np.float64(0.05019851034574761),
    ...np.float64(0.012813139189347619))
CPT: P(a->b) = P(bbar->abar); CP asymmetry P(mu->e) - P(mubar->ebar) proportional to J
    ...(vanishes for delta = 0, pi) : True
NUM V_CC = sqrt2 G_F n_e for rho Y_e = 1 g/cm^3 [eV] (textbook 7.63e-14 eV) :
    ...7.632466872241316e-14
V_CC(rho Y_e = 1 g/cm^3) = 7.63e-14 eV : True
matter (constant density): unitary evolution, probabilities doubly stochastic : True
NUM DUNE in matter (rho = 2.848 g/cm^3, Y_e = 0.5): P(mu->e), P(mubar->ebar) :
    ...(np.float64(0.07232693782692827), np.float64(0.029191799469459462))
```

```
matter: a flavour-universal potential has no effect (only differences of potentials matter) :
    ...True
two-flavour MSW: sin^2 theta_m from diagonalization = formula; maximal mixing (sin^2 2theta_m =
    ...1) at 2 E V = dm^2 cos 2theta : True
NUM Sun: P_ee (numerical through the solar density profile, adiabatic formula) for E = 0.3, 1,
    ...5, 10 MeV : {0.3: (np.float64(0.5251624291065616), 0.5251645055880894), 1.0:
    ...(np.float64(0.4584114174121251), 0.4584122490661078), 5.0:
    ...(np.float64(0.3103377102301538), 0.3103335111666083), 10.0:
    ...(np.float64(0.2979217906988696), 0.29791748878645763)}
Sun: numerical P_ee = adiabatic MSW formula to 2e-3 at E = 0.3, 1, 5, 10 MeV : True
Sun: P_ee falls from near the vacuum average c13^4 (1 - sin^2(2 theta12)/2) + s13^4 (within
    ...0.03 at 0.3 MeV) to about 0.31 at 10 MeV : True
NUM supernova MSW resonance densities for E = 20 MeV, Y_e = 0.5: H (dm31, theta13), L (dm21,
    ...theta12) [g/cm^3] : (1587.1391520947036, 18.939748107619376)
NUM cosmic neutrino background: T_nu = (4/11)^(1/3) T_gamma [K], number density per flavour
    ...[cm^-3] : (1.9453688391750839, 112.01641316574336)
CNB: T_nu = 1.945 K and n = 112 cm^-3 per flavour : True
brane frame metric diag(+1,-1,-1,-1) = eta17 on x0..x3 = squares of the quaternion basis 1,
    ...e_1, e_2, e_3 (Re(e_a e_b)); H closed : True
brane representation: gamma_mu (x) 1_4 (mu = 0..3) and gamma5 (x) rho_k (k = 4..7) satisfy
    ...{R_A, R_B} = 2 eta_AB : True
intertwiner U (Schur average over the 256 Clifford products): Gamma17[A] U = U R_A for A =
    ...0..7, U unitary : True
8D chirality at the brane: chi17 = U (gamma5 (x) rho4567) U^dag, rho4567 Hermitian with
    ...eigenvalues +1, +1, -1, -1 : True
bilinears at the brane: Psibar Psi' = sum_a psibar_a psi'_a and Psibar Gamma^mu Psi' = sum_a
    ...psibar_a gamma^mu psi'_a (mu = 0..3) : True
Kaluza-Klein: the 16x16 operator gamma.p (x) 1 + gamma5 (x) rho.q/b0 - m is singular exactly on
    ...p0^2 - |p|^2 = m^2 + |q|^2/b0^2 : True
KK operator: X^2 = -|q|^2/b0^2 and X anticommutes with gamma^mu (x) 1 : True
portal mixing: light masses (eV) and heavy masses (GeV) of the 7x7 Dirac mass matrix : ([0.0,
    ...0.008654465673872413, 0.05033886016084743], [1.0, 1.0, 1.00000000025,
    ...1.0000000007499998])
portal mixing: N N^dag = 1 - mu mu^dag/m^2 to second order (non-unitary light mixing), (1 - N
    ...N^dag)_ee : (True, np.float64(9.99999860695766e-10))
non-unitary oscillations: sum_b P(a->b) = (N N^dag)_aa^2-type deficit < 1 (zero-distance
    ...effect), complemented by the heavy states : True
portal decoupling: for mu -> 0 the light block becomes the PMNS matrix and the oscillation
    ...probabilities the three-flavour vacuum ones (section 19) : True
NUM vector portal (V_P,e = 0.2 V_CC): P(mu->e) at DUNE with and without the dirac16complex
    ...potential : (np.float64(0.07704998293140915), np.float64(0.07232693782692827))
NUM age of the universe from the brane Friedmann equation (Planck 2018: H0 = 67.4, Omega_m =
    ...0.315) [Gyr] : 13.790728126409226
age 13.79 Gyr (Planck 2018: 13.787 +- 0.020) : True
ADD relation: 3 Ha^2 = kappa8 rho4/V_perp = 8 pi G4 rho4 with G4 = G8/V_perp : True
NUM ADD example (reduced 8D Planck mass 10 TeV): V_perp [GeV^-4], V_perp^(1/4) [m], lowest KK
    ...mass 2 pi/V_perp^(1/4) [MeV] : (5929225000000.0, 3.079186208511254e-13,
    ...4.026524867292262)
boolean checks: 33   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_coupled.py (independent checks of sections 20-23; exit code 0)

```
stabilizing potential: S_U = -Int sqrt|g| U(b) gives eps = U, pa = -U, pb = -U - (b/4) U'(b) :
    ...True
brane tension, smeared over the torus (delta^4(y) -> 1/Lc^4): eps = f_W/V_perp, pa =
    ...-f_W/V_perp, pb = 0, V_perp = b^4 Lc^4 : True
brane fluid (Standard Model plasma, rho_4 = N_W a^(-3(1+w))), smeared: eps = rho_4/V_perp, pa =
    ...w eps, pb = 0 : True
massless dirac16complex particle (comoving momenta k_a, k_b): eps V = omega, pa V = k_a^2/(3
    ...a^2 omega), pb V = k_b^2/(4 b^2 omega), V = a^3 b^4 : True
every such matter is conserved: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for arbitrary W(a, b),
    ...a(x0), b(x0) : True
```

```
field equations from S_grav + S_m: EL_n = (a^3 b^4/kappa)(S00 - kappa eps), EL_a = (3a^2
    ...b^4/kappa)(S11 + kappa pa), EL_b = (4a^3 b^3/kappa)(S44 + kappa pb), S = Sum alpha_k
    ...E^(k) (minisuperspace, lapse n = 1) : True
radion equation (Einstein): Hb' + 3 Ha Hb + 4 Hb^2 = kappa (eps/6 - pa/2 + pb/3) + Lambda/3 (Hb
    ...= beta', exact) : True
radion mass: with U = U0 + (K/2) ln^2(b/b0) the restoring term is -m_r^2 beta, m_r^2 = kappa
    ...K/12 : True
static auxiliary space: (kappa/4) b U'(b) = Lambda + kappa U + 2 kappa f_W/V_perp +
    ...(kappa/2)(eps - 3pa) + kappa pb - 12 alpha2 Ha^2 (Ha' + Ha^2) : True
static auxiliary space: E^(3) does not contribute, and Ha' = -(kappa/2)(eps + pa) (brane
    ...Friedmann acceleration) : True
four-dimensional Friedmann equation on the brane (b static): 3 Ha^2 = 8 pi G4 rho4 + Lambda4
    ...with 8 pi G4 = kappa/V_perp and Lambda4 = Lambda + kappa U0 + kappa f_W/V_perp : True
cubic Lovelock density: L_3 = R^3 - 12 R Ric.Ric + 16 Ric^3 + 24 R^ac R^bd R_abcd + 3 R
    ...Riem.Riem - 24 R^ab R_a^cde R_bcde + 4 R^ab_cd R^cd_ef R^ef_ab - 8 R_a^c_b^d R_c^e_d^f
    ...R_e^a_f^b (exact, four random integer algebraic curvature tensors) : True
Gauss-Bonnet density: L_2 = R^2 - 4 Ric.Ric + Riem.Riem (same tensors) : True
vacuum constraint in u = Hb/Ha, X = Ha^2 (alpha2 = 1, alpha3 = xi): E(u) + X G(u) + xi X^2 K(u)
    ...= 0 with E = 3(2u^2 + 4u + 1), G = -12u(u^3 + 12u^2 + 18u + 4), K = 360 u^3 (3u + 4) :
    ...True
Einstein regime: u1 = Hb/Ha = -1 + 1/sqrt(2), root of E(u); Kasner p1 = 1/(3 + 4u1) = (1 + 2
    ...sqrt 2)/7, q1 = u1 p1 = (2 - 3 sqrt 2)/14 : True
Gauss-Bonnet regime: u2 = Hb/Ha, root of u^3 + 12u^2 + 18u + 4 in (-1/3, 0); Kasner p2 = 3/(3 +
    ...4u2), q2 = u2 p2 (3p + 4q = 3) : True
NUM u1, u2, p1, q1, p2, q2 : (-0.2928932188134525, -0.26958419031043745, 0.5469181606780271,
    ...-0.16018862050852037, 1.5611476243561566, -0.4208607182671175)
exact vacuum solution: for every u in (u1, 0) the curve Ha = sqrt(X(u)), Hb = u Ha, X(u) = [G +
    ...sqrt(G^2 - 4 xi K E)]/(2 xi |K|), solves the constraint (residual < 1e-20 relative) :
    ...True
NUM effective Kasner exponent p_eff = -Ha^2/Ha' along the curve at u -> u1, u = u2, u -> 0 (xi
    ...= 1e-6) : (0.5469181624462031, 1.553736173053791, 1.0000018125008945)
the effective exponent goes from p1 (Einstein, u -> u1) through about p2 (Gauss-Bonnet, u = u2)
    ...to 1 (cubic, u -> 0) : True
cubic regime (tau -> 0): Ha = 1/tau + A0 + A1 tau, Hb = h0 + h1 tau + h2 tau^2 solves all three
    ...equations to the orders used; h0 = -sqrt(alpha2/(30 alpha3)) : True
cubic regime: h1 = alpha2/(16 alpha3) - 1/(32 alpha2), A0 = sqrt(30)(58 alpha2^2 - 45
    ...alpha3)/(1920 alpha2^(3/2) alpha3^(1/2)) : True
cubic regime: A1 = -3647 alpha2/(92160 alpha3) - 223/(6144 alpha2) + 645 alpha3/(8192
    ...alpha2^3), h2 = sqrt(30)(-1212 alpha2^4 + 1180 alpha2^2 alpha3 + 225 alpha3^2)/(307200
    ...alpha2^(5/2) alpha3^(3/2)) : True
NUM cubic-regime coefficients (xi = 1e-6): h0, h1, h2, A0, A1 : (-182.57418583505537,
    ...62499.96875, -21609345.4877448, 165.4577275405445, -39572.51893438307)
Einstein regime (Ha -> 0): u = u1 + d1 Ha^2 + ..., d1 = alpha2 (16 - 11 sqrt 2)/4; p_eff = p1 +
    ...e1 Ha^2 + ..., e1 = alpha2 (1 + 9 sqrt 2)/7 : True
NUM Gauss-Bonnet regime: p_eff = p2 + f1/Ha^2 + f2 xi Ha^2 + ..., f1, f2 (alpha2 = 1) :
    ...(-0.6988734676355542, -0.7853407141875988)
cubic-series initial data at tau = 1e-8 satisfy the constraint to < 1e-12 (relative to 48
    ...alpha2 Ha^3 |Hb|) : True
NUM exact quadrature: tau(u2), Ha(u2), s1 = tau(u2) - p1/Ha(u2) + Int_0^Ha(u2) (1/F - p1/H^2)
    ...dH : (0.15935497980477448, 9.721560116243376, 2.0795884930371127)
the ODE solution and the quadrature agree: tau at Ha(u2) to 1e-10, s1 (ODE at Ha = 3e-3) to
    ...1e-6 : True
NUM time shifts: tau = p1/Ha - e1 Ha + s1 (Einstein), tau = p2/Ha + f1/(3 Ha^3) - f2 xi Ha + s2
    ...(Gauss-Bonnet, at Ha_mid = (f1/(f2 xi))^(1/4)): s1, s2, Ha_mid : (2.0795884930371127,
    ...np.float64(-0.0009864616381799337), 30.713910188158728)
NUM accuracy (max relative error of tau(Ha)): cubic for Ha >= 1e4, Gauss-Bonnet for 3 <= Ha <=
    ...300, Einstein for Ha <= 0.1 : (np.float64(6.651897785481253e-06),
    ...np.float64(0.0008521318063101635), np.float64(0.00032101971624165504))
regime approximations: cubic error < 1e-5 for Ha >= 1e4, Gauss-Bonnet < 1e-3 for 3 <= Ha <=
    ...300, Einstein < 1e-3 for Ha <= 0.1 : True
NUM worst error of the best regime approximation over 1e-3.5 <= Ha <= 1e8 (in the crossover
    ...windows) : 0.10466193313053529
NUM vacuum crossover times: tau at Ha = 1e3 (cubic -> GB) and Ha = 1 (GB -> Einstein) :
    ...(np.float64(0.0011471184489938008), np.float64(1.3947137801608218))
NUM coupled, near the singularity (Ha = 1e15): p_eff, Hb, kappa eps tau^3, alpha2 E2/(kappa
```

```
    ...eps), alpha3 E3/(kappa eps) : (np.float64(0.7500003270775101),
    ...np.float64(-136.08292174325447), np.float64(1224.7435263138136),
    ...np.float64(2.250006541529109), np.float64(-1.2500065415291277))
matter-dominated cubic regime: p_eff -> 3/4, Hb -> -sqrt(alpha2/(54 alpha3)), kappa eps tau^3
    ...-> 9 alpha2 |h*|, alpha2 E2 : alpha3 E3 : kappa eps -> 9/4 : -5/4 : 1 (to 1e-3) : True
NUM coupled: (p_eff, gas fraction kappa eps/(S1 + S2 + S3 + kappa eps)) at Ha = 3e5 (vacuum
    ...cubic), 30 (Gauss-Bonnet), 1e-2 (Einstein) : ((np.float64(0.9993182542368277),
    ...np.float64(0.0011887681564502635)), (np.float64(1.5596654174306679),
    ...np.float64(1.8450335287549852e-07)), (np.float64(0.5470861129173028),
    ...np.float64(2.8320201318030544e-05)))
NUM shares (Einstein S1, Gauss-Bonnet S2, cubic S3, gas kappa eps) of the 00 equation at Ha =
    ...3e5, 30, 1e-2, 1e15 : ((np.float64(5.730063146835421e-10),
    ...np.float64(0.5011389710000743), np.float64(0.49767226027046907),
    ...np.float64(0.0011887681564502635)), (np.float64(0.00023534899632421242),
    ...np.float64(0.9988000625274751), np.float64(0.0009644039728478138),
    ...np.float64(1.8450335287549852e-07)), (np.float64(0.99945569394049),
    ...np.float64(0.0005159858581290555), np.float64(6.272576290401881e-14),
    ...np.float64(2.8320201318030544e-05)), (np.float64(2.296393963305256e-19),
    ...np.float64(0.5000000000002758), np.float64(0.2777784238527222),
    ...np.float64(0.22222157614700191)))
shares: Einstein > 0.99 at Ha = 1e-2, Gauss-Bonnet > 0.9 at Ha = 30, Gauss-Bonnet and cubic
    ...about 1/2 each at Ha = 3e5 (to 1e-2), gas > 0.1 at Ha = 1e15 : True
coupled: where the gas is a small fraction, the effective exponents are those of the vacuum
    ...solution (1, p2, p1 within 1e-2) : True
NUM coupled, gas-dominated late regime (Ha = 1e-14): tau Ha, tau Hb, pa/eps, pb/eps :
    ...(np.float64(0.00031495953861375523), np.float64(0.3999952443921748),
    ...np.float64(1.873595711619992e-06), np.float64(0.24999859480321623))
without a stabilizing potential the gas-dominated Einstein regime decompactifies: a -> const
    ...(tau Ha -> 0), b ~ tau^(2/5) (tau Hb -> 2/5), pb/eps -> 1/4 : True
NUM gas-fraction growth exponent in the Einstein regime (numerical, between Ha = 1e-2 and
    ...1e-4), and 2 - 3p1 - 5q1 : (1.1698331157808215, 1.1601886205085203)
gas fraction grows like tau^(2 - 3p1 - 5q1) = tau^1.160 in the Einstein regime (to 2e-2) : True
Einstein + fluid power laws a ~ tau^P, b ~ tau^Q: (wa, wb) = (0, 1/4) -> (P, Q) = (0, 2/5)
    ...[decompactification]; (1/3, 0) -> (1/2, 0) [brane radiation, b frozen]; (1/7, 1/7) ->
    ...(1/4, 1/4) [isotropic] : True
matter-dominated cubic regime (any fluid pa = w_a eps, b finite): leading 44 equation p^3 (p -
    ...1)(648 alpha3 h^2 - 12 alpha2) = 0 -> h^2 = alpha2/(54 alpha3); 11/00 -> p = 1/(1 +
    ...w_a); kappa eps tau^3 = (64/3) p^3 alpha2 |h| : True
NUM h* = -sqrt(alpha2/(54 alpha3)), h0 = -sqrt(alpha2/(30 alpha3)) (xi = 1e-6) :
    ...(-136.08276348795434, -182.57418583505537)
NUM Einstein-Kasner pair creation |beta|^2 = F(K) at K = 0.3, 1, 3, 8, 16, and the tail
    ...exponent d ln F/d ln K between 8 and 16 : ((0.13558706527933204, 0.010749346167248614,
    ...0.0001493265403051232, 7.104066464566534e-06, 8.157299133414563e-07),
    ...-3.1224816166636584)
pair creation has a power-law tail F ~ K^(-2(p1 - q1)/(1 - p1)) = K^(-(3 sqrt 2 + 2)/2) =
    ...K^(-3.121) (to 2e-2): it comes from the singularity, so the number of created pairs is
    ...dominated by the earliest times : True
F(1) agrees with the value of section 14.3 of 7+1.nb (|beta|^2 for (|p|, |q|) = (1, 1)) : True
NUM brane radiation fraction kappa eps_W/(3 H^2) at T = M8, H = M8 (M8 = 10 TeV, reduced M_Pl =
    ...2.435e18 GeV, g* = 106.75) : 1.9743638405361938e-28
the Standard Model plasma on the brane (T <= M8) is gravitationally negligible in the bulk
    ...while H ~ M8: fraction < 1e-27 : True
NUM capture: start tau, beta(start) = ln(b/b0), m_r : (np.float64(56.750492072167184),
    ...np.float64(0.4595957216796004), 0.001)
NUM after capture: tau_E from <H_E> over the period after tau = 2.9e5, then beta = A sin(m_r
    ...tau + vartheta)/(m_r (tau - tau_E)) fitted on 1e5 <= tau <= 3e5: A, vartheta, tau_E, max
    ...residual/amplitude : (0.46933049563194473, 2.136009270851999,
    ...np.float64(-1564.3580565850134), np.float64(0.012786248665086845))
NUM after capture: <H_E>(tau - tau_E) over one period at tau = 1e5, 2e5 (H_E = Ha + 2Hb,
    ...four-dimensional matter domination: 2/3) : (np.float64(0.6669420716555031),
    ...np.float64(0.6667362091499506))
NUM after capture: gas fraction kappa eps/(3 H_E^2) at tau = 1e5 and 3e5 (constant: both are
    ...four-dimensional matter) : (np.float64(0.008595792799709226),
    ...np.float64(0.008728032079226138))
capture: the radion oscillates as a four-dimensional massive field in a matter-dominated
```

```
    ...universe: j0 form to 2e-2, <H_E>(tau - tau_E) -> 2/3 to 1e-2, the gas fraction stays
    ...constant to 2e-2 : True
NUM after capture: radion share of the energy, 1 - gas fraction (the cosmological moduli
    ...problem) : 0.9912719679207739
four-dimensional (Einstein-frame) form of the 00 equation: 3 (Ha + 2Hb)^2 = 6 Hb^2 + kappa
    ...eps_total, i.e. 3 H_E^2 M_Pl^2 = (1/2) phi'^2 + rho with phi = sqrt(12) M_Pl ln(b/b0) :
    ...True
NUM radion of mass 10 GeV (M8 = 10 TeV): decay rate into brane fields (b, c, tau, mu pairs and
    ...two gluons) in GeV, lifetime in s, reheating temperature in GeV :
    ...(4.572168576618335e-37, 1439605617925.0205, 1.012614577110346e-09)
through brane fields alone a radion of 10 GeV would decay far too late (T_RH < 1 keV < 4 MeV
    ...needed by nucleosynthesis): the moduli problem of the minimal model : True
NUM number of Kaluza-Klein momenta of the auxiliary torus with |q| < m_r/2 (m_r = 10 GeV):
    ...V_perp (pi^2/2)(m_r/2)^4/(2 pi)^4 : 11733517482065.24
Jacobi's four-square formula r_4(k) = 8 sigma(k) - 32 sigma(k/4) equals the direct count of
    ...lattice vectors for 1 <= k <= 60 : True
NUM radion into bulk dirac16complex pairs (m << m_r/2): Gamma_bulk lattice, continuum (GeV),
    ...lifetime (s), Gamma_bulk/Gamma_brane : (5.9991693514127285e-24,
    ...np.float64(5.9991693514135954e-24), 0.10971718222040179, 13121058969898.768)
the radion decays into the bulk: lattice sum = continuum limit to 1e-6, number of momenta =
    ...N_KK to 1e-5, Gamma_bulk/Gamma_brane > 1e12 : True
NUM M8, M_Pl (reduced), V_perp = M_Pl^2/M8^6 (GeV^-4), l_perp = V_perp^(1/4) (m), lowest
    ...Kaluza-Klein mass 2 pi/l_perp (GeV) : (10000.0, 2.435e+18, 5929225000000.0,
    ...3.079186208511254e-13, 0.004026524867292262)
NUM g*(T), g*s(T) at T = 1 TeV, 10 GeV, 1 GeV, 300 MeV, 100 MeV, 10 MeV, 1 MeV, 100 keV, 1 keV
    ...: ((106.79411834061365, 106.77847184722867), (86.29628742994949, 86.20451456036336),
    ...(76.41793064429221, 75.57705711354103), (64.64582342667666, 64.2172398064234),
    ...(18.315240842717653, 17.94958998859558), (10.837168710423134, 10.835178439973458),
    ...(10.633559462970378, 10.636104471804657), (4.327020814133464, 4.814383599639942),
    ...(3.383002311703709, 3.9374638708790815))
g* limits: 106.75 above the electroweak scale (to 0.1), 10.75 near 3 MeV (to 0.1), and after
    ...e+e- annihilation 2 + (21/4)(4/11)^(4/3)(N_eff/3) = 3.3855 (to 1e-3) : True
NUM radiation era: t(T) in s (entropy conservation with g*(T), g*s(T)) at T = 1 GeV, 100 MeV,
    ...10 MeV, 1 MeV, 100 keV, 10 keV : (np.float64(2.794095659821038e-07),
    ...np.float64(6.123536906169378e-05), np.float64(0.007393002790464219),
    ...np.float64(0.7431868178969353), np.float64(118.83951905287026),
    ...np.float64(13161.662106152166))
NUM radiation era: ratio of the instantaneous approximation t = 1/(2H(T)) to t(T) :
    ...(np.float64(0.9907571426533383), np.float64(0.9234161431082585),
    ...np.float64(0.9943226486914915), np.float64(0.9985476394981425),
    ...np.float64(0.978927979398272), np.float64(0.9996413686006429))
NUM radiation era: relative change of t(10 keV) when the grid is halved (error estimate) :
    ...3.579397018782515e-07
radiation era: the t(T) quadrature is converged (halving the grid changes t(10 keV) by < 1e-5)
    ...: True
radiation era: t(1 MeV) = 0.74 s within 5% (standard value 1/(2H) with g* = 10.75) : True
NUM late universe: Omega_r, Omega_Lambda, a_eq = Omega_r/Omega_m, z_eq, age (Gyr) :
    ...(9.209579083825086e-05, 0.6849079042091618, 0.00029236758996270117, 3419.351756935764,
    ...13.790686840671615)
age of the universe 13.79 Gyr (Planck 2018: 13.787 +- 0.020) : True
NUM composite solution t(a) = t_rm(a) + t_mL(a) - t_m(a): maximal relative error for a =
    ...10^(-8), 10^(-7.75), ..., 10^0.5 : 0.000220157276312416
the composite analytic solution (exact radiation+matter, exact matter+Lambda, minus their
    ...common matter part) is accurate to < 3e-4 at all times : True
NUM radion displacement today (m_r = 10 GeV): from Lambda4 + matter, from a brane tension f_W =
    ...M8^4 : (1.741296272562016e-86, 5.6218701994499e-24)
NUM Gauss-Bonnet and cubic corrections today: alpha2 H0^2 and alpha3 H0^4 with alpha2 = 1/M8^2,
    ...alpha3 = 1e-6/M8^4 : (2.0670464555964908e-92, 4.2726810495940146e-190)
NUM normalcy temperature (order of magnitude) T* = (M8^6/M_Pl)^(1/5) in GeV : 13.264784746562139
NUM far future: H_inf = H0 sqrt(Omega_Lambda) (1/s and GeV), Gibbons-Hawking temperature H/(2
    ...pi) in K : (1.80769618063081e-18, 1.1898472405336613e-42, 2.1975484073507053e-30)
far future: H(t) = H_inf coth(3 H_inf t/2) (exact for matter + Lambda) agrees with the full
    ...solution at a = 3, 10 to 1e-4 : True
far future: the radion obeys beta'' + gamma beta' + m_eff^2 beta = 0 around de Sitter
    ...(constraint used for delta Ha); alpha2 = alpha3 = 0: gamma = 3H, m_eff^2 = m_r^2 at
```

```
    ...beta_eq -> 0 : True
NUM far future (beta_eq -> 0): gamma/(3H) and m_eff^2/m_r^2 with Gauss-Bonnet and cubic terms
    ...(sympy) : ('-(1024*H**8*alpha2**4 - 512*H**6*alpha2**3 - 192*H**6*alpha2*alpha3 -
    ...48*H**4*alpha2**2 + 48*H**4*alpha3 + 16*H**2*alpha2 - 1)/(32*H**4*alpha2**2 -
    ...4*H**2*alpha2 + 1)**2', '1/(32*H**4*alpha2**2 - 4*H**2*alpha2 + 1)')
far future: equilibrium of the radion in de Sitter: 3 m_r^2 beta_eq = 3H^2 - 12 alpha2 H^4 (+
    ...O(beta_eq^2)) : True
NUM far future: 2 pi M/H_inf for M = 1 eV and 1 GeV (|beta|^2 = 1/(exp(2 pi M/H) + 1)) :
    ...(5.280665528426573e+33, 5.280665528426573e+42)
eight-dimensional de Sitter (no stabilizer): 21 H^2 - 420 alpha2 H^4 + 2520 alpha3 H^6 = Lambda
    ...(all three equations) : True
NUM de Sitter radion test (l2 = 1, Lambda = 3e-4, m_r = 0.1, start 0.05 above): beta(3000),
    ...Ha(3000), and the equilibrium (beta_eq, H_eq) of the two algebraic equations :
    ...(np.float64(0.010204078087430525), np.float64(0.010103586699399756),
    ...0.010204078087430532, 0.010103586699399753)
far future: the radion relaxes to its shifted equilibrium (to 1e-6 relative) and Ha to H_eq (to
    ...1e-8) : True
boolean checks: 50   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_dark.py (independent checks of section 24; exit code 0)

```
NUM Omega_c, rho_c/h^2 (GeV/cm^3), s0 (1/cm^3), m Y (GeV) : (0.264156592027754,
    ...1.0533924943202557e-05, 2891.2844014437223, 4.372005025009337e-10)
NUM M_D (GeV) = (2 pi)^(2/3) M8 : 34050.219214767545
NUM Pauli bound for g = 16 internal states, 0.13 keV (2/16)^(1/4) (keV) : 0.07729846247517687
required abundance m Y = 4.37e-10 GeV (rho_c/h^2 = 1.053e-5 GeV/cm^3, s0 = 2891/cm^3, to 1e-3)
    ...: True
M_Pl^2 = R^4 M_D^6 with V = (2 pi R)^4 = M_Pl^2/M8^6 gives M_D = (2 pi)^(2/3) M8 = 34 TeV >
    ...11.2 TeV (strongest collider bound) : True
massive particle: eps V = omega, pa V = k_a^2/(3 a^2 omega), pb V = k_b^2/(4 b^2 omega) : True
lightest mode (k_b = 0): pa/eps = k_a^2/(3 (m^2 a^2 + k_a^2)), pb = 0, eps ~ a^-3 at fixed b
    ...for k_a = 0 : True
NUM radion displacement by dark matter today H0^2 Omega_c/(2 m_r^2) (m_r = 10 GeV) :
    ...2.730119736367085e-87
dark matter leaves G4 unchanged: displacement < 1e-80 : True
regime III: Omega_m a^-3/Omega_L = csch^2(3 H t/2) for a = (Omega_m/Omega_L)^(1/3) sinh^(2/3)(3
    ...H t/2), -> 4 exp(-3 H t) : True
NUM from tools/ref_coupled.py: gas share at tau = 1e5, 3e5; Gamma_brane, Gamma_bulk (GeV); tau
    ...Hb at Ha = 1e-14 : (0.008595792799709226, 0.008728032079226138, 4.572168576618335e-37,
    ...5.9991693514127285e-24, 0.3999952443921748)
zero-mode couplings: lambda_P/V = c_P M8/M_Pl^2, y/sqrt(V) = c_y M8/M_Pl, G_P/V = c_V/M_Pl^2,
    ...g_r/V = c_r/M_Pl^2 : True
NUM lambda_eff, y_eff, G_r per unit coupling (GeV^-1, 1, GeV^-2) : (1.68656105983497e-33,
    ...4.106776180698152e-15, 1.68656105983497e-37)
NUM sigma_SI (cm^2) at c_P = 1, m = 40 GeV : 9.556076962877839e-104
NUM sigma/m (cm^2/g) at c_r = 1, m = 1 GeV; c_r allowed by the Bullet Cluster at m = 1 GeV :
    ...(4.9442250036435816e-79, 1.1898710736040751e+39)
NUM mu_P (GeV) at c_y = 1; lifetimes chi -> nu gamma, chi -> 3 nu (s) at c_y = 1, m = 1 GeV and
    ...10 keV : (7.150044241252333e-13, 3593359688030266.0, 28170199696474.52,
    ...3.5933596880302647e+30, 2.8170199696474517e+28)
NUM largest m (GeV) with lifetime chi -> 3 nu > age for c_y = 1 : 0.04015133785290378
dark and collisionless: sigma_SI < 1e-50 x 2.2e-48 cm^2, sigma/m < 1e-70 cm^2/g : True
stability: lifetime = age at m = 40 MeV (c_y = 1); lifetime ~ 1/(c_y^2 m^3); Gamma(nu
    ...gamma)/Gamma(3 nu) = 27 alpha/(8 pi) : True
NUM gas share after capture f, overabundance O at T_RH = 1 GeV and 4 MeV :
    ...(0.008728032079226138, 15104422.311749911, 60417.689246999646)
NUM fraction of radion decays into the bulk; brane share Gamma_brane/Gamma_bulk :
    ...(0.9999999999999238, 7.621336069703795e-14)
primordial gas overabundance 1.5e7 T_RH/GeV (6e4 at 4 MeV) : True
radion decays: more than 1 - 1e-12 into the bulk : True
NUM T* (GeV), S(m_h/T*) : (13.264784746562139, 0.0078461320088985)
Boltzmann factor: int_0^oo t^3 K_1(t) dt = 3 pi/2 (sympy), S(0) = 1 (mpmath), S(m_h/T*) < 1e-2,
```

```
    ...13 < T* < 14 GeV : True
NUM zero mode: Yukawa-portal Y at c_y = 1 (N = 1, S = 1), Higgs-portal Y at c_P = 1 (S = 1) :
    ...(9.726257996829745e-18, 1.9889471111127847e-49)
NUM m* (GeV) for N = 1 and N = 12 : (1.2000037341243723e-06, 4.156934873551575e-06)
NUM c_y needed and allowed at m = 10 keV, 10 MeV : (2120154.126594499, 254419.2868832596,
    ...67045.16030643586, 8.045444272269018)
zero mode, Yukawa portal: abundance and stability compatible only below m* < 5.7 keV (N = 1,
    ...12) : True
NUM lattice sums for R = 100: Sum r_4(k)(1 - k/R^2)^2 / (pi^2 R^4/12) - 1, Sum r_4(k) sqrt(k)/R
    ...(1 - k/R^2)^2 / (16 pi^2 R^4/315) - 1 : (5.981171113944583e-10, 9.233644959749654e-10)
Kaluza-Klein tower: int 2 pi^2 u^3 (1-u^2)^2 = pi^2/12, int 2 pi^2 u^4 (1-u^2)^2 = 16 pi^2/315
    ...(sympy); lattice sums agree to 1e-8 (R = 100) : True
NUM Kaluza-Klein tower below m_h: number, mass sum (GeV), Y_tower and mass per entropy (GeV) at
    ...c_y = 1 (N = 1, S = 1); c_y bound at T* : (7.688015893943899e+17, 5.866907976666065e+19,
    ...7.477562606822603, 570.6306062471253, 9.881775656983552e-06)
NUM tau(chi_n -> 3 nu) (s) at mu_n = 10 GeV, c_y = 1; largest m (GeV) with tau(chi -> 3 nu) >
    ...age for c_y = 1e-5 : (28170199696.47452, 86.50343512177517)
tower through the Yukawa portal: 7.7e17 states below m_h, mean mass (192/315) m_h, c_y bound <
    ...2e-5 at T* : True
NUM Higgs portal pairs: I_hp, I_hp^E, number of pairs, mass per entropy (GeV) at c_P = 1 (N =
    ...4, S = 1); c_P needed for S = 1 and at T* : (9.538467456842521e-05,
    ...7.380786601565824e-05, 3.40448189100676e+38, 2.5028869407639637e-12, 13.216598897717843,
    ...149.20792978255014)
NUM Higgs portal pairs at c_P = 1, T_RH = T*: fraction of the dark matter :
    ...4.491756351661614e-05
Higgs portal: zero mode Y < 1e-40; lambda(1, u1^2, u2^2) = (1 - (u1+u2)^2)(1 - (u1-u2)^2);
    ...pairs need 10 < c_P < 20 (S = 1), 100 < c_P < 200 (T*) : True
NUM vector portal: T_V (GeV), (T*/T_V)^11 : (492.19416145801307, 5.4475502039269596e-18)
vector portal: 480 < T_V < 500 GeV, (T*/T_V)^11 < 1e-17 : True
NUM T_RH (GeV) = 5, 8, 10, T*: S(m_h/T_RH), c_y bound of the tower (N = 1), c_P needed by the
    ...Higgs-portal pairs (N = 4) : [(5.0, 1.248895314095822e-08, 0.007832488705773653,
    ...118265.1241437344), (8.0, 4.9440196437621006e-05, 0.00012448662751176695,
    ...1879.6613707290032), (10.0, 0.0006775813090297954, 3.3626532542268916e-05,
    ...507.73722057246755), (13.264784746562139, 0.0078461320088985, 9.881775656983552e-06,
    ...149.20792978255014)]
NUM w_min of Kohn-Sham matter and its kF (units m = 1); saturation kF0 : (-0.05801598575399315,
    ...0.67, 0.7566361507014929)
Kohn-Sham matter: p > 0 at low density, p < 0 just below kF0, p = 0 at kF0, a spinodal region,
    ...w >= -0.06 : True
Epstein sum: Sum over Z^4\{0} of |n|^-8 = 8 (1 - 4^-3) zeta(4) zeta(3) = shell sum to k = 20000
    ...(tail < 3e-8) : True
NUM rho_Lambda (GeV^4) : 2.518265996152132e-47
NUM Casimir energy / rho_Lambda: dirac16complex (m = 0), graviton (20 dof) :
    ...(6.762461109529047e+34, -8.45307638691131e+34)
NUM m l_perp and m (GeV) at which the dirac16complex Casimir energy equals rho_Lambda :
    ...(92.22624853874979, 0.05910239233816993)
NUM kF, p, w at kF = 0.1, 0.3, 0.5, 0.6, 0.67, 0.7, 0.75 (units m = 1) : [(0.1,
    ...1.0815057495004936e-14, 0.0011018832731922797), (0.3, 1.8769661856714918e-10,
    ...0.00849359466975192), (0.5, -4.3081067785473656e-10, -0.0005296332340018398), (0.6,
    ...-1.0277578600831483e-07, -0.035970422554837775), (0.67, -3.4548242506780997e-07,
    ...-0.05801598575399315), (0.7, -4.109652096104839e-07, -0.05166870169668898), (0.75,
    ...-1.0713521089577265e-07, -0.008446078650216108)]
NUM dirac16complex Casimir energy / rho_Lambda at m l_perp = 1, 10, 30, 60 :
    ...[6.05689541888988e+34, 4.1996195140622416e+32, 2.4693584837075517e+25,
    ...23006760008666.074]
Casimir: massless > 1e34 rho_Lambda; massive sum -> massless (m l = 1e-3, to 1e-4); =
    ...rho_Lambda at m l = 92.2 : True
boolean checks: 20   failed: 0
VERDICT: ALL CHECKS PASSED
```

4.4 Hashes of the files used to build 7+1.nb and its documentation

file	SHA-256
7+1.nb	39f64ee81d35c0c9fb94188fd7f8c8e75c96ea198fbf72a5ca6b3c0027426018
4+4.nb	740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5
tools/cayley_dickson_formulas.py	bdde8c4c8199b09ef36ae4e15d5d53bc62744fc85b4c5ba52eab5516f77b5d5f
tools/cayley_dickson_match.py	73f2cc584bbe1a3620c50890ffe5fb299857b04a14bc34fed2f9eaefd2e36c4e
tools/classify_clifford.py	42291276f46b046183fdbb7b62be7c5cedfd695fe9da96ac256afa23580930a9
tools/doc_common.py	1e37b140f8334aa813418f4a79b7d0a8e67740b2c4a3fe4d5f4e34abde690d3e
tools/doc_coupled.py	fe0d9cdd4c157aaf3ab0bd386fb1a47350c3d0d9eb8e5de24411cf12b82d22c2
tools/doc_coupled_provenance.py	9a6ceba936407689ac30ab70070ea397afc352ba42cef4b77d3d045aaf12e497
tools/doc_coupled_symbols.py	e0557a5249c59e43087326d2d5d9f77937a38bf52845374daca4e00485981f82
tools/doc_dark.py	9eea2c461cd1a7bf353c0e2aec43a22be2072acc44e54f4e6c84593c5e86f653
tools/doc_dark_provenance.py	89956664f5d6992c054917d452c3e973ae8c1f36001fe83a5fb7e8c6f8f2ea14
tools/doc_dark_symbols.py	ee662066ae88d927c4d85991256e845b67b3e08ea5d6ba9406bb1d421c0d3481
tools/doc_dirac16.py	3f81df5fe2237843edd29116e5676679527b80e946623890e6289a1487f754f1
tools/doc_dirac16_common.py	9324b4fae59e51019125ebe09157f3c8c9489560e1383a697fe6c22a3a93e208
tools/doc_dirac16_provenance.py	daf12ff5757d0c4ca8857840d027907f21429e7340b2f02c3b50fbf8edb2d663
tools/doc_dirac16_symbols.py	69224cd370206759e31abd225fc556630455d6d99226db0bc46068eae9b27db0
tools/doc_gravity.py	184c7aa38742a2057449a98037a8521842fc81072613611c228c63c17083af0c
tools/doc_gravity_provenance.py	3845317cdef374106b60524416eda8e3a81fd5d2a8e35c327c3f1ae32925b0f4
tools/doc_gravity_symbols.py	fcff3b16ea23fa62fe205e66310552f7d3e119c619c150379e67bd17150b5d22
tools/doc_ks_gravity.py	9aa73527f90318ee3444fcf97af41d2255a5d8fb1697d9fa0128ca671e94e51f
tools/doc_ks_gravity_provenance.py	112b51528ab67acd8ed110b57c347352c602c984eae1408d7dd36e41c74fc0ea
tools/doc_ks_gravity_symbols.py	2acdf7f901b9cb7c3047b6de6895d32385be37d16789df9d9124f976c6d5a515
tools/doc_provenance.py	381ce86378ab4f8903fe9898373df8f850d761a17ed183e1a1f8088e2b7d1c15
tools/doc_readme.py	27eb16b51bf5d119664edfc46400a200877a346e39188212e8267f0cbbfec2ba
tools/doc_readme_stage3.py	9d52a48e1d9d20a19b0874f144cf5bfe5204a81a3c0f448ef44f510f48a1cf91
tools/doc_readme_stage4.py	86bc4089deb0eaa1f7e13d9fdfcb5924541d9573bdb2a3ab577ab52584b8035f
tools/doc_readme_stage5.py	a9186a213bbe4c2c39e81c899d0216bb829dc37217c3f43b5efaee1928a1b564
tools/doc_sm.py	d7245d8c5f5dfd2687e987d43d9d544d0a1a46538901066e244fe0261bd427c3
tools/doc_sm_provenance.py	742dec7e9ea7214a8921777a260838830c18204223b5cbb37b57962ab522d01c
tools/doc_sm_symbols.py	a019dcbf5e68f3f00f41613f5c592b84eafa8b48d4023208516fd5d93b8893f8
tools/doc_stage3_common.py	3491a5c8ae7897b402871f5015b9f48637d0d9c207a337bc4bdadc0f9405fa76
tools/doc_symbols.py	e64a2e24d850890853db104198f3aef00bd5583718d19e1e64a8e424871ec038
tools/docgen.py	5db6f40cb1eb10e48da5b69c3893829737c3d8d2e90edcb28311a980c1aa5fea
tools/eval_cells_log.wl	833b6bd854cab253e5bb935c28f28ab50b286f409d130f433fa5dfb9b9236bbe
tools/extra71.py	9935ef99cd9300f2a7d4f06d36cf7e173912ebae61b6feff448ebe0588853168
tools/git_push.wl	ac05a6d0256834b48cc9fa943fa247bccc574f723e68f053d00bfba9aca2e21a
tools/lovelock71.py	d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af95a1
tools/make_7p1_nb.py	bd6ef1b2311770b6e0f361d640e727dec1beb617c6f99a0b788774d19aefe3f9
tools/make_docs.py	c7061d397cdbdfc1f4a20aa0604d02f5e500adf15a2301db5aa8b474e683eb31
tools/nb_part_coupled.py	e630521d64132e71afc861114f699eb06454557cf14a57ad8ec1eef96247aae0
tools/nb_part_dark.py	10ad8fb8ae25fa07bbee758c653995acd4e2c80733059a8b3648bd1593e46db0
tools/nb_part_gravity.py	c4caf2ad5c932e8f603341edade5f4b98fa784feb05906c051eab7c884cd68f8
tools/nb_part_ks_gravity.py	a656c0ad040b4576b75b17e029545109e531c8e656ad99bfdb5cba946e967772
tools/nb_part_sm.py	43e92beed57453a4b57931de6d603ac6c218efba6e4005133b69b8054a975c5b
tools/nbparse.py	ad2255a8864be7447a55023840b327f789cb8c9d868d42261b19261653847423
tools/nbwrite.py	dcbe8ce5b3203c89a1d17fbe946e1b6b7bb7f8895afa944743d651cfefadeffb
tools/octalg.py	36b50120b69a963e252e07d60ec007755220822b4ad339716ea7aed93d58eb24
tools/ref44.py	ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064faf5
tools/ref71.py	1a7a06eb0add3f459611776dc9c474a5bb959d947a9db949db4612b1ab9434e0
tools/ref_coupled.py	695dbd9375753653472d9c37502f0067c4a64fe01a64ba8c05e8b2c5da85aa58
tools/ref_dark.py	1600e6b57ab0383b4f2b664e2796d3864c9afa77a9ed59f007f66e1139453ff5
tools/ref_dirac16.py	32e8c50de1c480bc4ef50d254d04a9aaa0e56b486198edf0355905c0dda25674
tools/ref_gravity.py	f07cf8adaff585af89c87d19532f1aaa7db606ad88ae90483b2d1f41341edf36
tools/ref_ks_gravity.py	b6488ef24b1c16aacd8ff6cf6a5a1ba3f316e082449ad97480022d1a2638327f
tools/ref_sm.py	f3f80806893f42bbf166e1f1de896ffd611dbfe847685da8e3ee367e9cde6aba
tools/run_eval.wl	85ba54f18a8362a69c991e66a207230dd001ba18e52639fc1edf7cac761aac30
tools/verify_7p1_nb.py	cb36356d5f8d22d1f0b74ce015e8add1566eff5c353c4a082439265a8f328d01
tools/watch_requests.wl	d9ceedb0b45ac3cf9b4dd8519ac164953f2c80e5188b1557fe20ef5107ae352d

5. Process and environment

- Workspace: `D:\Developer\github\7+1` on Windows; Python and LaTeX ran in a Linux shell with that folder mounted. Intermediate files go to `build/`, which the repository's `.gitignore` excludes.
- The notebook is generated by `tools/make_7p1_nb.py` (cells written with `tools/nbwrite.py`), evaluated in place by `tools/run_eval.wl` in Wolfram 15.0.1, and cross-checked by `tools/verify_7p1_nb.py`.
- Python 3.13.17 with sympy 1.14.0 and numpy 2.5.3 for the documentation run of 2026-10-27 (exact rational arithmetic for every identity; the first stage was checked with Python 3.10.12, sympy 1.14.0 and numpy 2.2.6).
- The documentation is generated by `tools/make_docs.py` from the modules `docgen.py`, `doc_common.py`, `doc_dirac16_common.py`, `doc_readme.py`, `doc_symbols.py` and `doc_provenance.py` of the folder tools/ (and the number formatting of `doc_dirac16.py`): one source renders Markdown and LaTeX, and every matrix, table and check list is computed or parsed, never typed. The LaTeX is compiled with XeLaTeX (TeX Live 2022, `latexmk -xelatex`, fonts Latin Modern and DejaVu Sans Mono).
- History: the first algebraic experiments were made in a temporary cloud sandbox and then moved into `tools/` of the workspace, where everything was re-run. A Wolfram 15.0 kernel window was opened once by mistake before Wolfram 15.0.1 was available; no result comes from it. Only Wolfram 15.0.1 evaluated 7+1.nb.
- Corrections made during the review: an earlier draft called the real on-shell amplitudes of notebook section 5.4 "Majorana spinors"; that label was removed, and section 5.5 was added to show which real spinor FIELDS exist (massless with 16 components, massive only with the 32×32 matrices).
- The evaluated notebook was rendered section by section in the Wolfram 15.0.1 front end (`Rasterize`) and inspected. Characters that the front end's text font shows badly are written differently in text cells: the double-struck O and the arrow of $\vec{x}$ as small inline formulas, the script ℓ in the font Cambria Math, and $\not{p}$, $\not{k}$ as `pslash`, `kslash`.
- Second stage (sections 7–10 of 7+1.nb): all Wolfram Language work ran in the front end `C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`. Jobs were requested without keyboard input through `tools/watch_requests.wl` (loaded once in that front end; every 10 seconds it reads `build/request.txt` and evaluates `tools/run_eval.wl`, `tools/git_push.wl` or a development file); new cells were developed with `tools/eval_cells_log.wl`, which evaluates a generated notebook expression by expression, logs messages and checks, and stops at the first failure. `tools/run_eval.wl` evaluates with a time limit of one hour. Pushes run the local Git for Windows from the front end (`tools/git_push.wl`, no login prompt) to `https://github.com/once-ere/7plus1.git`, the new name of the repository.
- Second stage, Python: `tools/ref_dirac16.py` (sympy, mpmath, numpy) is the independent reference; `tools/verify_7p1_nb.py` was extended to sections 7–9 and to the comparison of sections 0–6 with the backup of the first stage.
- Third stage (sections 11–19 and the summary of all checks, then section 20): the new sections were first evaluated cell by cell in a fresh parallel kernel of the same front end (a development file runs the input cells of sections 0–6, 7.1, 7.10 and 11–19 and writes a progress line around every cell and the list of failed checks); this exposed a name clash (a helper named like the zero vector `zero8` of section 1, renamed `asMat8`) and a formulation of the exponential solutions that did not finish (xAct with symbolic exponentials; replaced by the substitution into the xAct components of the cosmological metric). The complete notebook was then evaluated by `tools/run_eval.wl` as before.
- Third stage, Python: `tools/lovelock71.py` (curvature and pair-set Lovelock engine), `tools/ref_gravity.py`, `tools/ref_ks_gravity.py`, `tools/ref_sm.py` (sympy, numpy, scipy, mpmath) are the independent references; the documentation of the third stage is generated by `tools/doc_stage3_common.py`, `tools/doc_gravity*.py`, `tools/doc_ks_gravity*.py`, `tools/doc_sm*.py` and `tools/doc_readme_stage3.py`.
- Fourth stage (sections 20–23, appended on 27 October 2026; the summary of all checks became section 24): the new sections were developed in a development notebook — the definitions of sections 1, 11, 12 and 14 that they use, followed by sections 20–23 — evaluated cell by cell with `tools/eval_cells_log.wl`

in a fresh kernel. This exposed a name clash (the unknowns of `NDSolveValue` were first named like the down-type Yukawa matrix `yD` of section 17; they, and every symbol that could clash with an abstract index of xAct, were renamed, e.g. `rIyV`, `rIyC`, `rIIyK`, `rIIIyD`) and machine-precision data passed to functions with a `WorkingPrecision` (removed). The complete notebook was then evaluated by `tools/run_eval.wl`.

- Fourth stage, Python: `tools/ref_coupled.py` (numpy, scipy, sympy, mpmath) is the independent reference; the documentation of the fourth stage is generated by `tools/doc_coupled.py`, `tools/doc_coupled_symbols.py`, `tools/doc_coupled_provenance.py` and `tools/doc_readme_stage4.py`.
- Section 24 (dirac16complex as dark matter and dark energy, appended on 27 October 2026; the summary of all checks became section 25): the data were taken from the papers of the collaborations (their abstracts and tables, read on 27 October 2026); the new section was evaluated with the complete notebook by `tools/run_eval.wl`. A first evaluation produced one message (a line-search warning of `FindRoot` for the largest stable mass, whose closed form $(K/(c_y^2 t_0))^{1/3}$ is now used); the second and later evaluations have no message. `tools/ref_dark.py` is the independent reference; the documents are generated by `tools/doc_dark.py`, `tools/doc_dark_symbols.py`, `tools/doc_dark_provenance.py` and `tools/doc_readme_stage5.py`.

6. The second stage: sections 7–10 of 7+1.nb

This section records the provenance of the Dirac equation and the Kohn–Sham theories of the field dirac16complex (sections 7–10 of 7+1.nb). Notation: $\Gamma_A = \Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A$, $\Gamma^0 = \Gamma_0$, $\chi = \chi^{(1,7)} = i\Gamma_8^{(1,7)}$, Ψ the field dirac16complex, $\bar\Psi = \Psi^\dagger\Gamma^0$, $\vec{x} = (x_1, \ldots, x_7)$, $\vec{p} = (p^1, \ldots, p^7)$.

6.1 The task

1. Back up 7+1.nb and append to it; the first part stays unchanged.
2. A checked Dirac equation on the real 7+1 spacetime for a 16-component quantized fermion field named dirac16complex whose components are complex-valued Grassmann operators, with eight PURELY REAL anticommuting 16×16 Dirac matrices taken from 7+1.nb.
3. Chiral projections writing the 16-component equation as two 8-component equations, coupled when the mass is not zero.
4. A 16-component Kohn–Sham theory of the field coupled to itself, and a chiral version using the chiral projections.
5. Complete, checked documentation (every document with a provenance file and a symbol file, Markdown and LaTeX, compiled to PDF, each self-contained); every completed stage pushed to github.com/once-ere/7plus1 (the repository was renamed from 7+1 to 7plus1); only the Wolfram front end 15.0.1 is used.

6.2 The first part of the notebook

The evaluated notebook of the first stage was copied, in the workspace, to the file below (the folder backups/ is excluded from the repository by its .gitignore). `tools/verify_7p1_nb.py` compares every input, text and section cell of sections 0–6 of the new 7+1.nb with that copy: every cell is present, and each is identical except the cells changed or added by the explanation of the signature of 27 October 2026 (section 1.4).

```
backups/7+1_2026-10-26_before_dirac16complex.nb
306963 bytes
SHA-256 f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
```

6.3 Derivations and decisions

Dirac matrices. The real generators $\Gamma_A^{(1,7)}$ of section 2.5 are used unchanged: $\Gamma_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$, $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}^{(1,7)} 1_{16}$. The opposite sign convention $\eta^{(7,1)}$ has no real 16×16 realization (section 2.7), so this is the only real choice. **The factor i.** With real Γ_A, $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \partial_0^2 - \partial_1^2 - \cdots - \partial_7^2$, hence

$$\big(i\Gamma^A\partial_A + m\big)\big(i\Gamma^A\partial_A - m\big) = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big)1_{16},$$
$$\Gamma^0\big(i\Gamma^A\partial_A - m\big) = i\partial_0 - H_{\rm D}, \qquad H_{\rm D} = -i\Gamma^0\Gamma^i\partial_i + m\Gamma^0.$$

Real amplitudes. For $e^{-i(Ex_0 - \vec{p}\cdot\vec{x})}$, with $P_\tau = p^i\tau_i^O$ ($P_\tau^{\mathsf T} = -P_\tau$, $P_\tau^2 = -\vec{p}^{\,2}1_8$):

$$H_{\rm D}(\vec{p}) = \begin{pmatrix} 0 & m1_8 + P_\tau \\ m1_8 - P_\tau & 0 \end{pmatrix}, \qquad (m1_8 + P_\tau)(m1_8 - P_\tau) = (m^2 + \vec{p}^{\,2})1_8 = E^2 1_8,$$
$$H_{\rm D}(\vec{p}) \begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s \end{pmatrix} = \begin{pmatrix} E^2 e_s \\ E(m1_8 - P_\tau)e_s \end{pmatrix} = E \begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s \end{pmatrix},$$
$$w_s = \frac{1}{\sqrt{E}} \begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s \end{pmatrix}, \qquad w_s^{\mathsf T} w_r = 2E\,\delta_{sr}.$$

Chirality. The real matrices anticommuting with every Γ_A are $c\,\Gamma_8^{(1,7)}$ (section 2.9), with square $-c^2 1_{16}$; the chirality is therefore the complex $\chi = i\Gamma_8^{(1,7)} = \left(\begin{smallmatrix} 0 & -i1_8 \\ i1_8 & 0 \end{smallmatrix}\right)$. With $E_\pm$ the 16×8 matrices $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1_8 \\ -i1_8 \end{smallmatrix}\right)$ and $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} -i1_8 \\ 1_8 \end{smallmatrix}\right)$ ($\chi E_- = -E_-$, $\chi E_+ = E_+$, $E_-E_-^\dagger = P_L^{(1,7)}$, $E_+E_+^\dagger = P_R^{(1,7)}$), $\Psi_L = E_-^\dagger\Psi$, $\Psi_R = E_+^\dagger\Psi$ and

$$E_-^\dagger\Gamma^0 E_+ = E_+^\dagger\Gamma^0 E_- = 1_8, \qquad E_\mp^\dagger\Gamma^0 E_\mp = 0, \qquad E_\mp^\dagger\Gamma^0\Gamma^i E_\mp = \mp i\tau_i^O, \qquad E_-^\dagger\Gamma^0\Gamma^i E_+ = 0,$$
$$E_-^\dagger\Gamma^0\big(i\Gamma^A\partial_A - m\big)\Psi = \big(i\partial_0 + \tau_i^O\partial_i\big)\Psi_L - m\Psi_R, \qquad E_+^\dagger\Gamma^0\big(i\Gamma^A\partial_A - m\big)\Psi = \big(i\partial_0 - \tau_i^O\partial_i\big)\Psi_R - m\Psi_L.$$

Self-coupling. The four Lorentz-invariant contact interactions without derivatives used in relativistic point-coupling models (scalar, pseudoscalar, vector, axial vector) with real couplings, and real sources that define the densities:

$$\mathcal{L} = \bar\Psi\big(i\Gamma^A\partial_A - m\big)\Psi - \bar\Psi\big(\mathsf{s} + i\chi\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\mathsf{a}_A)\big)\Psi$$
$$- \frac{g_S}{2}(\bar\Psi\Psi)^2 - \frac{g_P}{2}(\bar\Psi i\chi\Psi)^2 - \frac{g_V}{2}(\bar\Psi\Gamma^A\Psi)(\bar\Psi\Gamma_A\Psi) - \frac{g_A}{2}(\bar\Psi\Gamma^A\chi\Psi)(\bar\Psi\Gamma_A\chi\Psi).$$

The tensor and higher-rank channels are not included. With $\hbar = c = 1$ every g has mass dimension -6 (an effective theory: no-sea approximation, or a momentum cutoff for the sea). **Hohenberg–Kohn.** For $H[w] = H_0 + H_{\rm int} + \int w\cdot n$ with $w\cdot n = \mathsf{s}\rho_S + \mathsf{p}\rho_P + \mathsf{v}_A j^A + \mathsf{a}_A j_5^A$, two sources $w \neq w'$ with different ground states and the same densities would give $E < E' + \int(w - w')\cdot n$ and $E' < E + \int(w' - w)\cdot n$, so $E + E' < E + E'$; hence $E[w] = \min_n(F[n] + \int w\cdot n)$. **Kohn–Sham.** Varying $T_s + \int w\cdot n + E_H + E_{xc}$ with respect to $\Phi_k^\dagger$ under $\int\Phi_k^\dagger\Phi_l = \delta_{kl}$ (multipliers λ_k), with $\delta\rho_S/\delta\Phi^\dagger = \Gamma^0\Phi$, $\delta\rho_P/\delta\Phi^\dagger = \Gamma^0 i\chi\Phi$, $\delta j^A/\delta\Phi^\dagger = \Gamma^0\Gamma^A\Phi$, $\delta j_5^A/\delta\Phi^\dagger = \Gamma^0\Gamma^A\chi\Phi$, gives $H_{\rm KS}\Phi_k = \lambda_k\Phi_k$ with

$$H_{\rm KS} = \Gamma^0\Gamma^i(-i\partial_i + \mathcal{V}_i) + \Gamma^0(m + \mathcal{S}) + \mathcal{V}_0 + \Gamma^0 i\chi\mathcal{P} + \chi\mathcal{A}_0 + \Gamma^0\Gamma^i\chi\mathcal{A}_i,$$
$$\mathcal{S} = \mathsf{s} + g_S\rho_S + \frac{\delta E_{xc}}{\delta\rho_S}, \qquad \mathcal{P} = \mathsf{p} + g_P\rho_P + \frac{\delta E_{xc}}{\delta\rho_P},$$
$$\mathcal{V}_A = \mathsf{v}_A + g_V j_A + \frac{\delta E_{xc}}{\delta j^A}, \qquad \mathcal{A}_A = \mathsf{a}_A + g_A j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A},$$

where $E_H = \int[\frac{g_S}{2}\rho_S^2 + \frac{g_P}{2}\rho_P^2 + \frac{g_V}{2}j^A j_A + \frac{g_A}{2}j_5^A j_{5A}]$ and $T_s = \sum_k\int\Phi_k^\dagger H_{\rm D}\Phi_k$. **Decision (no sea):** the orbitals are the occupied positive-energy states. **Exchange.** Wick's theorem for a Slater determinant, $\langle c_a^\dagger c_c^\dagger c_d c_b\rangle = \varrho_{ba}\varrho_{dc} - \varrho_{da}\varrho_{bc}$ with $\varrho_{xy} = \sum_k \Phi_{k,x}\Phi_{k,y}^*$ and $\mathcal{R} = \varrho\,\Gamma^0$, gives $\langle{:}\ (\bar\Psi O_1\Psi)(\bar\Psi O_2\Psi)\ {:}\rangle = \mathrm{tr}(O_1\mathcal{R})\mathrm{tr}(O_2\mathcal{R}) - \mathrm{tr}(O_1\mathcal{R}O_2\mathcal{R})$, hence the exact exchange energy

$$E_x = -\frac{g_S}{2}\mathrm{tr}(\mathcal{R}\mathcal{R}) - \frac{g_P}{2}\mathrm{tr}(i\chi\mathcal{R}\,i\chi\mathcal{R}) - \frac{g_V}{2}\mathrm{tr}(\Gamma^A\mathcal{R}\Gamma_A\mathcal{R}) - \frac{g_A}{2}\mathrm{tr}(\Gamma^A\chi\mathcal{R}\Gamma_A\chi\mathcal{R}).$$

Chiral Kohn–Sham. With $\Phi_{kL} = E_-^\dagger\Phi_k$, $\Phi_{kR} = E_+^\dagger\Phi_k$ and the blocks above, $\kappa = \sum_k\Phi_{kL}^\dagger\Phi_{kR}$, $\rho_S = \kappa + \kappa^*$, $\rho_P = i(\kappa - \kappa^*)$, $\mathcal{M} = m + \mathcal{S} + i\mathcal{P}$, $\mathcal{V}_A^{L,R} = \mathcal{V}_A \mp \mathcal{A}_A$:

$$\big[\mathcal{V}_0^L - \tau_i^O\partial_i - i\tau_i^O\mathcal{V}_i^L\big]\Phi_L + \mathcal{M}\Phi_R = \lambda\Phi_L,$$

$$\mathcal{M}^{*}\Phi_L + \big[\mathcal{V}_0^R + \tau_i^O\partial_i + i\tau_i^O\mathcal{V}_i^R\big]\Phi_R = \lambda\Phi_R,$$
$$e^{i\vartheta\chi} = \cos\vartheta\,1_{16} - \sin\vartheta\,\Gamma_8^{(1,7)}: \qquad \kappa \to e^{2i\vartheta}\kappa,$$
$$\Delta E_H = \frac{g_S - g_P}{2}\big[(e^{4i\vartheta}-1)\kappa^2 + (e^{-4i\vartheta}-1)\kappa^{*2}\big].$$

Uniform matter. Units $m = 1$; all 8 spin states filled to k_F; $|S^6| = 16\pi^3/15$ is the area of the unit sphere in $\mathbb{R}^7$ and $c_7 = 8|S^6|/(2\pi)^7 = 1/(15\pi^4)$. With the energy density $\mathcal{E}$, its kinetic part $\mathcal{E}_{\rm kin}$, ρ_S and the density ρ:

$$\rho = \frac{k_F^7}{105\pi^4}, \qquad \rho_S = c_7\int_0^{k_F}\frac{k^6 m^*\,dk}{\sqrt{k^2+m^{*2}}}, \qquad \mathcal{E}_{\rm kin} = c_7\int_0^{k_F} k^6\sqrt{k^2+m^{*2}}\,dk,$$
$$m^* = m + g_S\rho_S(m^*), \qquad \mathcal{E} = \mathcal{E}_{\rm kin} - \frac{(m^*-m)^2}{2g_S} + \frac{g_V}{2}\rho^2,$$
$$E/A = \frac{\mathcal{E}}{\rho}, \qquad E_F = g_V\rho + \sqrt{k_F^2+m^{*2}}.$$

Because $\partial\mathcal{E}/\partial m^* = 0$ is the gap equation, $d\mathcal{E}/d\rho = E_F$ at every density, and $\frac{d}{dk_F}\frac{\mathcal{E}}{\rho} = \frac{c_7 k_F^6}{\rho}(E_F - \frac{\mathcal{E}}{\rho})$ vanishes at saturation: $E/A = E_F$ (Hugenholtz–Van Hove). **Decisions:** $g_S = -10^5$, $g_V = 10^4$, $g_P = g_A = 0$ (binding and saturation); root finding by Brent's method with 50-digit (gap equation) and 40-digit (saturation) working precision. **Dirac sea.** With a cutoff $\Lambda_{\rm c}$, $g_S = g_P = g$ and a uniform $\mathcal{M}$:

$$J(M) = \frac{1}{120\pi^4}\int_0^{\Lambda_{\rm c}}\frac{q^6 dq}{\sqrt{q^2+M^2}}, \qquad K(M) = \frac{1}{120\pi^4}\int_0^{\Lambda_{\rm c}} q^6\sqrt{q^2+M^2}\,dq,$$
$$\kappa = -4\mathcal{M}^* J(|\mathcal{M}|), \qquad \mathcal{M} = m - 8gJ(|\mathcal{M}|)\mathcal{M}, \qquad g_c = -\frac{1}{8J(0)} = -\frac{90\pi^4}{\Lambda_{\rm c}^6},$$
$$\mathcal{E}_{\rm vac} = -8K + 8J\,\mathrm{Re}[(\mathcal{M}-m)\mathcal{M}^*] + 32g|\mathcal{M}|^2J^2.$$

Decisions: $\Lambda_{\rm c} = 1$, $g = \frac{6}{5}g_c$, and $m = \frac{1}{100}$ for the explicit-breaking example.

6.4 Literature used in the second stage

- P. A. M. Dirac, Proc. R. Soc. Lond. A 117 (1928) 610 — $(i\gamma^\mu\partial_\mu - m)\psi = 0$, $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$.
- P. Hohenberg and W. Kohn, Phys. Rev. 136 (1964) B864 — $E[v] = \min_n(F[n] + \int vn)$; W. Kohn and L. J. Sham, Phys. Rev. 140 (1965) A1133 — $[-\frac{1}{2}\nabla^2 + v + v_H + v_{xc}]\varphi_k = \varepsilon_k\varphi_k$, $n = \sum_k|\varphi_k|^2$ (atomic units).
- A. K. Rajagopal and J. Callaway, Phys. Rev. B 7 (1973) 1912; R. M. Dreizler and E. K. U. Gross, *Density Functional Theory*, Springer (1990) — the energy as a functional of the current j^A, the Dirac form of the Kohn–Sham equations, the no-sea approximation.
- G. C. Wick, Phys. Rev. 80 (1950) 268; M. Fierz, Z. Phys. 104 (1937) 553 — contractions in a Slater determinant (above), and $\sum_K(\Gamma_K)_{ab}(\Gamma_K^{-1})_{cd} = 16\delta_{ad}\delta_{cb}$ for the 256 products Γ_K of the generators.
- J. D. Walecka, Ann. Phys. 83 (1974) 491 — mean-field saturation of uniform matter, $m^* = m - \frac{g_\sigma^2}{m_\sigma^2}\rho_S$, $V_0 = \frac{g_\omega^2}{m_\omega^2}\rho$ (here $g_S = -g_\sigma^2/m_\sigma^2$, $g_V = g_\omega^2/m_\omega^2$); N. M. Hugenholtz and L. Van Hove, Physica 24 (1958) 363 — $E/A = E_F$ at saturation.
- Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345 — $M = m - 2G\langle\bar\psi\psi\rangle$ with a cutoff and a critical coupling (here $G = -g/2$).

6.5 Verification of the second stage

Sections 7–9 of 7+1.nb contain 74 checks, all True; they are rows of section 4.2 above, and the evaluation is the one of section 4.1. `tools/ref_dirac16.py` recomputes the constructions independently with 90 boolean checks, all True, and its output is printed in section 4.3, together with that of `tools/verify_7p1_nb.py`, which compares the printed spinors, the closed-form integrals and 23 printed numbers with it (at least 17 digits) and the first part of the notebook with the backup.

7. The third stage: sections 11–19 of 7+1.nb

This section records the provenance of the gravity, Kohn–Sham and Standard Model sections (11–19) and of the final summary (then section 20, now section 25). Notation: $\Gamma_A = \Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A$, $\eta = \mathrm{diag}(+1,-1,\ldots,-1)$ (simply the squares of the octonion basis, $\eta_{AB} = \mathrm{Re}(e_A e_B)$), Ψ the field dirac16complex.

7.1 The task

1. Rebuild the ideas of github.com/once-ere/Dirac_claude on the 7+1 spacetime of this project (one time coordinate, seven space coordinates; never the 4+4 metric of Dirac_claude): the Einstein and Lovelock tensors; generalized frame matrices; a canonical spin connection, with a vanishing total covariant derivative, that couples gravity to dirac16complex; the Kohn–Sham theory of dirac16complex in gravitational fields in the primordial (near $x_0 = 0$), modern and far-future epochs; a generalized self-coupling; and the totality of the Standard Model.

2. Compute the Lovelock tensors as in the author's corrected reference notebook Einstein-Lovelock-7+1-Nash.nb, and do not use its section "FAILED Model metric g71; DO NOT USE!".

3. Document every stage (documentation, provenance and symbol files; Markdown, LaTeX, PDF; self-contained) and push every completed stage to github.com/once-ere/7plus1.

4. Make clear in all documentation, examples and provenance records that the xAct declaration $\{7,1,0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1,-1,\ldots,-1)$ is simply the squares of the octonion basis (the author's instruction of 27 October 2026).

7.2 Inputs and how they were used

The reference notebook Einstein-Lovelock-7+1-Nash.nb (P. L. Nash, 2022, GPL; the author's corrected version) supplied: the xAct packages xCoba and xTras; the manifold with abstract indices a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y; the chart with coordinate scalars `x0[]`, …, `x7[]`; `MyArrayComponents`; the generalized delta `kδ`; the three Lovelock tensors with their normalization

$$\mathrm{Lovelock}_k = -\frac{1}{2^{k+1}}\, R_{c_1 d_1}{}^{f_1 i_1} \cdots R_{c_k d_k}{}^{f_k i_k}\, \delta_a^{b\, c_1 d_1 \cdots c_k d_k}_{f_1 i_1 \cdots f_k i_k} \quad (k = 1,2,3);$$

the evaluation of components by `CD = LC[g]`; and the vacuum field equations $0 = -\Lambda\delta + H^{-2}w_1\mathrm{Lovelock}_1 + H^{-4}w_2\mathrm{Lovelock}_2 + H^{-6}w_3\mathrm{Lovelock}_3$ with the procedure of solving them for w_2, w_3. The model metric of its FAILED section was not used anywhere. Taken unchanged as well: the signature declaration `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]` — in xAct the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension, exactly the 7+1 spacetime of this project. It is not to be confused with the matrix $\eta = \mathrm{diag}(+1,-1,\ldots,-1)$, which is simply the squares of the octonion basis ($e_0^2 = +1$, $e_i^2 = -1$, $\eta_{AB} = \mathrm{Re}(e_A e_B)$) and gives the signs of the metric components ($g_{00} > 0$). **A correction (27 October 2026):** an earlier version of 7+1.nb declared $\{1,7,0\}$, reading the list as the signs of η; the author pointed out that $\{7,1,0\}$ declares seven space-like and one time-like dimension while $\mathrm{diag}(+1,-1,\ldots,-1)$ is simply the squares of the octonion basis, and 7+1.nb now declares $\{7,1,0\}$ as the reference notebook does. No Lovelock component changed: they follow from the component matrix of the metric, and `SignDetOfMetric` is -1 for both lists (checked with xTensor 1.2.0). xAct versions as printed by the notebook: xPerm 1.2.3, xTensor 1.2.0, xCoba 0.8.6, xTras 1.4.2.

```
Einstein-Lovelock-7+1-Nash.nb   SHA-256 b2dcb7b9f435d644da085fdd9e738a0cc80d3268ef28606ff9f367ed
    ...6828fdfc
Einstein-Lovelock-7+1-Nash.pdf  SHA-256 6580f9fcaff9c7f1337df35a7fce6b4b04c022d0e2d68954e8b3954b
    ...30f45125
```

Data of the Standard Model part: Particle Data Group 2024/2025 (G_F, boson masses, quark masses, CKM parameters), NuFit-6.0 (PMNS parameters, normal ordering, without Super-Kamiokande atmospheric data), Planck 2018 (cosmological parameters).

7.3 Decisions

#	decision	reason
1	Christoffel symbols written $\{^{\lambda}_{\mu\nu}\}$; Landau–Lifshitz curvature	Γ is reserved for Dirac matrices; with the signs of η ($g_{00} > 0$), $G^{\mu}{}_{\nu} - \bar{\Lambda}\delta^{\mu}_{\nu} = \kappa T^{\mu}{}_{\nu}$, $T^{0}{}_{0} > 0$
2	the Lovelock tensors are computed twice (xAct; pair-set sums)	exact agreement is an independent check of the reference method
3	the exponential solutions are obtained from the xAct components of the cosmological metric	a new xAct run with symbolic exponentials does not finish in reasonable time
4	the canonical (Levi-Civita) spin connection, $\Omega_{\mu} = \frac{1}{4}\omega_{\mu AB}\Gamma^{A}\Gamma^{B}$	it is the connection with $\nabla_{\mu}\Gamma_{\nu} = 0$
5	the self-coupling is organized by the nine ranks of the Clifford algebra	each rank is one Lorentz-invariant class; the Fierz matrix closes on them
6	Kohn–Sham matter: homogeneous, spin-saturated, no Dirac sea, adiabatic	the cosmological spacetime is homogeneous; the expansion rates are small compared with the orbital energies
7	the Standard Model on a 3+1 brane of a torus of the auxiliary coordinates	the 7+1 analogue of the fold of Dirac_claude
8	portals are invented local couplings at the brane	the only contact of dirac16complex with the Standard Model besides gravity

7.4 Literature

- D. Lovelock, J. Math. Phys. 12 (1971) 498; D. Lovelock and H. Rund, *Tensors, Differential Forms, and Variational Principles*, Dover (1989), Remark 2 after Eq. (4.38) — the Lovelock tensors (section 12.1).
- J. M. Martín-García et al., xAct: efficient tensor computer algebra for the Wolfram Language (xPerm, xTensor, xCoba), GPL, www.xact.es; T. Nutma, Comput. Phys. Commun. 185 (2014) 1719 (xTras) — the abstract-index computation of section 12.1.
- L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields*, Pergamon — curvature conventions; E. Kasner, Am. J. Math. 43 (1921) 217 — power-law solutions; G. W. Gibbons and S. W. Hawking, Phys. Rev. D 15 (1977) 2738 — the de Sitter temperature $H/2\pi$.
- S. Navas et al. (Particle Data Group), Phys. Rev. D 110 (2024) 030001 and the 2025 update; I. Esteban et al., JHEP 12 (2024) 216 (NuFit-6.0); Planck Collaboration, Astron. Astrophys. 641 (2020) A6 — the data of section 12.5–12.6.
- N. Arkani-Hamed, S. Dimopoulos, G. Dvali, Phys. Lett. B 429 (1998) 263; L. Wolfenstein, Phys. Rev. D 17 (1978) 2369; S. P. Mikheyev and A. Yu. Smirnov, Sov. J. Nucl. Phys. 42 (1985) 913 — extra dimensions, the MSW effect (section 12.5–12.6).

7.5 Verification of the third stage

Sections 11–19 of 7+1.nb contain 148 checks (gravity 100, Kohn–Sham 24, Standard Model 24), all True; they are rows of section 4.2 above. The independent programs recompute everything: `tools/ref_gravity.py` (101 boolean checks, 101 True), `tools/ref_ks_gravity.py` (29, 29), `tools/ref_sm.py` (33, 33); their complete outputs are printed in section 4.3.

8. The fourth stage: sections 20–23 of 7+1.nb

This section records the provenance of the coupled field equations and their solutions in the three gravity regimes (sections 20–23 of 7+1.nb, appended on 27 October 2026; the summary of all checks became section 24).

8.1 The task

1. Compute accurate approximate solutions of the coupled field equations of the model for all three gravity regimes; write each and every interaction in equation form; define all symbols; explain and analyze the solutions (summarized; the original wording is the author's private working material).
2. Document them in a documentation file with a provenance file and a symbol file, in Markdown and LaTeX, the LaTeX compiled to PDF; no document refers the reader to another file, and every equation used is written in the file that uses it.
3. Then analyze whether dirac16complex can be the dark matter and the dark energy, on the basis of these solutions (a separate stage).
4. Push every completed stage to github.com/once-ere/7plus1 and verify the repository.

Interpretation. The "three gravity regimes" are the three epochs of the model: primordial (near $x_0 = 0$; inside it the cubic, Gauss–Bonnet and Einstein terms dominate in turn), modern (a stabilized auxiliary space, four-dimensional gravity on the brane) and far future (de Sitter).

8.2 Inputs, data and decisions

Inputs from the earlier sections: the closed forms of the Lovelock tensors of the cosmological metric (section 12.6), the pair-set densities (section 12.5), the Kasner forms (section 14.1). Data: Particle Data Group 2024/2025 (masses), Planck 2018 ($H_0 = 67.4$ km/s/Mpc, $\Omega_m = 0.315$), $T_{\rm CMB} = 2.7255$ K, $N_{\rm eff} = 3.044$, $M_{\rm Pl} = 2.435 \times 10^{18}$ GeV. Choices: $M_8 = 10$ TeV, $\alpha_2 = 1/M_8^2$, $\alpha_3 = 10^{-6}/M_8^4$, $m_r = 10$ GeV, $f_W = M_8^4$; in regime I units $\ell_2 = 1$, $\xi = 10^{-6}$, and a massless collisionless dirac16complex gas normalized to $\kappa\varepsilon = 10^{-6}\times 3H_a^2$ at $a = b = 1$, $H_a = 1$; the size of a curvature term is the sum of the absolute values of its monomials (in a Kasner regime the net value cancels). Before the new sections were appended the evaluated notebook was copied to `backups/7+1_2026-10-27_before_coupled.nb` (excluded by .gitignore).

8.3 Verification of the fourth stage

Sections 20–23 of 7+1.nb contain 48 checks (48 True); they are rows of section 4.2 above. The independent program `tools/ref_coupled.py` (49 boolean checks, 49 True) repeats every derivation and numerical solution with its own code; the 34 groups of numbers printed by both programs agree to a largest relative difference of 0.00032 (tolerances from 10^{-14} for exact constants to 2×10^{-3} for quantities read off long numerical integrations); its complete output is printed in section 4.3.

9. Section 24 of 7+1.nb: dirac16complex as dark matter and dark energy

This section records the provenance of section 24 of 7+1.nb (appended on 27 October 2026; the summary of all checks became section 25).

9.1 The task

1. On the basis of the accurate approximate solutions of the coupled field equations in the three gravity regimes, analyze the validity of associating dark matter (and dark energy) with dirac16complex (summarized; the original wording is the author's private working material).
2. Document the analysis in a documentation file with a provenance file and a symbol file, in Markdown and LaTeX, the LaTeX compiled to PDF; no document refers the reader to another file, and every equation used is written in the file that uses it.
3. Push every completed stage to github.com/once-ere/7plus1 and verify the repository.

9.2 Inputs, data and decisions

Inputs from the earlier sections: the stress tensor from the comoving energy (section 20.2), the radion equation (20.3), the gas-dominated regime without stabilizer (21.3), the gas share after capture (22.1), the radion decay rates into brane fields and into the bulk with Jacobi's $r_4(k)$ (22.2), the late universe (22.4), the Standard Model constants (17.3), the uniform Kohn–Sham matter (8.7). Data: Planck 2018; DESI DR2 (arXiv:2503.14738) and Garcia-Quintero et al. (arXiv:2504.18464); LZ (arXiv:2410.17036); Randall et al. (Bullet Cluster); Iršič et al. (Lyman-α); Ballesteros, Garcia and Pierre (decay-produced dark matter); Alvey et al. (Pauli bound, $g = 2$, scaling $g^{-1/4}$); Roach et al. (NuSTAR); Particle Data Group 2024 (large extra dimensions). Decisions: natural couplings at M_8; Majorana normalization of the sterile-neutrino decay rates; instantaneous reheating at $T_{\rm RH} \le T_*$; production by Higgs decays as a lower bound; continuum Kaluza–Klein sums checked against lattice sums; Pauli bound for $g = 16$. Before section 24 was appended the evaluated notebook was copied to `backups/7+1_2026-10-27_before_dark.nb` (excluded by .gitignore). One correction to section 23.3 was made: the cold-dark-matter density in its table read "0.265" and now reads "Omega_c = 0.264" ($0.120/0.674^2 = 0.26416$).

9.3 Verification

Section 24 of 7+1.nb contains 20 checks (20 True); they are rows of section 4.2 above. The independent program `tools/ref_dark.py` (20 boolean checks, 20 True) recomputes every number with its own code; the 29 groups of numbers printed by both programs agree to a largest relative difference of 1.4×10^{-7}; its complete output is printed in section 4.3. `tools/verify_7p1_nb.py` confirms that, with section 24 removed, the notebook equals the backup except the documented correction.

10. Authorship

Notebook, tools and documentation written by Claude (Anthropic, model claude-opus-5-5) for Nash: the first stage (sections 0–6 of 7+1.nb) from the task statement and the notebook 4+4.nb described in section 1, the second stage (sections 7–10) from the task of section 6.1, the third stage (sections 11–19) from the task of section 7.1 and the reference notebook Einstein-Lovelock-7+1-Nash.nb of its author, the fourth stage (sections 20–23) from the task of section 8.1, section 24 from the task of section 9.1; this document was generated on 2026-10-27.